GNBF5020

The Foundation Course

to Molecular Biology and Genetics

Introduction to Molecular Biology and Genetics · Dr. Tony Wong · GNBF5020

The foundation of the course, taught as a course. Eight modules in a fixed order, from the atom to the working cell: chemistry, biomolecules, proteins, DNA and genes, the cell, and how a cell actually works. Each module carries a reading budget, the Cantonese narration to listen to, the short video guides that reinforce it, and the questions that check it.

Every technical term is defined the moment it is used, and the whole 905-term vocabulary of the course is in this book. Technical terms stay in English, because that is how you will be examined.

About this edition

This is the foundation half of the Beginner's Guide, standing on its own. It covers Part 0 and Chapters 1 to 7 - the part of the guide that assumes no biology and no chemistry - and it adds the two things the guide does not have: a learning path, and a glossary that lives with the foundation instead of at the end of the exam pack. Course: GNBF5020 Introduction to Molecular Biology and Genetics. Lecturer: Dr. Tony Wong.

The learning path

Work through the modules in this order. Each one ends with a "check yourself" block in the book; the quiz questions listed here are the same material in a form you can mark yourself. The times are a reading budget, not a deadline.

#	Chapter	What you will be able to do afterwards	Read
M0	Part 0 · Read this first	You know how the course is marked, what an exam hook is, and why every answer has to be in English.	25 min
M1	Ch 1 · Chemistry from zero: atoms to molecules	You can say what an atom, a bond and a molecule are, and why carbon builds all the rest of biology.	60 min
M2	Ch 2 · The four kinds of biomolecule	You can name the four kinds of biomolecule, say what each one is built from, and give one job for each.	60 min
M3	Ch 3 · Proteins: what they are and how they work	You can explain how a chain of amino acids becomes a folded machine, and why shape is function.	75 min
M4	Ch 4 · DNA, genes and chromosomes	You can explain what DNA, a gene and a chromosome each are, and how a typo in the manual becomes a disease.	75 min
M5	Ch 5 · The cell and its parts	You can name the parts of a cell and say what each one does.	60 min
M6	Ch 6 · How a cell actually works	You can follow the path from DNA to protein, say how damage is repaired, and explain how a gene is switched on.	75 min
M7	Ch 7 · The foundation glossary	Every term of the course is defined, with a plain-English meaning and the place it does its work.	—

Audio, video and quiz for each module are listed with the module itself, in the chapter. The Cantonese narration of the whole book is in **_cantonese/index.html**; the quiz is in **_site/quiz.html**.

Contents

Eight Parts are the big dividers of the book; thirteen Chapters are the numbered sections you actually read. Each of the fourteen figures gets a full page of its own, placed at the end of the section that introduces it.

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Chapter 4 - DNA, genes and chromosomes [the recipe, the printed manual, and the typo] 38

Chapter 5 - The cell and its parts 48

Chapter 6 - How a cell actually works 58

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Video guides

Sixteen short video guides go with this course: English narration, Traditional Chinese subtitles. Six of them sit on the foundation chapters below and the rest on the lecture chapters. The MP4 files live in the project folder **_video/assets/** - the full path is printed next to every link, so you can still find a file after this PDF has been moved. **_video/index.html** plays them all on one page.

Video	Term	Chapter	Video file
V01	**Kaplan–Meier curve**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V01_Kaplan_Meier_curve.mp4
V02	**Contractile ring → binucleated cell**	Ch 8 · Cell biology	_video/assets/V02_Contractile_ring_binucleated_cell.mp4
V03	**NHEJ defect → SCID**	Ch 6 · DNA damage and repair	_video/assets/V03_NHEJ_defect_SCID.mp4
V04	**Mitochondrial pedigree**	Ch 8 · Cell biology	_video/assets/V04_Mitochondrial_pedigree.mp4
V05	**Meiosis I vs Meiosis II (M1 vs M2)**	Ch 8 · Cell biology	_video/assets/V05_Meiosis_I_vs_Meiosis_II_M1_vs_M2.mp4
V06	**BER vs NER**	Ch 6 · DNA damage and repair	_video/assets/V06_BER_vs_NER.mp4
V07	**Hot tumour vs cold tumour**	Ch 9 · Immunology and cancer immunity	_video/assets/V07_Hot_tumour_vs_cold_tumour.mp4
V08	**Driver vs passenger**	Ch 10 · Cancer biology and targeted therapy	_video/assets/V08_Driver_vs_passenger.mp4
V09	**IgG1 vs IgG4**	Ch 9 · Immunology and cancer immunity	_video/assets/V09_IgG1_vs_IgG4.mp4
V10	**Predictive vs prognostic biomarker**	Ch 11 · Assay techniques and omics	_video/assets/V10_Predictive_vs_prognostic_biomarker.mp4
V11	**cis vs trans**	Ch 7 · Control of gene expression	_video/assets/V11_cis_vs_trans.mp4
V12	**de novo vs maintenance methylation**	Ch 4 · Chromatin and epigenetics	_video/assets/V12_de_novo_vs_maintenance_methylation.mp4
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Open the video index page ->

PART 0

Read this first

0.1 How to use this guide

This guide is written for a reader who has **zero prior knowledge of biology and zero prior knowledge of chemistry**. That is not a polite assumption. It is the stated audience.

You do not need to know what an atom is. You do not need to know what a cell is. You do not need to have met the word "gene" before. This book starts at the very bottom and builds up.

Here is how the book is put together.

- **Part 0** is the page you are reading now. It tells you the one rule, what the course covers, and how to read the rest.
- **Part 1 (Foundations, Chapters 1 to 7)** teaches the background the lectures assumed you already had. Chapters 1 and 2 are here, in this file. Chapter 3 is proteins, Chapter 4 is DNA and chromosomes, Chapter 5 is the cell, Chapter 6 is how a cell actually works, Chapter 7 shows you where the plain-English glossary is.
- **Parts 2 to 6** follow the five lectures in order. One part per lecture.
- **Part 7** is the exam pack: the master glossary, the five questions Dr. Wong named as exam-type, the key numbers, the "will not be examined" list, and a self-test.

Every chapter follows the same shape. Learn the shape once and you never have to hunt for anything again.

Section	What it is	How long it takes you
n.0 What this chapter is about	Three to five sentences saying why the lecturer taught this.	1 minute
n.1 Words to know before we start	A table of the terms the chapter uses, in plain English.	3 minutes
n.2 The big picture	The chapter told as a story, in flowing prose. This part stands on its own.	5 minutes
n.3 Walkthrough	Topic by topic, following the order the lecturer taught.	The main work
n.4 The N things to remember	The short list you will actually be able to recite in the exam.	5 minutes
n.5 Check yourself	Five questions, then five answers that show the reasoning.	10 minutes

If you read only one thing in each chapter, read the Big Picture. That is the whole orientation in one sentence.

Two house rules run through everything below.

First, **every technical term is defined the moment it is used**. When you meet a new word, the sentence you are in also tells you what it means. You never have to look anything up.

Second, **technical terms stay in English**. The Chinese edition of this book translates the explanation and keeps the term in English. That is deliberate, because that is how you will be examined.

0.2 The one rule: write every answer in English

[EXAM HOOK] THE ONE RULE THAT MATTERS MOST

Dr. Wong said this three separate times across the lectures. He said it on day one, and he said he gave zero marks because of it.

> **Lecture 1**
> *"every assignments and exam should be done in English, even this is a Chinese University of Hong Kong."*

> **Lecture 1**
> *"I'm not kidding, because last year there are some student they answered the question in Chinese, and I have no choice that I give the zero mark, because everything should be in English."*

> **Lecture 4**
> *"you have to answer in English, although this is a CUHK, haha, okay."*

He was joking the third time. He was not joking the first two.

What this means for you, in plain terms.

1. **The midterm is graded in English.** Answer in English, even in complete sentences that are short and simple. Simple English scores better than no English.
2. **A Chinese answer scores zero**, not "partial credit for the idea". This is the specific thing he said he did last year.
3. **Write full sentences.** His advice for the short questions is to defend your reasoning. A page of bullet points is not an answer.
4. **Keep the technical terms in English** inside those sentences. Write "the KRAS mutation means cetuximab will not work" in English, not in Chinese with one English word in it.
5. **This book is in English for that reason.** It is the language you will be graded in, so this is the language to revise in.

One more sentence from him, because it tells you what he is actually grading:

> **Lecture 1**
> *"multiple choice is for you to pass the exam. But if you want to get a full mark or a better result, you'd better perform well for the short questions, which emphasize on data analysis and also the scenarios questions."*

So: the multiple choice questions (MCQs) are the floor. The short questions are where the marks are. Both are in English.

0.3 What this course is, and how the five lectures build on each other

GNBF5020 is an introduction to molecular biology and genetics. The name sounds like two subjects. It is really one subject with two halves.

The first half is **molecular biology**. What are living things made of? How do cells take in fuel and turn it into usable energy? How do they build machines out of small pieces? How do they switch those machines on and off?

The second half is **genetics**. How is the recipe for those machines stored? How is it copied? How is it repaired? How is it changed, and what happens when it is changed wrongly?

Dr. Wong set the course goal himself, on the first lecture.

His stated goal for the course: to translate genomics and molecular biology into **clinical and real-world applications** - diagnostic, prognostic and predictive biomarkers.

A **biomarker** is a measurable thing in a sample - blood, tissue, urine - that tells you something about disease. A **diagnostic biomarker** tells you the patient has the disease. A **prognostic biomarker** tells you how the disease is likely to go. A **predictive biomarker** tells you which drug will work. He came back to this framing in the first two lectures.

There are also two running themes, which he put on his overview slide (Slide 4) next to a portrait of Gregor Mendel, a multi-generation family tree, and a comparison of sperm and egg formation.

- **Diversity** - genetic variation. Why do we look different from each other? How do new combinations of genes arise? How do tumours become genetically different from one another?
- **Stability** - reliable inheritance. How is the recipe copied without errors? How is damage repaired? How does the cell remember what it was doing last week?

Every lecture touches both. The memorable ones are the moments where they collide. Recombination creates diversity but can also break a gene. Epigenetic marks give stability but can lock a cell into the wrong state.

Here is the map of the five lectures.

No.	Date	Title	What you will be able to do after it
1	12 Sep 2026	Introduction to Biomolecules	Name the four classes of biomolecule, and explain how cancer changes cell metabolism. Read a metabolic or proteomic dataset and see a biomarker.
2	19 Sep 2026	Basic Cell Biology	Use the core tools (FACS, microscopy, FISH, IHC). Explain what organelles do and what breaks when they fail. Run the cell cycle, apoptosis, stem cells, CAR-T and liquid biopsy.
3	3 Oct 2026	DNA and Chromatin Structure	Explain how an enormously long piece of DNA fits inside a nucleus. Read a karyotype. Understand chromatin, methylation, the histone code, telomeres, copy number, driver and passenger.
4	10 Oct 2026	DNA Replication, Repair and Recombination	Follow the replication fork. Match each kind of DNA damage to its repair pathway. Use PCR, cloning and CRISPR. Read a clinical trial design.
5	17 Oct 2026	Control of Gene Expression	Explain how the cell decides what to make and when. Work with the lac operon, eukaryotic promoters, splicing, nuclear receptors, signalling pathways and microRNA.

Here is the same map as one chain, which is the version to keep in your head.

1. **Lecture 1 asks: what are we made of?** Four families of molecules. Sugar, fat, protein, DNA. Each one is also a fuel, a building material, or a machine. This is the vocabulary for the whole course.
2. **Lecture 2 asks: where does the work happen?** Inside a cell. A cell is a factory. It has a control room (the nucleus), power stations (mitochondria), a factory floor (ribosomes), and a set of tools (the methods: FACS, microscopy, IHC, FISH).
3. **Lecture 3 asks: where is the recipe, and how is it switched on and off?** DNA is the recipe. Chromatin is the filing cabinet and the on/off switch. Methylation and histone marks are the labels on the folders.
4. **Lecture 4 asks: how is the recipe copied and repaired?** This is the maintenance half. Copying has errors, damage happens, repair systems fix most of it, and when they fail you get cancer. This is also where CRISPR comes from.
5. **Lecture 5 asks: how does the cell control the output?** Having the recipe is not enough. The cell decides when to use it, which version to use, and when to stop. This is the control half, and it is where most drug targets live.

Chapters 1 and 2 of this book sit underneath that entire chain. Chapter 1 gives you the chemistry - atoms, bonds, water, energy, ATP. Chapter 2 gives you the four molecule families from Lecture 1, but taught from zero.

0.4 How the marks work, and how to read the exam-hook boxes

First, how the marks work. From the exam-focus notes built from his own statements:

Item	What he said	Where
Midterm weight	**30 %** of the final grade. The final exam is **40 %**. Two "super-easy" assignments make up the other **30 %**.	Lecture 1, Slide 2
Duration	**1.5 hours (90 minutes)**	Lecture 5
Part A	**Multiple choice, single best option only.** Last year students lost marks by trying to select multiple options.	Lectures 4, 5
Part B	**Data-analysis or scenario questions.** Choose one (Lecture 4) or two out of three (Lecture 5).	Lectures 4, 5
Language	**English only. Zero marks for Chinese answers.**	Lectures 1, 4, 5

There is one inconsistency you should know about, because he flagged it himself. In Lecture 4 he described "about 25 questions, one mark each" with two short questions where you choose one. In Lecture 5, which was later and was written on his own slide, he described "20 multiple choice questions, and two out of three short question ... 20 marks ... short question is account for 10 marks ... 1.5 hours".

Plan for the Lecture 5 version:

20 MCQ (20 marks) + choose 2 of 3 short questions (10 marks) = 30 marks, in 90 minutes.

Either way the ratio is identical, so the strategy is identical. MCQs are pure recall and they get you over the line. The scenario questions are where the remaining marks are.

He named only **two families** of scenario question, and they are worth practising:

1. **A pedigree or genetic-counselling report.** "If you are a genetic consultant and I give you a medical report, how would you advise the client whether he or she will have a baby with a certain genetic disease?"
2. **A survival dataset.** "I give you some survival data, and then you tell me how this survival data can really benefit the treatment of the patient."

Second, how to read the boxes in this book. There are two kinds of marker, and the difference between them is the whole point.

EXAM HOOK	means Dr. Wong's own words, his own numbers, or something he said he would ask. Where his exact wording matters, you will get his exact wording in quotation marks.

- **He did not flag this as examinable** means he never mentioned that point as a possible question. It is **not** a claim that the topic is off the paper. It is a statement that he gave you no signal, so do not spend your time memorising it.

Everything that is not in a hook is **background explanation**. That is teaching, not invention. In plain terms, it is the part of science the lectures assumed you already had. Chapter 1 of this book is almost entirely that kind of material, because he told the class not to worry about it:

> **Lecture 1**
> *"Please don't memorize all these chemical structure ... They would not be included in my exam."*

So you will notice that Chapter 1 says he did not flag this very often. That is honest, and it is good news: it means the chemistry is something you understand once rather than memorise.

Third, the numbers. Every number Dr. Wong stated is reproduced exactly, and followed immediately by a plain-English sentence saying what it means and why it matters. Numbers are the easiest MCQ fodder there is, and they are also the easiest thing to get wrong by a factor of ten.

Fourth, a habit worth forming now. He said he is emphasising on how you analyse the scientific questions, not on memorised facts. The exam notes turn that into a four-move shape - state the observation with the number, name the mechanism, interpret the clinical meaning, then state the action or the limit. That is a sensible reading of what he said, not a quotation of it, and it is a good answer shape to practise. Read the answer templates in Part 7 before you sit the exam.

0.5 A sixty-second orientation

If you read only one thing in each chapter, read the Big Picture. That sentence is the entire navigation system of this book.

Here is the sixty-second version of the whole course.

1. **Everything is atoms.** Atoms join into bonds, bonds make molecules, molecules join into long chains called macromolecules, and macromolecules fill a cell. A cell is a factory, and the factory runs on energy.
2. **There are four families of macromolecule.** Carbohydrates (sugar and starch), lipids (fat and membranes), proteins (the machines), nucleic acids (the information). Everything else in the course is a detail of one of these four.
3. **Information flows one way.** DNA is the recipe. mRNA is a photocopy of one recipe. A ribosome reads the photocopy and builds a protein, which is a folded machine made from a bead necklace. This is the central dogma, and Lecture 5 is about how the cell controls it.
4. **Most of the course is about damage and repair.** DNA gets damaged. Repair systems fix most of it. When repair fails, you get mutations, and when the mutation hits the wrong word, you get cancer. Mutation means a typo. A driver is a typo in an important word. A passenger is a typo on page 900.
5. **Almost everything is a drug target or a biomarker.** If you can measure it, you can test for it. If you can switch it on or off, you can drug it. That is why the course is built around clinical application rather than around taxonomy.
6. **And you answer in English, in full sentences, with the number, the mechanism, the clinical meaning, and the limit.** Zero marks for a Chinese answer. He said so three times.

PART 1

Foundations: biology and chemistry from zero

IN THIS PART

Chapter 1 · Chemistry from zero: atoms to molecules

Chapter 2 · The four kinds of biomolecule

Chapter 3 · Proteins: what they are and how they work [the folded machines that do the cell's work]

Chapter 4 · DNA, genes and chromosomes [the recipe, the printed manual, and the typo]

Chapter 5 · The cell and its parts

Chapter 6 · How a cell actually works

Chapter 7 · The plain-English glossary

About this part

Part 1 - Foundations - Chapters 3 and 4 - Dr. Tony Wong

Part 1 - Foundations (continued) - Chapters 5, 6 and 7

Course: GNBF5020 Introduction to Molecular Biology and Genetics - **Instructor:** Dr. Tony Wong - **Dates:** Sep-Oct 2026

This file continues Part 1 of the beginner guide. Chapter 5 gives you the cell and its parts. Chapter 6 shows you the cell at work. Chapter 7 shows you where the big alphabetical glossary lives - the one that turns every technical word in this book into plain English.

CHAPTER 1

Chemistry from zero: atoms to molecules

1.0 What this chapter is about

You cannot understand a cell, a gene or a drug without knowing what atoms and bonds are. This chapter builds that from nothing.

It starts with the smallest piece of matter, and ends with the question that drives the whole course: where does a cell get its energy, and how does it spend it?

Dr. Wong did not teach this material. He taught the biology on top of it and assumed you already had it. He even told you to skip the details. That is why this chapter exists.

1.1 Words to know before we start

Term	In plain English
atom	The smallest piece of an element that still behaves like that element. The unit that everything is built from.
proton	A particle with a positive charge. The number of protons decides which element an atom is.
neutron	A particle with no charge, sitting in the nucleus. It adds mass without changing the element.
electron	A particle with a negative charge that moves around the nucleus. Electrons do the bonding.
electron shell	A fixed energy layer around the nucleus. Each shell holds a set number of electrons.
covalent bond	Two atoms sharing a pair of electrons. This is the bond that builds most of biology.
ionic bond	A strong attraction between atoms with opposite charges, with no sharing.
hydrogen bond	A weak, temporary tug between a slightly positive hydrogen and a slightly negative atom.
molecule	Two or more atoms joined together. A molecule has a shape and its own properties.
functional group	A small cluster of atoms on a molecule that gives that molecule its chemical personality and its reactivity.
solvent	The liquid that something else dissolves in. For a cell, that liquid is water.
pH	A number from 0 to 14 that says how acidic a liquid is. 7 is neutral, below 7 is acidic, above 7 is alkaline.
ATP	Adenosine triphosphate. The molecule a cell spends as energy, the way you spend money.
oxidation and reduction	Losing electrons (oxidation) and gaining electrons (reduction). They always happen together.

1.2 The big picture

Everything you will meet in this course - a protein, a piece of DNA, a sugar, a cell - is built out of atoms. Start there and molecular biology stops being frightening. It becomes an assembly problem whose rules you can learn.

An **atom** is the smallest piece of an element that still behaves like that element. "Element" means a substance made of only one kind of atom. Carbon is an element, and carbon is the one this whole course is built on.

Every atom has a dense centre called the **atomic nucleus**, holding **protons** and **neutrons**. Protons carry a positive charge, and their number is the atom's identity: six protons means carbon. Neutrons carry no charge and mostly add bulk. Around the nucleus orbit **electrons**, which carry a negative charge. They occupy **shells**, fixed energy layers, and the outermost shell decides what an atom can do.

Here is the rule that does most of the work. An atom is happy when its outer shell is full. For carbon a full shell means four electrons in the outer layer. Carbon has four, so it wants four more, and will share with one atom here and two atoms there. That single fact is why carbon can build chains, rings, branches and cages. It is why there is organic chemistry at all, and why there is you.

Three kinds of attraction hold atoms together, and you must keep them apart. A **covalent bond** is two atoms sharing a pair of electrons, so both get a fuller outer shell. This is the bond that builds molecules: in water, in sugar, in a protein backbone. An **ionic bond** is a strong pull between atoms with opposite charges, with nothing shared - sodium and chlorine. A **hydrogen bond** is much weaker: the tug between a slightly positive hydrogen and a slightly negative atom such as oxygen. They break and reform on demand, which is exactly what makes them useful. They are the folding pins in proteins, the pairing rules in DNA, and the reason water behaves so usefully.

When atoms join they make a **molecule**. One oxygen and two hydrogens make one water molecule. A molecule has a definite shape, and the shape decides what it can touch. Water is not a straight line. It is bent, because the oxygen pulls the two hydrogens to one side. That bend makes water **polar**, one end slightly positive and the other slightly negative. Polar attracts polar and ignores non-polar. That asymmetry explains why water dissolves salt and sugar while oil refuses to mix, and why a phospholipid builds a membrane.

Life needs water for four structural reasons. It is the **solvent** in which every reaction happens, and a **participant** in the chemistry of proteins and ATP. Its **hydrogen bonds** give large molecules their three-dimensional shape. And its **high specific heat** keeps body temperature steady.

Acidity is measured on the **pH** scale, from 0 to 14. Seven is neutral, below seven is acidic, above seven is alkaline. Each step is a ten-fold jump, so the scale is logarithmic. pH matters here for one reason: it changes protein shape. Charge attracts charge, so pH changes which bits of a protein stick to which. His slides use this. He showed the **deamidation** pathway, in which an asparagine (Asn) side chain becomes an aspartate (Asp) side chain at high pH, and the mass moves from 114.04 Da to 115.03 Da. One unit of mass in a drug can change how well it binds its target.

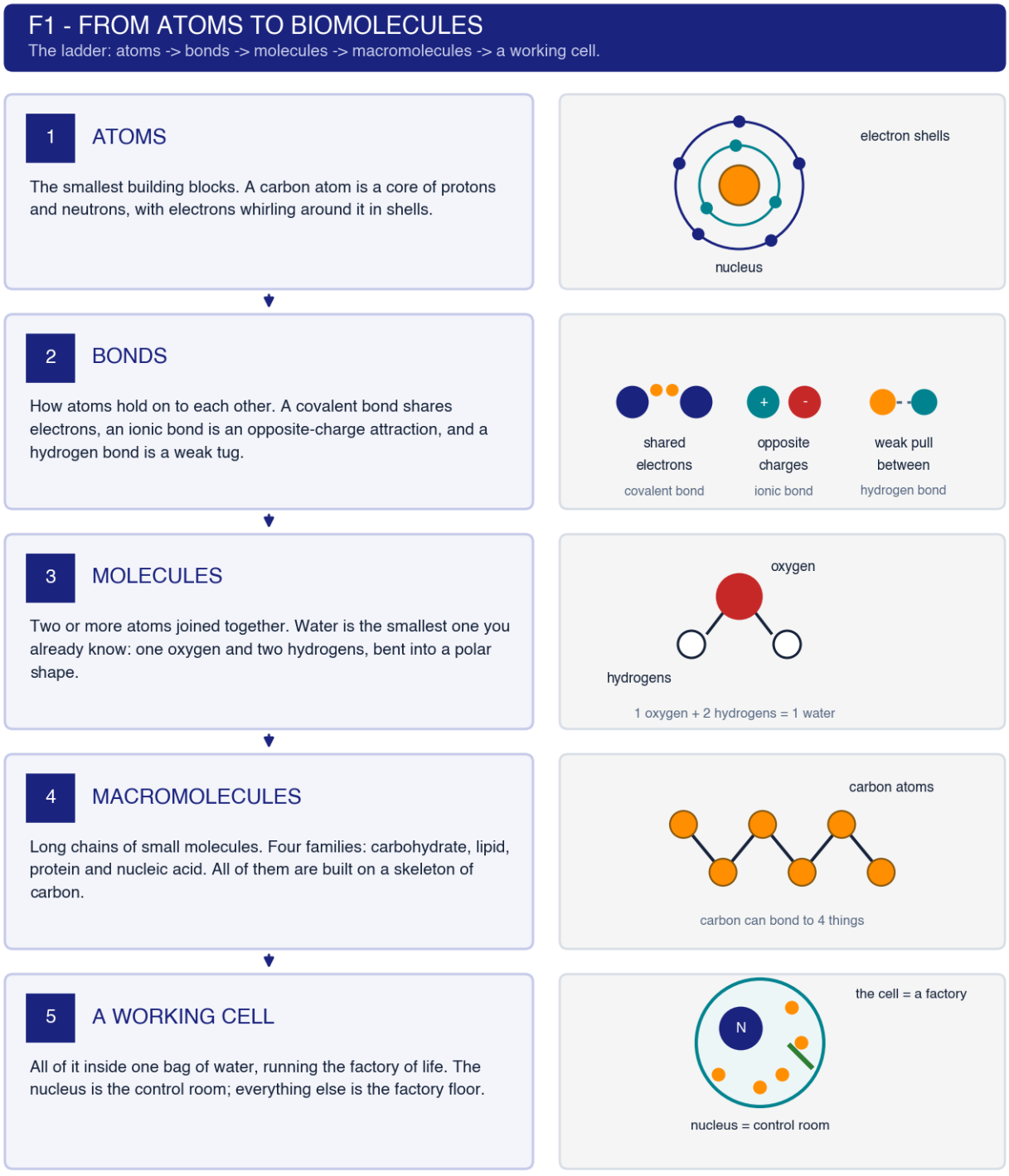
Now the part that drives everything else: **energy**. In a cell, energy is not a vague warmth. It is the capacity to do work - to move something, build something, change a chemical. And work is almost always paid for in one currency: **ATP**, adenosine triphosphate. ATP is the spendable unit. A kinase takes a phosphate from ATP and sticks it onto a protein; a phosphatase takes it off again. That one switch is how the cell turns things on and off. In his Hippo pathway example, MST1 and LATS1 add phosphate to YAP, and phosphorylated YAP is pushed out of the nucleus so the cell stops dividing. Remove it and YAP walks in and the cell grows.

Finally, **oxidation and reduction**. Oxidation means losing electrons. Reduction means gaining electrons. They are two halves of one transfer and always happen together. A useful handle: oxidation often means losing hydrogen, reduction often means gaining it. When a cell burns sugar with oxygen, the sugar is oxidised, the oxygen reduced, and the released energy is captured as ATP.

That is the whole chapter. Atoms want to be satisfied. Bonds satisfy them. Molecules with shapes make membranes. Water holds the reaction together. pH tunes the machinery. ATP pays for the work. And electrons moving is what energy is.

1.3 Walkthrough

Figure F1 - From atoms to biomolecules



The ladder, in five rungs: atoms, bonds, molecules, macromolecules, and finally a working cell. Each rung on the right carries the picture for that step - an atom with its shells, the three ways atoms hold on to each other, one water molecule, a chain of carbons, and a cell that is a factory.

1.3.1 The building block of everything - atom, proton, neutron, electron

WHAT IT IS	An **atom** is the smallest piece of an element that still behaves like that element. Its centre, the atomic nucleus, holds **protons** (positive) and **neutrons** (no charge). **Electrons** (negative) orbit outside it in shells. The count of protons is the element's identity: 6 protons is carbon, 8 is oxygen, 1 is hydrogen.
HOW IT WORKS / WHY IT MATTERS	Every molecule in this course is built from atoms, so every mechanism in this course is really a story about electrons. Change an electron and you change the chemistry. That is the whole reason chemistry belongs in a molecular biology course. Also: the atoms in your body are not special. They are the same carbon, oxygen, hydrogen and nitrogen that are in the air and in a textbook. What is special is which atoms are joined to which, and in what order.
EXAM HOOK	He did not flag this as examinable. He did say on Lecture 1: "Please don't memorize all these chemical structure ... They would not be included in my exam." Learn the idea so the rest of the course makes sense. Do not memorise atomic numbers.

1.3.2 The layers around the nucleus - electron shells

WHAT IT IS	**Electrons** do not float anywhere. They sit in **electron shells**, which are fixed energy layers around the nucleus. Each shell holds a maximum number of electrons: two in the innermost, then eight in the second.
HOW IT WORKS / WHY IT MATTERS	The outermost shell, the valence shell, decides what an atom can bond to. An atom with a full outer shell is stable and does not react. An atom with a partly filled outer shell is looking for a partner. This single rule explains why carbon is so useful: it has four electrons in its outer shell and needs four more, so it always has four hands to hold on to other atoms.
EXAM HOOK	He did not flag shell structure as examinable. No number he stated in the exam-focus document depends on it.

1.3.3 How atoms hold on to each other - covalent, ionic and hydrogen bonds

WHAT IT IS	A **covalent bond** is two atoms sharing a pair of electrons. An **ionic bond** is a strong attraction between oppositely charged atoms, with no sharing. A **hydrogen bond** is a weak, temporary attraction between a slightly positive hydrogen and a slightly negative atom such as oxygen.
HOW IT WORKS / WHY IT MATTERS	All three appear in the course, and they do different jobs. Covalent bonds are strong and permanent; they build the skeleton of every molecule. Ionic bonds hold salts and are used in batteries and in mineral bone. Hydrogen bonds are weak but reversible, and reversibility is the point: they let a protein fold one way and then change shape, they let the two strands of DNA unzip so the cell can copy them, and they let the base pairs of the double helix be swapped for the wrong letter.
EXAM HOOK	He did state the hydrogen bond numbers, and they are the numbers most likely to appear as an MCQ stem. **Guanine pairs with cytosine using 3 hydrogen bonds. Adenine pairs with thymine using 2.** In plain English: a GC-rich piece of DNA needs more heat to pull the two strands apart, because there are more cross-links holding them together. That is why melting temperature depends on GC content. His slide gave the direction, not a number: more GC content means a higher melting temperature, and that is what matters when you design a PCR primer. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and those two do not reconcile with each other - so learn the direction and the reason, and do not put a number on it.

1.3.4 Two or more atoms joined - molecule

WHAT IT IS	A **molecule** is two or more atoms joined together. The atoms are held in a fixed arrangement, and the molecule has its own shape and its own properties. It is not a bag of separate atoms any more. Water is one oxygen and two hydrogens. Glucose is six carbons, twelve hydrogens and six oxygens.
HOW IT WORKS / WHY IT MATTERS	The shape of a molecule decides everything it does, because shape decides which parts touch what. This is the idea you will meet again as "structure determines function" in proteins, and it is the same idea again when a DNA base pairs with its partner. Dr. Wong's slides use the two simplest sugars as examples: glucose and fructose have exactly the same formula, C6H12O6, yet they behave differently because their atoms are arranged differently. Glucose forms a six-membered ring. Fructose forms a five-membered ring. Same parts, different shape, different job.
EXAM HOOK	He did not flag the formula or the ring shapes as examinable, and he explicitly excluded memorising chemical structures. The idea that shape decides function is examinable in the protein form, and that is Chapter 3.

1.3.5 The element that builds life - carbon

WHAT IT IS	**Carbon** is the element at the centre of all organic chemistry. It has four electrons in its outer shell, so it forms four covalent bonds. It bonds to itself in chains, rings and branches, and it bonds to hydrogen, oxygen, nitrogen and sulphur.
HOW IT WORKS / WHY IT MATTERS	Because carbon can make chains, and chains can be any length and any shape, one element can build an enormous variety of different molecules. Sugars, fats, proteins and DNA are all carbon-based. That is why we call them "organic" molecules even though there is no organic chemistry of the supermarket kind involved. A useful mental picture: carbon is the steel of biology, and the other atoms are the rivets and panels.
EXAM HOOK	He did not flag carbon itself as examinable. It matters here because it is the reason the four molecule families in Chapter 2 can all exist at the same time.

1.3.6 The reactive part of a molecule - functional groups

WHAT IT IS	A **functional group** is a small cluster of atoms on a molecule that gives that molecule its chemical personality. Common ones are the hydroxyl group (an oxygen with a hydrogen attached, written -OH), the carboxyl group (a carbon with two oxygens, written -COOH), the amino group (written -NH2) and the phosphate group (written -PO4).
HOW IT WORKS / WHY IT MATTERS	The long carbon chain of a molecule is mostly a handle. The interesting chemistry happens at the functional groups, and it is the functional group that tells you how a molecule will behave. A carboxyl group makes an acid. An amino group can accept a hydrogen, which is how amino acids join into proteins. A phosphate group carries charge, which is why adding and removing phosphates is the cell's favourite on/off switch. The famous exception is carbon dioxide: it is what you are left with after a carboxyl group is removed, so it has no carbon chain of its own.
EXAM HOOK	He did not flag functional groups as examinable, and he excluded memorising chemical structures. But he did build one functional group into a clinical teaching point: the phosphate group, used by kinases and phosphatases. That is covered next.

1.3.7 The liquid life runs in - water

WHAT IT IS	**Water** (H2O) is one oxygen and two hydrogens. Because oxygen is much bigger and pulls the shared electrons harder, the molecule is bent and **polar**: one end is slightly negative, the other slightly positive.
HOW IT WORKS / WHY IT MATTERS	Four jobs, all structural. (1) It is the **solvent** - the liquid everything dissolves in, so every reaction in the cell happens in it. (2) It is a **reactant** in key reactions, including the reactions that make and spend ATP. (3) Its **hydrogen bonds** give proteins their three-dimensional shape, because they hold one part of a chain against another. (4) Its **high specific heat** buffers temperature changes, so the cell does not swing wildly when it is working. Water is also the reason a phospholipid membrane works, because a polar molecule dissolves in polar water and a non-polar one does not. That is the entire trick behind the bilayer.
EXAM HOOK	He did not teach water as chemistry. He used it as scenery on the slides, for example when he showed the phosphatase in the phosphorylation switch cartoon using water to release inorganic phosphate. The examinable version is the membrane consequence, and that is the next chapter.

1.3.8 How acidic a liquid is - pH

WHAT IT IS	**pH** is a number that says how acidic a liquid is, running from 0 to 14. Seven is neutral, lower is acidic, higher is alkaline. Each step is a ten-fold change in acidity, so the scale is logarithmic, not linear.
HOW IT WORKS / WHY IT MATTERS	pH changes the charge on molecules, and charge decides which parts of a protein stick together. So pH changes protein shape, and protein shape decides function. His slide on **deamidation** makes this concrete: the side chain of an asparagine (Asn, 114.04 Da) is converted first into a succinimide intermediate (Asu, 97.02 Da), then into either aspartate (Asp, 115.03 Da) or isoaspartate (isoAsp, 115.03 Da). This is a therapeutic protein liability, because a change like that can alter how well a drug binds its target.
EXAM HOOK	He did not list deamidation among the numbers or named question types in the exam-focus document, but it is the only worked example of a chemical change on a drug in Lecture 1, so it is worth knowing. What he did flag: the amino-acid R-group table is **not** in the exam. His words: "don't memorize this ... definitely not include in the exam."

1.3.9 What a cell spends - energy and metabolism

WHAT IT IS	**Energy** in a cell means the capacity to do work: move something against a gradient, build a structure, or drive a chemical reaction. **Metabolism** is the whole set of chemical reactions in a cell, and the reactions that release energy are **catabolic**, while the reactions that spend energy to build are **anabolic**.
HOW IT WORKS / WHY IT MATTERS	The trade is simple and it runs the whole course. Catabolism breaks big molecules down into small ones and harvests energy. Anabolism spends that energy to build big molecules. A cell that is growing is running anabolism hard, and it needs a constant supply of both fuel and raw material. This is exactly the argument behind the **Warburg effect**, his Lecture 1 cancer case: a proliferating cancer cell diverts most of its pyruvate to lactate even when oxygen is available, which yields less ATP but leaves the half-broken sugar molecules behind as perfect building blocks for nucleotides, lipids and amino acids. The inefficient switch is a growth strategy.
EXAM HOOK	He used the Warburg effect as the entry point to Lecture 1, because it is the reason metabolomics became a useful way to find cancer biomarkers. He did **not** ask students to memorise the details of the glycolysis pathway. On his comparison slide, a quiescent normal cell with oxygen makes about **36 ATP molecules per glucose** by oxidative phosphorylation, and only **2 ATP molecules per glucose** by anaerobic glycolysis without oxygen; a proliferating cancer cell makes about **4 ATP molecules per glucose** by aerobic glycolysis either way. Those three numbers were on the slide, but they are not in his list of stated numbers, so treat them as background rather than as exam fodder.

1.3.10 The cell's rechargeable battery - ATP

WHAT IT IS	**ATP** (adenosine triphosphate) is a molecule made of adenine, ribose sugar and three phosphate groups. The bond between the last two phosphates is the one the cell actually spends.
HOW IT WORKS / WHY IT MATTERS	A **kinase** (an enzyme that adds a phosphate, using ATP) takes the terminal phosphate from ATP and attaches it to a protein. The added phosphate carries charge, and the mutual repulsion of the charges changes the protein's shape. A **phosphatase** removes it again, using water and releasing inorganic phosphate, and the protein snaps back. So ATP is not just energy for building; it is also the currency of the cell's fastest on/off switch. His slide showed it as two Pac-Man shapes, one adding the phosphate and one taking it away.
EXAM HOOK	ATP is one of the two things he named when he listed what biomolecules are for: they serve as "disease biomarkers, energy sources, structural components, and signalling molecules". He did not ask you to draw ATP. He did build it into the Hippo pathway example, where MST1 and LATS1 phosphorylate YAP and push it out of the nucleus. That pathway is examinable, and it is Lecture 1 topic 9.

1.3.11 Losing and gaining electrons - oxidation and reduction

WHAT IT IS	**Oxidation** is losing electrons. **Reduction** is gaining electrons. They are two halves of one transfer, so they always occur together. In biological systems, oxidation very often means losing hydrogen, and reduction very often means gaining hydrogen.
HOW IT WORKS / WHY IT MATTERS	Energy capture is an oxidation-reduction reaction. When a cell burns sugar with oxygen, the sugar gives up electrons and hydrogen and the oxygen accepts them. The released energy is trapped as ATP. Reverse the idea and you get the other half of the course: when repair enzymes work by adding or removing hydrogen and electrons, they are running the same chemistry in a smaller box. It also explains a piece of vocabulary you will meet in metabolism: pyruvate can be converted to lactate, or it can go on to acetyl-CoA with carbon dioxide released.
EXAM HOOK	He did not teach oxidation and reduction as a named concept in Lecture 1, and the phrase does not appear in his exam-focus list. He used the underlying chemistry in the Warburg slide, where pyruvate is converted to lactate in one direction and to acetyl-CoA with release of carbon dioxide in the other. Learn the plain definition here so that when you meet the words in Lecture 2 (oxidative phosphorylation in mitochondria) they mean something.

1.4 The 12 things to remember

1. An atom's identity is set by its number of protons. Electrons do the bonding.
2. Electrons sit in shells. The outermost shell decides what an atom can bond to.
3. A covalent bond is shared electrons. An ionic bond is opposite charges attracting. A hydrogen bond is a weak tug.
4. Hydrogen bonds are weak but reversible, and reversibility is why proteins change shape and DNA can be copied.
5. A molecule has a shape, and the shape decides what it can touch and what it can do.
6. Carbon has four hands. It bonds to itself and to everything else, which is why one element builds four families of molecules.
7. Functional groups are where the chemistry happens. The long carbon chain is only the handle.
8. Water is polar, and that single fact explains why it dissolves salt, shapes proteins, and lets a membrane exist.
9. pH changes charge, charge changes shape, shape changes function. The scale is logarithmic.
10. Energy in a cell is ATP, and ATP is the spendable unit. A phosphate added or removed is the cell's fastest switch.
11. Oxidation is losing electrons; reduction is gaining them; they always happen together.
12. He said do not memorise chemical structures. Understand the logic and move on.

1.5 Check yourself

Question 1. A water molecule is drawn as a bent shape, not a straight line. Why does that matter to a cell?

Answer. The bend comes from the oxygen atom being larger and pulling the shared electrons more strongly than hydrogen does. That makes the whole molecule **polar**, meaning one end is slightly positive and the other slightly negative. Polar attracts polar. So water can hold salt and sugar in solution while oil stays separate. It also means that a phospholipid, which has a polar head and a non-polar tail, packs itself into a membrane with the heads facing the water and the tails hiding from it. Without the bend, water would behave like oil, and membranes and cells would not exist. **The reasoning to hold on to: shape causes polarity, polarity causes chemistry, chemistry causes the cell.**

Question 2. Two DNA sequences are identical in length, but one has more guanine-cytosine pairs than the other. Which one needs more heat to unzip, and why?

> **Answer.** The GC-rich one needs more heat. Dr. Wong's number is that G pairs with C using **3 hydrogen bonds** while A pairs with T using **2**. So the GC-rich sequence has more cross-links holding the two strands together, and more energy is needed to break them. That is why melting temperature depends on GC content, and why it matters when designing a PCR primer - a primer with the wrong melting temperature will not anneal properly. **The reasoning: more hydrogen bonds means more energy is needed to separate the strands.**

Question 3. A drug is a protein. The manufacturing process makes it slightly alkaline, and the drug then binds its target less well than it used to. Using only Chapter 1, what has probably happened?

> **Answer.** pH has changed the shape of the protein. The pH scale is logarithmic, so a small numerical change is a large chemical change. A change in pH alters the charge on certain side chains, and since opposite charges attract, that shifts which parts of the protein stick to which. He showed exactly this mechanism with the **deamidation** pathway: an asparagine side chain (114.04 Da) is converted through a succinimide intermediate (97.02 Da) into aspartate (115.03 Da) or isoaspartate (115.03 Da) at high pH. A changed side chain means a changed shape, and a changed shape can mean a changed binding site. **The reasoning: pH to charge to shape to function. That chain is the whole answer.**

Question 4. A cell has plenty of glucose and plenty of oxygen, yet it is not making much ATP. Meanwhile it is growing and dividing fast. What might explain this, and did Dr. Wong talk about it?

> **Answer.** The cell may be running the Warburg pattern: sending most of its pyruvate to lactate even though oxygen is present. On his comparison slide, a quiescent normal cell with oxygen makes about 36 ATP molecules per glucose by oxidative phosphorylation, while a proliferating cancer cell makes about 4 ATP molecules per glucose by aerobic glycolysis either way. That looks wasteful, but the half-broken sugar is left behind as raw material for nucleotides, lipids and amino acids, and a cell dividing a billion times needs building material more than it needs fuel. Dr. Wong did use this as his entry point to Lecture 1, because it is the reason metabolomics became a useful biomarker-discovery tool. He did **not** ask you to memorise the glycolysis pathway. **The reasoning: the "inefficient" choice is a growth strategy, not a mistake.**

Question 5. A kinase puts a phosphate on a protein and the protein changes shape. A phosphatase takes it away and the protein changes back. Where does the phosphate come from, and what does that make ATP?

> **Answer.** The phosphate comes from ATP. A kinase uses ATP to attach a phosphate group to a protein. The added phosphate carries charge, and the mutual repulsion of the added charges is what changes the protein's shape. The phosphatase reverses it, using water and releasing inorganic phosphate. This is why ATP is described as the cell's spendable unit of energy, and why the addition and removal of a single phosphate is the cell's fastest on-off switch. It is exactly the mechanism behind his Hippo pathway example, where MST1 and LATS1 phosphorylate YAP so that YAP is pushed out of the nucleus and the cell stops dividing. **The reasoning: ATP is not just fuel for building. It is also the currency of signalling.**

CHAPTER 2

The four kinds of biomolecule

2.0 What this chapter is about

Every living thing is made from four families of large molecules. Lecture 1 opened with them and the rest of the course is built on them.

This chapter teaches all four from zero: what they are made of, what shape they take, what job they do in the cell, and how their failure causes disease.

Two of the four - proteins and nucleic acids - get only a preview here, because Chapters 3 and 4 give them their own full chapters. This chapter is the map, and you will return to it every time a new molecule appears in the lectures.

2.1 Words to know before we start

Term	In plain English
biomolecule	A molecule that a living thing makes and uses. Carbohydrates, lipids, proteins and nucleic acids are the four families.
carbohydrate	A molecule built from carbon, hydrogen and oxygen, often with the formula CH2O. Sugars and starches.
monosaccharide	The simplest carbohydrate: a single sugar unit, such as glucose.
disaccharide	Two sugar units joined together, such as sucrose, which is table sugar.
polysaccharide	A long chain of sugar units joined together, such as glycogen, starch or cellulose.
glycosidic bond	The oxygen bridge that joins one sugar to the next.
lipid	A fat-like molecule that does not dissolve in water. It stores energy or builds membranes.
fatty acid	A long chain of carbons with a small acidic end. It is the building block of most lipids.
saturated / unsaturated fatty acid	Saturated means the carbon chain carries no double bonds and is straight. Unsaturated means it has one or more, which makes it kinked.
triglyceride	Three fatty acids joined to a glycerol. The body's energy store.
phospholipid	A lipid with a water-loving head and a water-fearing tail. This is what membranes are made of.
amphipathic	Having one end that likes water and one end that avoids it. The key word for understanding a cell membrane.
bilayer	Two layers of phospholipid, heads out, tails in. The wall of the cell.
nucleotide	A base plus a sugar plus a phosphate. It is the building block of DNA and RNA, and it is also ATP.

2.2 The big picture

Here is the whole of molecular biology in one idea: living things are made from four families of molecule, each has a natural shape, and that shape assigns it a job. When the job fails, you get disease. That is the logic of this chapter, and of the course.

A **biomolecule** is simply a molecule that a living thing makes and uses. Dr. Wong's own summary was that biomolecules serve four purposes: as **disease biomarkers** (a measurable molecule in blood, tissue or urine that tells you something about disease), as **energy sources**, as **structural components**, and as **signalling molecules**. Keep those four roles in mind, because every clinical example below sits in one of them.

The first family is the **carbohydrates**, built from carbon, hydrogen and oxygen. A single sugar unit is a **monosaccharide**, and the two you will meet most are glucose and fructose. Two units joined make a **disaccharide**, such as sucrose, which is table sugar. Long chains make a **polysaccharide**. The units are joined by a **glycosidic bond**, an oxygen bridge from one sugar to the next. Here is the interesting part: the same chain of

glucose can be built in different shapes, and each shape has a completely different job. **Glycogen** is a branched chain: compact, and branched so the cell can release sugar from many points at once, which is what you want when fuel is needed urgently. **Starch** is the plant version, and it is what you eat. **Cellulose** is an unbranched chain with the bonds flipped. It forms long straight fibres, and that is what plant cell walls are made of - which is why you cannot digest it and a cow can. So the job of a carbohydrate is set by its shape: energy currency, energy storage, or structure.

The second family is the **lipids**, and lipids are the odd ones out because they do not dissolve in water. Their building block is the **fatty acid**: a long chain of carbons with a small acidic end. If the chain carries no double bonds it is a **saturated fatty acid**, and it is straight and tightly packable. If it carries one or more double bonds it is an **unsaturated fatty acid**, and each double bond puts a kink in the chain so it cannot pack tightly. That difference is why a saturated fat is solid at room temperature and an unsaturated oil is liquid. Three fatty acids joined to a small three-carbon molecule called glycerol make a **triglyceride**, and that is the body's energy store: compact, waterproof and calorie-dense. The other major lipid is the **phospholipid**, which is **amphipathic** - a water-loving head and a water-fearing tail. This one property is why the cell has a membrane. Put amphipathic molecules in water and they arrange themselves, without any instruction, into a **bilayer**: heads out into the water, tails tucked away from it. Your cell wall is a self-assembled oil slick, and it is a wall because the water on both sides pushes it that way. **Cholesterol** is a small lipid that slots into the middle of the bilayer and controls how fluid it is, so the membrane works at every temperature. In the blood, fats cannot travel alone, so they are packaged into **lipoproteins** - particles of fat wrapped in protein, acting as delivery vans.

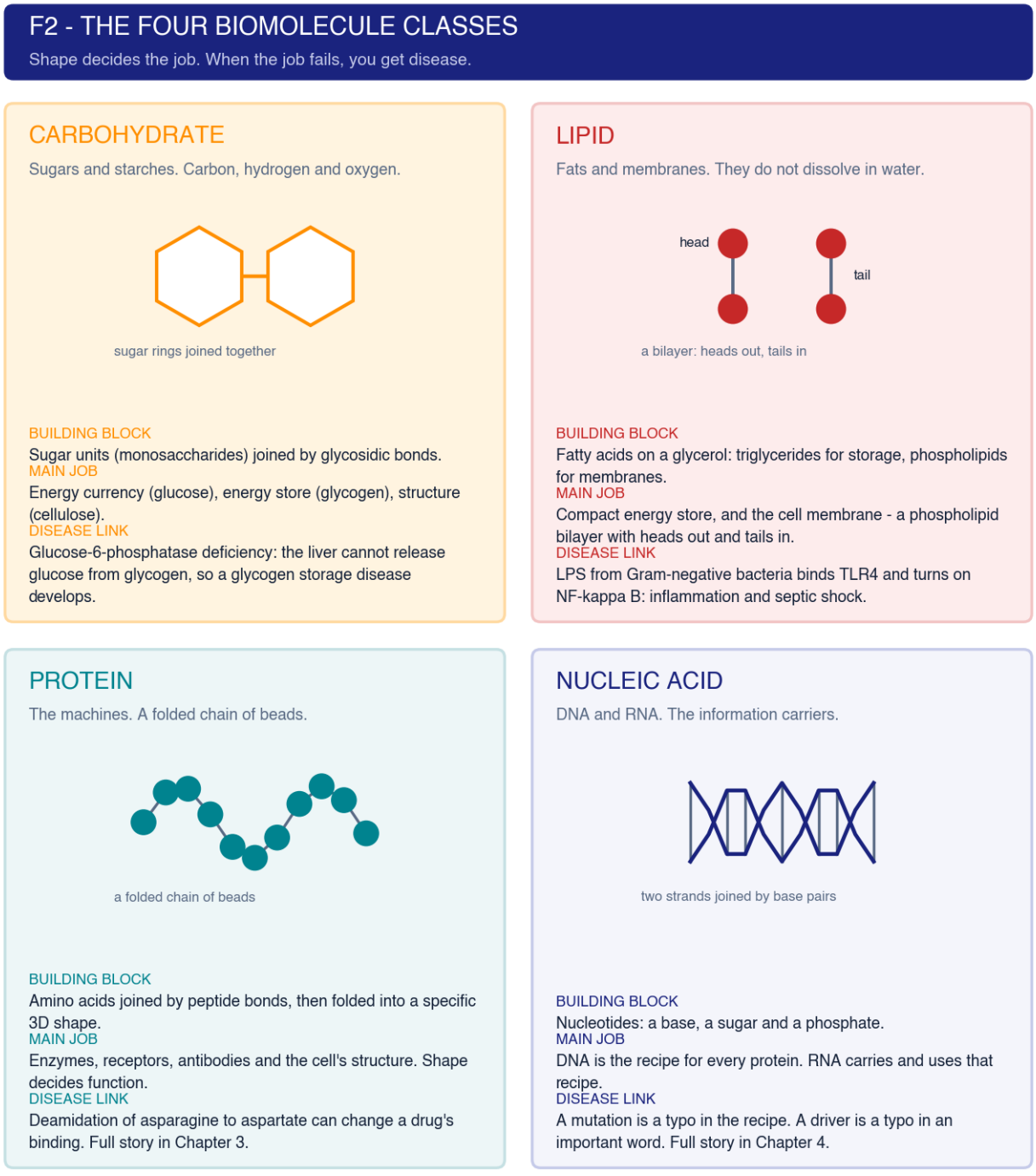
The third family is the **proteins**, and this is the family that does the work. A protein is a chain of **amino acids** (the 20 small building blocks from which every protein is made) joined by **peptide bonds** (the strong links between them), and the chain then folds into a specific three-dimensional shape. The folding is not decoration. The shape is the machine. Enzymes that speed up reactions, receptors that receive signals, antibodies that catch invaders and the cell's structural framework are all folded protein. Chapter 3 gives the full hierarchy, from the sequence of beads up to the assembled multi-part machine, and the methods for studying it.

The fourth family is the **nucleic acids**: DNA and RNA. A **nucleotide** is a nitrogenous base plus a sugar plus a phosphate; remove the phosphate and you have a **nucleoside**. Long chains of nucleotides make the double helix of DNA and the working forms of RNA. DNA is the recipe; RNA is a working copy or a tool. Chapter 4 covers the helix, the packing and what a mutation is.

One last idea carries through everything: **the shape of a molecule is set by its chemistry, and its chemistry is set by its shape.** A glycosidic bond flipped one way makes cellulose; flipped the other, glycogen. A double bond makes a fatty acid bend. A phosphate added to a protein changes what the protein does.

2.3 Walkthrough

Figure F2 - The four biomolecule classes



Four panels. For each class: a tiny schematic of the shape, the building block it is made from, the main job it does in the cell, and the disease link Dr. Wong used.

Figure F2 is the map for this whole chapter. If you can redraw it from memory, you can follow Lecture 1.

2.3.1 Molecules that living things make - biomolecule

WHAT IT IS	A **biomolecule** is a molecule synthesised naturally in a living system. Dr. Wong's own list has four: carbohydrates, lipids, proteins and nucleic acids. Everything else in the course is a detail, an example, or a method applied to one of these four.
HOW IT WORKS / WHY IT MATTERS	His slide listed the subcategories, and they matter because they are the vocabulary of the exams. Carbohydrates run monosaccharide, disaccharide, polysaccharide. Lipids run fatty acid, triglyceride, phospholipid. Proteins run amino acid, peptide, polypeptide, protein complex. Nucleic acids run nucleotide, DNA, RNA, mRNA, tRNA, microRNA, lncRNA. The reason this matters clinically is his central point about the course: the link between genotype and phenotype is mediated through biomolecular processes, so a measurement of a biomolecule in blood is a readout of what is happening inside cells.
EXAM HOOK	He did not frame "what is a biomolecule" as a standalone question, but he did use the four classes as the spine of Lecture 1, and the whole of Lecture 1 topic 1. He also stated clearly that chemical structures are **not** examinable.

2.3.2 Sugar and starch - carbohydrates

WHAT IT IS	**Carbohydrates** are molecules of carbon, hydrogen and oxygen, often with the formula CH2O. A single unit is a **monosaccharide** (glucose, fructose). Two units joined are a **disaccharide** (sucrose). Long chains are a **polysaccharide** (glycogen, starch, cellulose), and the units are joined by a **glycosidic bond**.
HOW IT WORKS / WHY IT MATTERS	Three jobs, decided by shape. **Glucose** is the cell's energy currency, the molecule that gets broken down to make ATP. **Glycogen** is the storage form, a branched chain so that it can be broken down fast and compactly. **Cellulose** is the structural form, a straight unbranched chain with the bonds oriented differently, which is what makes plant cell walls strong and why you cannot digest it. His slide showed the difference precisely: glycogen has **alpha-1,4** linkages along the chain with **alpha-1,6** branch points, and cellulose has the same sugar but a flipped bond orientation.
EXAM HOOK	He taught a disease that follows directly from the biochemistry. **Glucose-6-phosphatase deficiency** means the liver cannot release free glucose from glycogen, so the hepatocyte has an energy crisis and the patient develops a glycogen storage disease. His slide traced the whole path: glycogen, broken down by glycogen phosphorylase, to glucose-1-phosphate, to glucose-6-phosphate, and then only liver and kidney have the phosphatase to release free glucose. He did not flag this as a named exam-type question, and he did exclude "metabolic pathway details" from the exam, so do not memorise the enzyme names. The concept to keep is that a missing enzyme produces a metabolic block and a disease.

2.3.3 Fat and membranes - lipids

WHAT IT IS	**Lipids** do not dissolve in water. The building block is the **fatty acid**: a long carbon chain with a small acidic end. A **saturated fatty acid** has no double bonds and is straight. An **unsaturated fatty acid** has one or more double bonds and is kinked. Three fatty acids plus glycerol make a **triglyceride**, the energy store. A **phospholipid** is a lipid with a water-loving head and a water-fearing tail, so it is **amphipathic**.
HOW IT WORKS / WHY IT MATTERS	Two jobs, and the second one is the elegant one. Triglycerides are compact energy storage: fat has more energy per gram than carbohydrate. Phospholipids build the **bilayer**. Because they are amphipathic, in water they spontaneously arrange with the heads out and the tails in, and that self-assembly needs no energy and no builder. This is why every cell on earth has the same basic membrane. **Cholesterol** sits within the bilayer and controls its fluidity, so it stays workable in the cold and does not turn fluid in the heat. In blood, fats are carried as **lipoproteins**, particles of lipid wrapped in protein so they can move in plasma, which is water-based.
EXAM HOOK	Two things he did emphasise. First, **LPS**. The lipopolysaccharide of Gram-negative bacteria is a lipid that binds **TLR4**, and that triggers **NF-kappa B**, which turns on inflammation, and in overwhelming amounts causes septic shock - he noted on the slide that this is a common cause of death among surgical patients in intensive care. Second, and more exam-adjacent: lipids are also **signalling molecules**, not just structural ones. His slide 19 showed phospholipid signalling: PI becomes PI(4)P, then PI(4,5)P by ATP-driven phosphorylation, and the enzyme PLC splits it into DAG, which stays in the membrane and activates PKC, and soluble IP3, which binds the IP3 receptor on the endoplasmic reticulum and releases calcium. A phosphatase then inactivates IP3. He did not flag the second pathway as exam-type, but the first one, LPS to TLR4 to NF-kappa B, runs through both Lecture 1 and Lecture 5 and you will meet it again.

2.3.4 The machines - proteins (preview)

WHAT IT IS	A **protein** is a chain of **amino acids** joined by **peptide bonds**. Dr. Wong's slide listed the categories as amino acid, peptide, polypeptide, protein complex. The chain then folds into a specific three-dimensional shape.
HOW IT WORKS / WHY IT MATTERS	Shape decides function, and that is the whole of Chapter 3. The chain builds up in levels: primary (the sequence of beads), secondary (local folds such as alpha-helix and beta-sheet), tertiary (the overall three-dimensional fold), quaternary (several folded chains assembled together). A **domain** is a part of a protein that folds and works on its own. Two disease and drug hooks from Lecture 1 are worth previewing here. The first is a **therapeutic protein liability**: the deamidation reaction, where an asparagine becomes an aspartate or isoaspartate and the mass shifts by about one unit, which can change how well the drug binds its target. The second is the **antibody**, which he described as having Fab and Fc halves, disulphide bridges, and a set of hypervariable loops called CDR1, CDR2 and CDR3 that make each antibody different - the basis of antibody diversity in Lecture 1 topic 8.
EXAM HOOK	Two exclusions to note. The amino-acid R-group table is **not** in the exam: "don't memorize this ... definitely not include in the exam". And antibody-drug conjugate structures are **not** in the midterm: "all this would not be included in the midterm. I'm just want you to know more". The concepts stay; the structures do not.

2.3.5 The information carriers - nucleic acids (preview)

WHAT IT IS	A **nucleotide** is a nitrogenous base plus a sugar plus a phosphate. Without the phosphate it is a **nucleoside**. The bases are **A, T, C, G and U** - adenine, thymine, cytosine, guanine and uracil. Long chains of nucleotides form DNA and the several kinds of RNA.
HOW IT WORKS / WHY IT MATTERS	DNA holds the recipes; RNA does the work. His slide listed the RNA types: mRNA, tRNA, microRNA and lncRNA, and he later added siRNA. The history matters too, because he used it to make a logical point. Griffith found that something transformed one strain of pneumococcus into another. Avery, MacLeod and McCarty showed that the transforming principle was destroyed by **DNase** and not by **proteases**, which proved the hereditary material is DNA and not protein. Watson, Crick and Franklin produced the double-helix model, and that model is what makes PCR possible.
EXAM HOOK	He stated the pairing numbers: **G-C uses 3 hydrogen bonds, A-T uses 2**. He also stated the PCR amplification figure of roughly **2 to the n** copies after n cycles, and the codon detail that the start codon is AUG and the stop codons are UAA, UAG and UGA. The base-pairing numbers are the ones most likely to turn up as an MCQ stem, because they connect straight to PCR primer design. The full structural story of DNA is Chapter 4, and the replication machinery is Lecture 4.

2.4 The 12 things to remember

1. A biomolecule is a molecule a living thing makes. There are four families, and they cover everything in this course.
2. Carbohydrates, lipids, proteins and nucleic acids are the four. Everything else is an example or a method.
3. A carbohydrate is built from sugar units joined by glycosidic bonds. Same units, different shape, completely different job.
4. Glucose is the energy currency. Glycogen is the branched storage form. Cellulose is the unbranched structural form.
5. A missing enzyme creates a metabolic block. Glucose-6-phosphatase deficiency is his worked example, and it produces a glycogen storage disease.
6. Lipids do not dissolve in water. That single fact explains storage fat, blood lipoproteins, and the reason a membrane can exist.
7. Saturated fatty acid means straight. Unsaturated means kinked. Straight packs tight, so it is solid.
8. A triglyceride is three fatty acids on a glycerol. It is the compact energy store.
9. A phospholipid is amphipathic, so it self-assembles into a bilayer. Heads out, tails in. That is the cell wall.
10. Cholesterol controls membrane fluidity. Lipoproteins deliver fat through blood.
11. LPS is a lipid. It binds TLR4, turns on NF-kappa B, and drives inflammation and septic shock.
12. A protein is a folded bead necklace, and DNA is a recipe. Full treatment in Chapters 3 and 4.

2.5 Check yourself

Question 1. Why does a cell membrane form by itself, with no energy and no machinery to build it?

> **Answer.** Because the lipid that membranes are built from is **amphipathic**. A **phospholipid** has a water-loving head and a water-fearing tail. Put those molecules in water - which is exactly what a cell is, a bag of water - and the cheapest arrangement for each molecule puts the head in the water and the tail away from it. When a whole sheet of them does this together, you get a **bilayer**: heads facing out on both sides, tails packed in the middle. The cell does not spend energy building this. It gets the bilayer for free from the chemistry of the molecule and the polarity of the water around it. **The reasoning: amphipathic plus polar water forces the arrangement.**

Question 2. A plant cell wall and a storage glycogen granule are both made of glucose units joined by glycosidic bonds. Why can a cow eat one and not the other?

> **Answer.** Because the bonds point in different directions. Glycogen uses **alpha-1,4** linkages along the chain with **alpha-1,6** branch points, and that branching makes it compact and quickly breakable - it is an energy store, so the animal needs to release its glucose fast. Cellulose uses the same sugar but a flipped bond orientation, which makes straight, strong, parallel fibres - ideal for a plant cell wall. The cow has bacteria in its stomach carrying the enzyme that can cut cellulose's bonds. You do not, so cellulose passes through you as fibre. **The reasoning: same building block, different bond geometry, different job, different animal that can use it.**

Question 3. A patient has a Gram-negative bacterial infection and their blood pressure collapses. Dr. Wong's lecture gives a specific molecular chain for this. What is it, and why does a lipid appear in it?

> **Answer.** The chain is **LPS to TLR4 to NF-kappa B to inflammation**, and in overwhelming amounts of endotoxin in the blood this causes septic shock, which he noted on the slide is a common cause of death among surgical patients in intensive care. The lipid appears because **LPS**, lipopolysaccharide, is itself a lipid - it is the outer membrane material of Gram-negative bacteria. Its lipid portion is what TLR4 recognises. So the same chapter of biology you just learned - lipids as membrane building blocks - turns out to be the recognition signal that starts the whole inflammatory cascade. **The reasoning: the thing that is recognised by the immune system is a lipid, so the lipid chapter and the signalling chapter are the same chapter.**

Question 4. A colleague says a protein drug has changed behaviour between two batches made under slightly different conditions. Based on Chapter 2, what are the three candidate explanations you would check?

> **Answer.** Check the shape of the protein first, because shape decides function, and then work backwards. (1) **pH.** A change in pH alters the charge on side chains, which alters which parts of the protein stick to which. He showed exactly this as the **deamidation** pathway, where an asparagine (114.04 Da) is converted through a succinimide (97.02 Da) into aspartate or isoaspartate (115.03 Da) at high pH, and that can change how well the drug binds its target. (2) **Oxidation.** If a molecule has lost electrons, or gained hydrogen, its chemistry has changed. (3) **Contamination or truncation** - a protein that was cut short, or a batch that was not pure. Note that he explicitly excluded the amino-acid R-group table from the exam, so you would not be asked to recite the table. You would be asked to reason about which mechanism explains the change. **The reasoning: identify the class of chemical change, then reason from chemistry to shape to function.**

Question 5. Dr. Wong says a nucleotide is a nucleoside plus a phosphate. What useful things does that one statement let you predict?

> **Answer.** It tells you where the negative charge lives, and negative charge is what makes molecules bind to each other and what makes DNA stable. A chain of nucleotides is a **sugar-phosphate backbone** with bases sticking out, and the bases are what pair up. It also tells you that the phosphate group is the reactive handle, which is why the same phosphate group shows up as ATP, where it is the spendable unit of energy, and in phosphorylation, where adding and removing it is the cell's on-off switch. And because the backbone carries the charge, the two strands of the double helix have to run in opposite directions - the antiparallel arrangement, with one strand 5-prime to 3-prime and the other 3-prime to 5-prime, which is the fact Lecture 4 builds the whole replication fork around. **The reasoning: one structural statement, and it explains backbone geometry, charge, energy currency, and replication directionality.**

CHAPTER 3

Proteins: what they are and how they work [the folded machines that do the cell's work]

3.0 What this chapter is about

Your body is built mostly from water and from a few very large molecules. The busiest of those large molecules are proteins. A protein starts as a long chain of small building blocks, and then it folds itself into one precise shape. That shape is the reason the protein can do its one job well, whether that job is copying a piece of DNA, cutting a sugar in half, or grabbing a germ and holding it still. Dr. Wong teaches proteins early in Lecture 1, right after the simple biomolecules, because the rest of the course is about what these machines do and how we measure them. This chapter gives you the vocabulary you need before he uses it.

3.1 Words to know before we start

Term	In plain English
protein	A large folded machine, made of a chain of small building blocks called amino acids. It does its one job because of its shape.
amino acid	One of the small building blocks of a protein. There are 20 common kinds, and they all have the same middle part with a different side piece.
R-group (side chain)	The variable side piece of an amino acid. It is what gives each amino acid its own size, charge and behaviour.
peptide bond	The chemical link that joins one amino acid to the next. A chain of them is a polypeptide.
polypeptide	A chain of amino acids joined by peptide bonds. A long folded polypeptide is a protein.
primary structure	The exact order of the amino acids in a chain. This is the protein's own "one-letter" code.
secondary structure	Two regular local shapes that the chain takes up: the alpha-helix and the beta-sheet. Both are held by hydrogen bonds.
tertiary structure	The full 3D shape of one folded chain.
quaternary structure	The way two or more folded chains (called subunits) come together into one working unit.
domain	A stretch of a protein that folds on its own and does a job by itself.
hydrogen bond	A weak sideways attraction between a slightly positive and a slightly negative atom. Many of them together hold a protein's shape.
enzyme	A protein that makes a chemical reaction go faster. Almost every tool in this course is an enzyme.
codon	A group of three letters in mRNA. It tells the ribosome which amino acid to add next.
antibody	A protein shaped like a pair of tongs. It sticks to one specific target, called an antigen.

3.2 The big picture

A protein is a machine. Like any machine, it has parts, and the parts only work if they are arranged the right way. The best way to picture a protein is as a bead necklace that folds itself.

Start with the beads. A **protein** is a chain of **amino acids**, and there are only 20 common kinds. Yet these 20 units, in different orders and lengths, make every protein in your body. Each amino acid has the same central carbon with the same attachments: an amino group, a hydrogen atom, a carboxyl group, and one variable piece called the **R-group**. The R-group is the personality of the amino acid. Some are tiny. Some carry a negative charge. Some can link to each other. The 20 amino acids are the same 20 backbones wearing 20 different hats, and the hat decides how each bead behaves.

Joining the beads together is simple chemistry. The carboxyl group of one amino acid reacts with the amino group of the next, and a **peptide bond** forms, releasing a water molecule. Because every amino acid has both an amino

end and a carboxyl end, they join head to tail into a long **polypeptide**. The chain always has a beginning and an end. The first amino acid keeps a free amino group, so that end is the **N-terminus**. The last keeps a free carboxyl group, so that end is the **C-terminus**. Proteins are always written from the N-terminus to the C-terminus, which is why a position number like "position 114" is meaningful.

Now the folding. A chain of amino acids is floppy, but it does not stay floppy. Two forces tidy it up. First, the **backbone** settles into two regular local shapes. The **alpha-helix** is a spring, like a corkscrew. The **beta-sheet** is a flat, pleated sheet, made when the chain folds back on itself and lies alongside its own earlier part. Both are held in place by **hydrogen bonds**, weak sideways attractions between a slightly positive hydrogen atom and a slightly negative oxygen atom. One hydrogen bond is too weak to matter. Dozens, all pulling the same way, hold a shape steady. Together these two regular shapes are the **secondary structure**.

Second, the whole chain folds into one overall 3D shape, the **tertiary structure**. Now the R-groups decide where things go. R-groups that dislike water hide inside the protein. R-groups that like water stay on the outside. Two cysteine R-groups can form a strong **disulfide bond**, which acts like a staple across the fold. The final shape is the working shape.

Sometimes one chain is not enough. When two or more folded chains come together, the assembly is the **quaternary structure**, and each chain in it is a **subunit**. Haemoglobin is the classic example. Antibodies are another. Inside a single chain there is one more layer: a **domain** is a stretch that folds by itself and does one job.

Why does all this matter? Because the shape is the function. A small chemical change can alter the shape, and therefore alter the job. Dr. Wong uses this idea directly. A **phosphorylation** switch works this way: a **kinase** adds a phosphate group using ATP, and a **phosphatase** removes it again. Adding that charged phosphate pushes the protein open. Removing it lets the protein close. Open and closed, on and off, in seconds. The same logic explains why a single chemical change on a therapeutic protein, such as the **deamidation** of one asparagine into aspartate, can change how well a drug binds and how well it works.

Some proteins are **enzymes**. An enzyme makes a reaction go faster by lowering the energy hill it has to climb. The enzyme has a pocket called the **active site**, and the molecule it acts on is the **substrate**. Almost every tool in this course is an enzyme: helicase, primase, polymerase, ligase, kinase, phosphatase, and many more.

So how does a cell build a protein? It copies the gene into **mRNA**, a working copy of the recipe. A **ribosome** reads that mRNA three letters at a time. Three letters is a **codon**, and each codon names one amino acid. With four letters there are 4 x 4 x 4 = 64 possible codons, and 64 is more than enough for 20 amino acids, so most have more than one spelling.

Finally, how do we know what a protein looks like? Four methods answer four different questions. **Edman degradation** reads the sequence from the N-terminus, one amino acid at a time. **LC-MS/MS** cuts the protein into pieces, weighs them and reads their sequences, so it tells you which proteins are in a sample and exactly what they are. **SDS-PAGE** sorts proteins by size. **X-ray crystallography** works out the exact 3D atomic shape. Using the wrong one answers nothing at all.

3.3 Walkthrough

Figure F3 - How a protein folds



From a chain of beads, to the alpha-helix and the beta-sheet, to one folded 3D shape, to an assembled multi-subunit machine. Every level is labelled in plain language, so the figure can be read on its own.

3.3.1 Amino acids: the small units every protein is built from

- **What it is.** An **amino acid** is a small molecule with a central carbon atom. That carbon carries four things: an amino group (written NH2), a hydrogen atom, a carboxyl group (written COOH), and one variable piece called the **R-group**. Every one of the 20 common amino acids has this same central carbon and these same three fixed groups. Only the R-group changes. A **protein** is a long chain of these units joined together.

- **How it works / Why it matters.** Because each amino acid carries both an amino group and a carboxyl group, two amino acids can react with each other. That is why proteins are chains and not rings, and why every protein has a definite beginning and end. The beginning keeps the free amino group, and it is called the **N-terminus** (N for nitrogen). The end keeps the free carboxyl group, and it is called the **C-terminus** (C for carbon). Chains are always read and written from the N-terminus to the C-terminus, which is what makes a position number such as "residue 114" meaningful. This numbering is not decoration. In one of Dr. Wong's own examples, the asparagine at position 114 is exactly the residue that goes wrong. The mass of that residue changes by about one dalton as the reaction happens, and the two possible end products are mirror-image twins with the same mass but different chemistry. A dalton is the mass unit used for molecules, named after John Dalton.

- **Exam hook.** Dr. Wong showed the generic amino acid on a slide, with the central carbon carrying NH2, H, R and COOH, and used exactly this structure when he explained the deamidation pathway. He did not set out to examine the structure diagram itself. The concept you need is the one sentence: same backbone for all 20, different R-group.

3.3.2 The twenty common amino acids and the R-group idea

- **What it is.** There are 20 **amino acids** that build essentially every protein in every living thing. They differ only in their **R-group**, the side chain hanging off the central carbon. The R-group can be tiny (glycine has only a hydrogen atom), bulky (tryptophan has a large ring), negatively charged (aspartate and glutamate), positively charged (lysine and arginine), or able to form a special bond (cysteine forms disulfide bridges). Group the 20 by their R-groups and the list becomes short: uncharged and water-shy, uncharged and water-loving, negatively charged, positively charged.

- **How it works / Why it matters.** The R-group decides everything local. Dr. Wong's point in Lecture 1 is that R-group chemistry dictates how flexible a stretch of chain is. Glycine, with the smallest possible R-group, is the most flexible amino acid, because there is almost nothing beside the backbone. Bulky or charged R-groups such as tryptophan and arginine make the backbone rigid, because those groups are physically awkward to move. So the same chain of beads can be floppy in one place and stiff in another, depending only on which beads are next to each other. Two of the 20 have names you will meet again, and they behave differently. Cysteine is the one that makes disulfide bonds. Proline is different from the others because its side chain loops back and joins its own backbone, which makes it a kink: proline is where the chain bends sharply, and the chain cannot spin freely there either.

The 20 also fall into four families, and the family a residue belongs to decides how it behaves in water. Nine are uncharged and water-shy: glycine, alanine, valine, leucine, isoleucine, methionine, phenylalanine, proline and tryptophan. Six are uncharged but water-loving: serine, threonine, cysteine, tyrosine, asparagine and glutamine. Two carry a negative charge: aspartate and glutamate. Three carry a positive charge: lysine, arginine and histidine. The two charged families are called acidic and basic. Remember this grouping, because it is the reason a protein folds the way it does: the water-shy residues hide inside, and the water-loving and charged ones stay on the outside where they can meet water and other molecules.

- **Exam hook.** Dr. Wong displayed the full table of 20 standard amino acids, with three-letter and one-letter codes and each R-group drawn out, on a slide of Lecture 1. He then told the class not to memorise it: *"don't memorize this... definitely not include in the exam"* (Lecture 1). He excluded the chemical structures on the same grounds: *"Please don't memorize all these chemical structure... They would not be included in my exam"* (Lecture 1). So do not learn the table. Learn the idea it exists: 20 types, one shared backbone, one variable R-group, and the R-group sets both charge and bulk. He also said this will not be examined.

3.3.3 Peptide bonds: how the beads are joined

- **What it is.** A **peptide bond** is the covalent link that joins one amino acid to the next. It forms when the carboxyl group of one amino acid reacts with the amino group of the next, and a molecule of water is released in the process. A chain of amino acids joined by peptide bonds is a **polypeptide**. A short polypeptide is often called a peptide. A long one that folds into a working shape is a protein.

- **How it works / Why it matters.** The peptide bond is strong, but it is also flat and rather rigid. Each amino acid sits in a plane, and the chain can turn only at the joints between them. This restriction is not a nuisance; it is the reason regular shapes are possible at all. If the joints were free, the chain would be a random spaghetti and would never settle into a helix or a sheet. The water released at each joining is worth remembering too, because it is the same kind of reaction that joins sugars into starch and glycogen, and it is one of the reasons water matters so much to cells.

- **Exam hook.** He did not flag peptide-bond chemistry as an examinable topic in any of the five lectures. The idea to hold on to is that the chain is directional, and that its ends are named for the groups they keep free.

3.3.4 Primary structure: the order of the beads

- **What it is.** The **primary structure** of a protein is its sequence: the exact order in which the amino acids appear, read from the N-terminus to the C-terminus. It is the flat, one-dimensional description. Every later level of structure is built on top of it and is determined by it.

- **How it works / Why it matters.** The order of the beads decides the fold, and the fold decides the job. Change one bead in the middle of a chain and you have changed the packing of everything around it. This is the link between this chapter and Chapter 4: a change in the DNA sequence is a change in the primary structure, and a change in the primary structure is usually a change in the protein's behaviour. It is also why reading a protein's sequence is such a big deal in the clinic. If you know the sequence, you can predict the shape, and therefore predict what the protein does and whether a drug will fit into it.

- **Exam hook.** He did not flag "the primary structure is the sequence" as a numbered exam item, but it is the spine of the protein hierarchy he taught in Lecture 1, and it is the answer to the common question "which level of structure is the sequence?"

3.3.5 Secondary structure: the alpha-helix and the beta-sheet

- **What it is. Secondary structure** is the set of regular local shapes that the backbone of a chain takes up. There are two important ones. The **alpha-helix** is a right-handed corkscrew, about 3.6 amino acids per turn. The **beta-sheet** is a flat, pleated sheet, made when the chain folds back on itself and runs alongside its own earlier section, either in the same direction or in the opposite direction.

- **How it works / Why it matters.** Both shapes are held in place by **hydrogen bonds**. A hydrogen bond is a weak attraction between a hydrogen atom that is slightly positive and an oxygen atom that is slightly negative. In a protein, the carbonyl oxygen of the backbone on one turn hydrogen-bonds to the backbone N-H group a few residues further along. On his Lecture 1 slide of the hierarchy, Dr. Wong drew the beta-sheet with the bonds drawn as C=O to H-N dashes, exactly this. Because a hydrogen bond is weak on its own, a single one would not hold anything. But the helix can line up hydrogen bonds down its whole length, and the sheet can line them up across its whole width, and then the sum of many weak attractions becomes a strong hold. The parts of the chain that are in neither regular shape are called loops or coils, and they are usually the most mobile parts.

- **Exam hook.** He did not flag the alpha-helix and beta-sheet as separate exam items, and he did not ask you to remember the 3.6 residues per turn. What he did stress, in Lecture 3, is the neighbouring principle: **histone-DNA binding is non-covalent**, so chromatin can open and close on demand. Do not confuse the two - a protein is switched on and off by adding or removing a phosphate, not by breaking hydrogen bonds.

3.3.6 Tertiary structure: one chain, one 3D shape

- **What it is.** The **tertiary structure** is the complete three-dimensional shape of a single folded polypeptide chain. It is the second big folding step, after the local helices and sheets.

- **How it works / Why it matters.** The R-groups decide the fold. R-groups that fear water tuck into the inside of the protein, away from the water around it. R-groups that carry a charge or like water stay on the outside, where they can interact with water and with other molecules. On top of that, two cysteine R-groups can form a **disulfide bond**, a strong chemical link between two parts of the chain that acts like a staple. Dr. Wong pointed out in Lecture 1 that disulfide bonds are what stabilise the tertiary and quaternary structure of a protein. The result is a single, specific, stable shape, and that shape is the working shape of the protein.

- **Exam hook.** Disulfide bonds stabilising tertiary and quaternary structure is his own statement from Lecture 1, and the accompanying point is the one that matters for drug design: in a therapeutic protein, any chemical change that disturbs the fold is a liability. He did not ask you to memorise the folding rules.

3.3.7 Quaternary structure: several chains, one machine

- **What it is.** The **quaternary structure** is the arrangement of two or more separate folded chains, each called a **subunit**, into one functional assembly. Not every protein has one. A single-chain protein stops at tertiary structure.

- **How it works / Why it matters.** Putting subunits together gives a machine extra capacity and extra control. Haemoglobin, which carries oxygen in your blood, is built from four subunits. Antibodies are built from four. Because the subunits sit next to each other, one subunit can detect a signal and the other can act on it, so the protein can switch itself on and off. This is also where the therapeutic industry takes hold: an antibody-drug conjugate uses one part of the antibody to find the target and another part to carry the drug payload.

- **Exam hook.** Dr. Wong named haemoglobin and antibodies as the examples of quaternary structure in Lecture 1. He did not flag the term itself as an exam item; the point to take away is that a multi-chain protein has more than one folded shape, and that the way the pieces come together can be regulated.

3.3.8 Domains: the reusable working parts

- **What it is.** A **domain** is a stretch of a protein that folds into a stable shape by itself and carries out a recognisable job. Domains are the modular building blocks of larger proteins.

- **How it works / Why it matters.** Domains let one protein do several jobs, and let the cell re-use the same module in different proteins. The clearest example in this course comes in Lecture 5, with nuclear receptors. Every nuclear receptor has the same three-part architecture: an N-terminal regulatory domain, a **DBD**, the DNA-binding domain, and an **LBD**, the ligand-binding domain. The DBD and the LBD are separate domains with separate jobs, and separating them is exactly how you work out which part of the receptor a drug acts on. Antibody structure gives the same lesson in a different way. The antibody has an antigen-binding part and a recruiting part, each a separate domain with a separate job.

- **Exam hook.** Dr. Wong defined a domain in Lecture 1 as an independently folding functional unit, and he returned to the idea as the DBD versus LBD pair - one of the classic confusable pairs the exam notes list as **DBD vs LBD: DNA-binding domain versus ligand-binding domain**. The DBD versus LBD distinction is the exam-facing version of the domain idea.

3.3.9 Why shape decides function

- **What it is.** The general rule of protein chemistry: a protein does its job by having a particular three-dimensional shape. Change the shape and you change the job. Small chemical changes to a side chain are enough to do it.

- **How it works / Why it matters.** Dr. Wong uses two worked examples from Lecture 1, and both are worth knowing. The first is the phosphorylation switch. A **kinase** uses ATP to attach a phosphate group to a protein, and a **phosphatase** uses water to remove it. The phosphate carries charge, and the mutual repulsion between

two charged groups pushes the protein open. Removing the phosphate lets the protein close again. So a protein can be switched on and off within seconds, using nothing but the addition and removal of one small chemical group. His Hippo pathway example is built on this: when the kinases MST1 and LATS1 phosphorylate YAP, YAP is held outside the nucleus and the cell stops dividing. When that phosphorylation is lost, unphosphorylated YAP walks into the nucleus and switches on proliferation, and in double-knockout mice the liver grows enormously. The second example is **deamidation**, a chemical change in a therapeutic protein. At high pH, an asparagine can lose ammonia and become a succinimide, and then take on water to become aspartate or its mirror-image twin iso-aspartate. On the numbers he used, asparagine is 114.04 daltons, the succinimide intermediate is 97.02, and both final products are 115.03. So one side chain quietly changes chemistry, and the protein's binding and its potency can change with it.

- **Exam hook.** Deamidation of asparagine to aspartate at high pH, altering drug efficacy and binding, is his own example of a therapeutic protein liability in Lecture 1. The phosphorylation switch is the mechanism he drew on a slide of Lecture 1. He did not ask you to memorise the dalton figures.

3.3.10 Enzymes: proteins that speed up reactions

- **What it is.** An **enzyme** is a protein that makes a chemical reaction go faster. It does not get used up by the reaction, so one enzyme molecule can do the job many times.

- **How it works / Why it matters.** Every reaction has an energy hill to climb, called the **activation energy**. An enzyme lowers that hill, so the reaction runs faster at body temperature. The enzyme holds its reacting partner in a pocket called the **active site**, and the molecule being changed is the **substrate**. The fit between the pocket and the substrate is specific, which is why each enzyme acts on one type of substrate. Almost every tool in this course is an enzyme: helicase, topoisomerase, primase, polymerase, exonuclease, ligase, kinase, phosphatase, histone deacetylase, DNA glycosylase and the caspases. When the lectures mention an enzyme by name, you can assume it is a protein with a shape, and that blocking or changing its shape changes the chemistry.

- **Exam hook.** He did not teach enzyme kinetics or active-site chemistry as an examinable topic. Enzymes are background here, and the reason for the background is simple: every machine in the rest of the course is a protein with a pocket, so you cannot follow the lectures without the idea.

3.3.11 How a protein is made and folded

- **What it is.** Making a protein is a four-step factory process. First, the gene is **transcribed**, which means the DNA recipe is copied into **mRNA**, a working copy. Second, mRNA leaves the nucleus and goes to a **ribosome**, a molecular machine that reads it and joins amino acids together. Third, the ribosome reads the mRNA three letters at a time, and each group of three letters is a **codon** that names one amino acid. Fourth, the finished chain folds itself into its working shape.

- **How it works / Why it matters.** The codon arithmetic is worth doing carefully, because it is the kind of sentence students trip over. There are four letters in RNA: A, U, C and G. A codon is three letters long, so the number of possible codons is 4 x 4 x 4 = 64. There are only 20 amino acids, so 64 codons is comfortably enough. Most amino acids are named by more than one codon, and a few have only one. One codon, AUG, does double duty: it starts the chain and it codes for methionine. Three codons, UAA, UAG and UGA, are stop signals, and they mean "the protein ends here". Because the same amino acid can be spelled in several ways, different organisms prefer different spellings, and this preference is called **codon usage**. That matters in industry. When a company makes a human protein in a **CHO cell**, a hamster cell line used to manufacture medicines, the human gene may need rewriting so that the host cell reads it more happily. That rewriting is **codon optimisation**, and Dr. Wong showed a western blot of the result, comparing the original sequence against optimised versions for two proteins. After the chain is built, cells use helper proteins called **chaperones** to keep the new chain from tangling and to help it fold correctly. A folded protein can then be modified chemically, and these modifications are called **post-translational modifications**. Phosphorylation is the one that matters most in this course. Finally, many proteins carry a short address label at the N-terminus that tells the cell where to send them, which is the signal peptide you will meet in Lecture 2.

- **Exam hook.** The central dogma, the start codon AUG and the stop codons UAA, UAG and UGA come straight from his Lecture 1 slides on the central dogma. The codon-usage point is his: for CHO cells, *"when you use the CHO cell, you need to do the codon optimization"*, and he showed the western blot with FALVAC-1 at 38.7 kd and PTP4A3 at 24 kd, each compared against a GAPDH loading control at 39 kd. The loading control is the band that tells you the experiment worked. He did not ask you to memorise the codon table.

3.3.12 Antibodies: proteins shaped to grab things

- **What it is.** An **antibody** is a protein shaped like a Y, or like a pair of tongs. The tips grip a target, and the target is called an **antigen**. The exact patch of the antigen that the antibody touches is the **epitope**.

- **How it works / Why it matters.** An antibody is built from two **heavy chains** and two **light chains**, held together by disulfide bridges. Each arm of the Y is called a **Fab** fragment, the antigen-binding fragment. The stem is the **Fc** fragment, and it is the part that recruits other parts of the immune system. On his Lecture 1 slide he gave the masses, and they make the architecture easy to remember: the smallest piece, the variable fragment or **Fv**, is about 25 kDa; a **Fab** is about 50 kDa; and the **Fc** is about 50 kDa. At the tip of each arm sits the **variable region**, which contains three hypervariable loops called **CDR1**, **CDR2** and **CDR3**. Those three loops are the gripping surface. Because they can take so many shapes, the immune system can build antibodies against so many different things. Dr. Wong put a number on that: somatic recombination produces billions of antibody variants. The raw material is a set of gene segments held in the DNA, and he gave their counts: about 39 copies of the V segment, 27 of the D segment and 6 of the J segment. The factory that shuffles them is the subject of Lecture 4, where we will meet V(D)J recombination and the NHEJ repair machinery that carries it out. For now, the shape is the point: the tips are shaped to fit one target, the stem is shaped to talk to the rest of the immune system.

- **Exam hook.** Antibody anatomy, the masses and the three CDR loops are all from his Lecture 1 slides. He also used the antibody shape as the delivery vehicle for antibody-drug conjugates in Lecture 1 and again in Lecture 4. The antibody-diversity chain (V(D)J recombination, billions of variants) links directly into Lecture 4, where a defect in the repair machinery that builds the repertoire causes severe combined immunodeficiency, which he named as one of his five exam-type questions.

3.3.13 The four standard ways we study proteins, and what each one actually tells you

- **What it is.** Four techniques come up again and again when proteins are measured. Each answers a different question, and the exam loves to test whether you know which is which.

- **How it works / Why it matters. Edman degradation** is the classical method. It chemically peels off one amino acid at a time from the N-terminus and identifies each one, so it gives you the sequence read straight from the start of the chain. Its limitation is that it works on a single, clean chain, so a real protein has to be broken into pieces first. **LC-MS/MS** is the modern workhorse. The protein is cut into smaller peptides with an enzyme, the peptides are separated by liquid chromatography, and each one is weighed by mass spectrometry. The mass tells you which amino acids are in the piece, and breaking the piece further tells you their order. Matching the result against a database identifies the protein. Crucially, LC-MS/MS also tells you **how much** of each protein is present, which is why it is the method behind every proteomics biomarker in this course. **SDS-PAGE** is the sorting method. The protein is treated with the detergent SDS, which coats every protein with the same negative charge, and then it is run through a gel under an electric field. Small proteins travel further than large ones, so the gel reads out like a set of stairs. It tells you the sizes of the proteins and roughly how many are there. It cannot tell you the sequence. **X-ray crystallography** is the shape method. The protein is turned into a crystal, bombarded with X-rays, and the pattern of scattered rays is used to work out where every atom sits. It tells you the exact 3D atomic structure, which is the thing that explains function.

- **How the methods fit together.** These four are not alternatives. They answer four different questions about the same protein, so a real study usually uses more than one of them. The next block, 3.3.14, follows one protein all the way from a cell sample to a patient blood test, using every one of these methods in turn.

- **Exam hook.** Edman degradation as the classical method and LC-MS/MS as the modern one are his own words from Lecture 1, and 2D-PAGE separating by charge and then mass is the workflow he showed. The X-ray connection comes from Lecture 3: the nucleosome core particle was solved at 2.8 angstrom resolution, and he noted that below 2.0 angstrom is "very excellent". He also made the practical point that histones are difficult to crystallise because of their heavy positive charge. So the method-to-question mapping is worth being able to state: Edman gives sequence from the N-terminus, LC-MS/MS gives identity and amount, SDS-PAGE gives size, X-ray gives atomic shape.

3.3.14 Putting the methods together: how a protein becomes a blood test

- **What it is.** A **biomarker** is a molecule you can measure in a sample from a patient to say something about their disease. A protein can become a biomarker, but only if the lab can find it, identify it, and show that its level differs between sick and healthy people. That is why the methods in 3.3.13 are chained together rather than used one at a time.

- **How it works / Why it matters.** Dr. Wong's worked example is **vimentin**, a protein of the cytoskeleton, in liver cancer. Here is the chain, step by step. Step one: get a cell sample. Step two: run the two-dimensional gel. **Isoelectric focusing** sorts the proteins by charge along one axis, using a pH gradient running from 4.0 to 9.0, and SDS-PAGE sorts them by size along the other, from 220 kDa down to 10 kDa. So every protein lands in its own spot of a grid. Step three: cut out the spot that changed, and cut the protein inside it into small peptides with an enzyme. Step four: run the peptides through LC-MS/MS. The mass of each peptide identifies the protein, and the size of the peak tells you how much of it is there. Step five: check that the result is real, using other tools: a western blot, immunohistochemistry, or qPCR on the RNA. Step six: ask whether the finding is worth anything, by consulting the Human Protein Atlas, which records where in the body the protein is normally found, and whether its level predicts survival in a given cancer. Step seven: move the measurement into something you can take from a patient. The vimentin finding in liver cancer was eventually measured in serum, using ELISA, a method that detects one specific protein using an antibody. That final step is what turns a laboratory finding into a blood test. The disease groups were separated by a real margin: controls averaged 176.0 nanograms per millilitre, plus or minus 61.3; patients with tumours of 2 cm or less averaged 271.5, plus or minus 165.9; and patients with larger tumours averaged 231.3, plus or minus 132.3.

- **Exam hook.** The two-dimensional workflow, the Human Protein Atlas and the vimentin case are all his own Lecture 1 material, and the serum vimentin figures are on his slide. He also gave the worked example a clinical twist worth remembering: the tumour tissue next to a tumour is not the same thing as normal tissue. He warned that adjacent tissue is *"biochemically different from truly normal tissue"* and should not be treated as healthy. He did not ask you to memorise the workflow step by step. The exam-facing idea is the logic: separate, identify, quantify, validate, then make it measurable in a patient.

3.4 The twelve things to remember

1. A protein is a folded machine, and its shape is the reason it can do its job.
2. There are only 20 common amino acids, and they all share the same backbone. Only the R-group differs.
3. The R-group sets charge and bulk, and therefore sets how rigid the chain is at that point. Glycine is the most flexible; bulky or charged R-groups such as tryptophan and arginine are the most rigid.
4. Amino acids join head to tail by peptide bonds, releasing water. The chain has a named start (the N-terminus) and a named end (the C-terminus).
5. Primary structure is the order of the beads. Secondary structure is the alpha-helix and the beta-sheet, held by hydrogen bonds.
6. Tertiary structure is the whole folded chain, held by the packing of the R-groups and stapled by disulfide bonds.
7. Quaternary structure is several subunits working together, as in haemoglobin and antibodies.
8. A domain is a part of a protein that folds and works on its own. Nuclear receptor DBD versus LBD is the exam version of this.
9. A codon is three letters of mRNA. Four letters give 4 x 4 x 4 = 64 codons, which is more than enough for 20 amino acids. AUG starts the chain; UAA, UAG and UGA stop it.
10. Enzymes lower the activation energy of a reaction by holding the substrate in a pocket called the active site.
11. Four methods, four different answers: Edman gives sequence from the N-terminus, LC-MS/MS gives identity and amount, SDS-PAGE gives size, X-ray crystallography gives atomic shape.
12. Antibodies are Y-shaped, built from two heavy and two light chains, with an antigen-binding Fab and a recruiting Fc, and three hypervariable loops, CDR1, CDR2 and CDR3, that do the gripping.

3.5 Check yourself

Q1. Someone tells you that a protein's function is decided by its shape, and asks you to explain why that is not just a slogan.

A1. Start with the observation that a protein is a chain of amino acids with one variable piece, the R-group. Because the R-groups differ in charge and bulk, they cannot all sit comfortably in the same place, so the chain has to fold in a particular way. That fold puts some R-groups inside, away from water, and others outside, where they can reach out. Next, name the mechanism: the fold is held by hydrogen bonds in the backbone and by hydrophobic packing of the R-groups, and it is reinforced by disulfide bonds between cysteines. Then the interpretation: because the shape determines which surfaces are available, it determines which molecule the protein can bind and what it can do. Dr. Wong's own examples are the phosphorylation switch, where adding and removing one charged phosphate opens and closes the protein, and deamidation, where one asparagine turns into aspartate and the protein's binding and potency change with it. So the slogan is shorthand for a causal chain: chemical change, then change of shape, then change of job.

Q2. Why does a codon have to be three letters long, and why is 64 enough?

A2. Do the arithmetic. There are four letters in RNA. A codon of three letters can therefore be any combination of three choices from four, so 4 x 4 x 4 = 64. Now compare with what has to be encoded: 20 amino acids, plus start and stop signals. Sixty-four is more than enough, so the code is redundant. Most amino acids are named by more than one codon, and only a few are named by a single one. That redundancy has a practical consequence which Dr. Wong used in Lecture 1: because the same amino acid can be spelled in several ways, different organisms prefer different spellings, so a gene may have to be rewritten, or codon optimised, when it is expressed in a CHO cell. The answer to "why three" is simply that three is the shortest word that fits 20 amino acids comfortably.

Q3. A colleague ran a sample on SDS-PAGE, saw four bands, and says "I now know what proteins are in the sample." What is wrong with the statement, and what should the colleague have used?

A3. Separate what the method can and cannot do. SDS-PAGE treats the proteins with SDS, which gives them the same negative charge, and then separates them by size in a gel. So the four bands tell you there are about four abundant protein sizes. They cannot tell you the identity of any of them, because a band of the right size could be any protein of that size. To identify them you need a method that reads the molecule itself. That is LC-MS/MS: cut the protein into peptides, separate them by liquid chromatography, weigh them, and match the fragments against a database. If you also want the shape, you need X-ray crystallography. So the correction is not "SDS-PAGE is wrong" but "SDS-PAGE answers the size question, and you have used it for an identity question".

Q4. You are told that the deamidation of asparagine to aspartate is a "liability" for a therapeutic protein. Explain what that word means.

A4. The observation first: the change is chemical and happens on a side chain, without the protein being rebuilt. The mechanism: at high pH, an asparagine can lose ammonia to become a succinimide intermediate, and then take on water to become aspartate or iso-aspartate. On Dr. Wong's figures, asparagine is 114.04 daltons, the intermediate is 97.02, and both products are 115.03, so you can even see the change on a mass measurement. Then the clinical meaning: because the shape of a protein carries out its job, and the shape depends on the chemistry of the side chains, a side chain that quietly changes chemistry can change how well the protein binds its target and therefore how well the drug works. A liability is a weakness you have to watch, test for and control, not a defect that makes the protein useless. And the limit of the claim: he did not say that every deamidation ruins a protein; he said it can alter efficacy and binding, which is why you monitor it.

Q5. Why is an antibody a good shape to explain quaternary structure, and why does Dr. Wong return to it in Lecture 4?

A5. First the structure: an antibody is built from four chains, two heavy and two light, joined by disulfide bridges. So it has more than one folded chain, and the way those chains come together is the quaternary structure. The parts are also named for what they do: the two arms are Fab, the antigen-binding fragments, and the stem is Fc, which recruits other immune cells. Second the reason it returns in Lecture 4: the enormous variety of antibodies, billions of variants, is produced by shuffling gene segments, and that shuffling is carried out by the same NHEJ repair machinery the cell uses to fix broken DNA. Dr. Wong made the NHEJ chain one of his five named exam-type questions, with the clinical endpoint being severe combined immunodeficiency when the machinery fails. So the antibody is where protein structure, protein function and DNA repair all meet.

CHAPTER 4

DNA, genes and chromosomes [the recipe, the printed manual, and the typo]

4.0 What this chapter is about

DNA is the recipe book of the cell, and this chapter opens the book. You will learn what the letters are, how two of them are always stuck together, and how a single letter is copied into a working copy. Then you will meet the two words that carry most of the course: a **gene**, which is one recipe, and a **chromosome**, which is the printed manual that holds thousands of recipes and packs them into something small enough to fit inside a nucleus. Finally you will meet the **mutation**, which is a typo. Dr. Wong teaches this in Lecture 1 as the chemistry and history of nucleic acids, then spends Lectures 3 and 4 on how DNA is packed and how it is copied, so this chapter is the bridge to both.

4.1 Words to know before we start

Term	In plain English
nucleotide	The building block of DNA. It is a base plus a sugar plus a phosphate group.
base	The letter part of a nucleotide. DNA has four: A, T, G and C.
sugar-phosphate backbone	The two long rails of the DNA ladder, made of alternating sugar and phosphate.
double helix	Two such rails wound around each other, with the bases pairing across the middle like ladder rungs.
base pairing	The strict rule for the rungs. Adenine pairs with thymine, and guanine pairs with cytosine.
hydrogen bond	The weak sideways attraction that holds each base pair together. A-T makes two, G-C makes three.
antiparallel	The two DNA strands run in opposite directions, one from 5' to 3' and the other from 3' to 5'.
replication	The copying of DNA before a cell divides, so each daughter cell receives a complete set.
gene	One recipe. A stretch of DNA that carries the instructions to build one protein (or one functional RNA).
chromosome	One printed manual. A single, very long piece of DNA plus its packaging proteins, condensed so it fits in the nucleus.
centromere	The waist of a chromosome. It is where the spindle fibres grip the chromosome during cell division.
telomere	The protective cap at each end of a chromosome.
nucleosome	A spool of protein with DNA wound around it. Many spools make chromatin, and chromatin condenses into a chromosome.
mutation	A typo. A change in the DNA sequence.

4.2 The big picture

Start with the letters. A **nucleotide**, the building block of DNA, has three parts: a **base**, a sugar, and a phosphate group. The sugar and the phosphate join end to end to make a long **sugar-phosphate backbone**: the two rails of a ladder. The bases stick out sideways from the rails, one per rung. DNA uses only four bases: adenine, thymine, guanine and cytosine, written A, T, G and C. RNA, the working copy of a gene, uses uracil, U, in place of thymine.

The rule for the rungs is absolute. A always pairs with T, and G always pairs with C. These are the **base pairs**, and each one is held together by **hydrogen bonds**, weak sideways attractions between a slightly positive hydrogen and a slightly negative oxygen or nitrogen atom. An A-T pair is held by two hydrogen bonds. A G-C pair is held by three. Because the two strands point in opposite directions, one running from 5' to 3' and the other from 3' to 5', the ladder is **antiparallel**. Wind the two strands around each other and you have the **double helix**.

The discovery was an experiment, not a guess. In 1928 **Griffith** worked with the bacterium that causes pneumonia and found that material from dead virulent bacteria could turn live harmless bacteria into virulent ones. He called it the transforming principle and could not say what it was. In 1943 **Avery, MacLeod and McCarty** identified that principle as DNA, using an enzyme test: **DNase**, which cuts up DNA, destroyed the transforming activity, while **protease**, which cuts up protein, did not. The hereditary material was DNA and not protein. **Watson, Crick and Franklin** then built the double-helix model, using Rosalind Franklin's X-ray images and the rule that A and T appear in equal amounts, and so do G and C.

Why does base pairing matter for everything afterwards? Four reasons. First, copying. Each strand is a template, and because pairing is strict, copying is accurate. Second, heat. G-C-rich DNA needs more heat to pull the strands apart, because it has more hydrogen bonds to break, so its melting temperature is higher. Dr. Wong's slide gave the direction, not a number: more GC content means a higher melting temperature. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and the two do not reconcile - learn the direction and the reason, not a number. That direction decides the temperature for a PCR primer. Third, repair. A cell can compare a fresh strand against the old one and cut out a letter that does not match, because the pairing rules say what the correct letter should have been. Fourth, damage. When a chemical attaches to a base and changes its shape, the pairing rule breaks, and the next round of copying bakes the mistake into the DNA. That is the route by which cigarette smoke causes cancer.

Now the two big nouns. A **gene** is one recipe: a stretch of DNA on a chromosome that gets copied into RNA and then used to build one protein. A **chromosome** is one printed manual: a single very long piece of DNA, plus the proteins that package it, condensed small enough to fit inside the nucleus. A human body cell has 46 of them, in 23 matched pairs, one from each parent. Each chromosome has a **centromere**, the waist where the cell's division machinery grips it, and a **telomere** at each end, a protective cap.

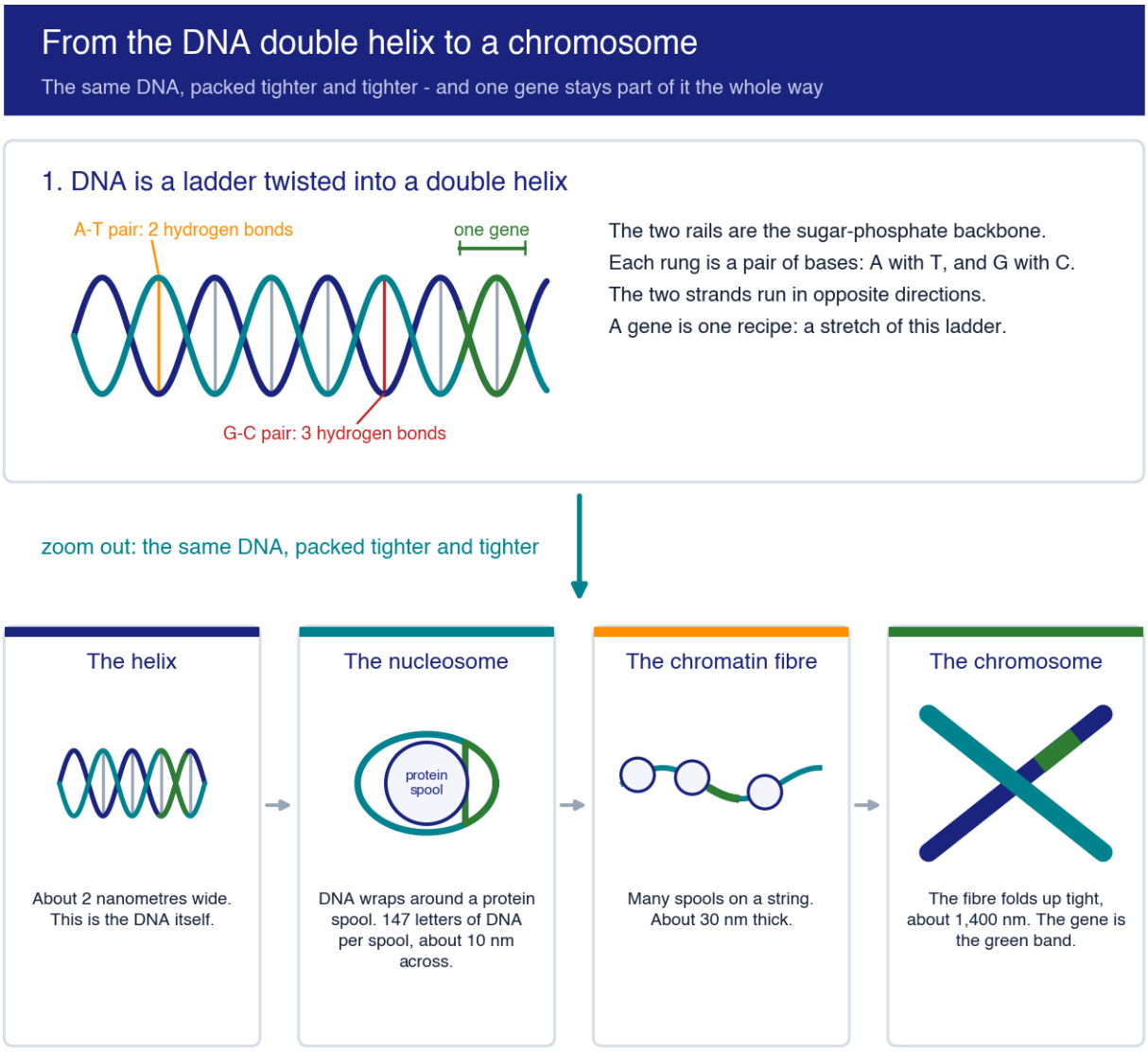
The manual is far too big to fit as it is. A human cell holds roughly a metre of DNA. So DNA is wrapped around protein spools called **histones**, and one spool with DNA wound round it is a **nucleosome**. Nucleosomes on a string form chromatin, the packaged manual, and chromatin folds until it becomes a chromosome. Loose, open chromatin is called **euchromatin** and is readable. Tightly packed chromatin is called **heterochromatin** and is switched off.

Before a cell divides, it copies all of this. **Replication** is **semi-conservative**, which means every new double helix ends up with one old strand and one new one. A team of enzymes does the work: one unwinds the helix, one relieves the twisting strain, one lays a short starting primer, one adds the letters and proofreads as it goes, and one seals the final gaps. Because DNA can only be built in one direction, one strand is built continuously and the other in short pieces that are later joined.

Finally, the **mutation**. A mutation is a typo in the recipe. Some typos change nothing. Some change one amino acid, and occasionally that turns a normal protein into one that drives a tumour. Some shift the reading frame, so everything after the typo is misread and the protein is cut short. Some are much bigger, involving whole genes or chromosomes. A typo that is copied and passed on becomes permanent, which is the basis of both cancer genetics and the inherited diseases in the exam scenarios.

4.3 Walkthrough

Figure F4 - From the DNA double helix to a chromosome



The green mark is one gene - a single recipe. It is the same piece of DNA in every picture.

Base pairing is labelled on the helix, then the same DNA is shown being packed step by step: helix, nucleosome, chromatin fibre, chromosome. A gene is marked on the helix and again on the chromosome, so you can see it is the same piece of DNA the whole way.

4.3.1 Nucleotides: the letters of DNA

- **What it is.** A **nucleotide** is the repeating unit of DNA and RNA. It is made of three parts: a **base** (the letter), a sugar, and one or more **phosphate** groups. Take the phosphate away and what is left, base plus sugar, is called a **nucleoside**. DNA's sugar is **deoxyribose**. RNA's sugar is **ribose**, which has one extra oxygen atom. DNA has four bases: adenine (A), thymine (T), guanine (G) and cytosine (C). RNA has the same three plus uracil (U) in place of thymine.

- **How it works / Why it matters.** The sugars and the phosphates join end to end to make the **sugar-phosphate backbone**, and the bases hang off the backbone and point inward. That arrangement is what allows two strands to pair up and form the helix.

Two small notes finish this off. First, the ends of a strand are numbered. Each sugar has numbered carbons, and the number 5 marks the carbon that carries the phosphate group, while the number 3 marks the carbon on the other side. So every strand has a 5' end and a 3' end. That is what the table in 4.1 is referring to, and it is why two strands can run in opposite directions without being mirror images of one another. Second, the phosphate group has a second job. Phosphates store and release energy, which is why ATP, a molecule with three phosphates, is the cell's spendable energy unit, and why kinases in this course use ATP to attach a phosphate to a protein. So the same small chemical group is doing two jobs in the cell: it is the structural backbone of DNA, and it is the wallet for energy.

- **Exam hook.** Dr. Wong defined nucleoside as base plus sugar and nucleotide as nucleoside plus phosphate in Lecture 1, and he listed the five bases A, T, C, G and U. This is one of the plainest definitional facts in the course, and it is the sort of thing that appears as a short definition in a multiple-choice question. He did not say the definition would be examined.

4.3.2 The double helix, and how it was discovered

- **What it is.** The **double helix** is DNA's final shape: two strands wound around each other like a twisted ladder, with the bases pairing across the middle and the sugar-phosphate backbones on the outside. The two strands run in opposite directions, which is called **antiparallel**, so one is read 5' to 3' and the other 3' to 5'.

- **How it works / Why it matters.** Three discoveries built the picture, and each one answers a different question. The first question was: what carries hereditary information? **Griffith** answered it in 1928 without knowing. He worked with *Streptococcus pneumoniae*, the bacterium that causes pneumonia, and found that something from killed virulent bacteria could convert live harmless bacteria into virulent ones. He named the active substance the transforming principle. The second question was: what is the transforming principle? **Avery, MacLeod and McCarty** answered it in 1943, and they answered it with an enzyme experiment. They showed that **DNase**, which cuts DNA, destroyed the transforming activity, whereas **protease**, which cuts protein, left it intact. The conclusion was that DNA, not protein, is the hereditary material. The third question was: what shape does it take? **Watson, Crick and Franklin** built the double-helix model, using Rosalind Franklin's X-ray diffraction images and the observed rule that the amount of A equals the amount of T, and the amount of G equals the amount of C. The model was immediately useful, because the two strands can be pulled apart by heat, and pulling them apart is exactly what PCR does.

- **Exam hook.** Dr. Wong presented this as the chemistry-and-history section of Lecture 1, with Griffith, Avery-MacLeod-McCarty and Watson-Crick-Franklin named on the slides, and he made one point sharply: the enzyme experiment is the proof. The dates on the slides, 1928 and 1943, were shown as part of the story and he did not flag them as examinable. He also stated that the double helix underpins PCR.

4.3.3 Base pairing: why A-T and G-C matters for everything downstream

- **What it is.** The pairing rule is that adenine pairs with thymine, and guanine pairs with cytosine. Each pair is held by **hydrogen bonds**. A-T pairs are held by **two** hydrogen bonds. G-C pairs are held by **three**.

- **How it works / Why it matters.** Four consequences follow, and all four matter later in the course. First, accurate copying. Because pairing is strict, a template strand tells the copying machinery exactly which letter

belongs opposite each letter. Second, melting temperature, written **Tm** for short. Separating two strands means breaking hydrogen bonds, so a G-C-rich stretch needs more heat than an A-T-rich stretch of the same length. On Dr. Wong's slide, the rule is the direction: more GC content means a higher melting temperature. The exam notes also carry a '~40 deg C drop' and a '60 deg C versus 55 deg C' board example, and the two do not reconcile - learn the direction and the reason, not a number. This is why PCR primer design has to look at GC content, as he stated in Lecture 3. Third, repair. The pairing rules tell the cell what the correct letter should have been, so a mismatch stands out and can be removed. The mismatch repair machinery of Lecture 4 works because of this. Fourth, damage and cancer. A chemical that attaches to a base, for example a tobacco-specific nitrosamine such as NNK, changes the shape of that base. The base then pairs incorrectly, the error is copied, and a mutation is fixed in the DNA. Dr. Wong used this route in Lecture 3 to explain how chemical carcinogenesis produces the KRAS mutations seen in lung cancer. Finally, the pairing rules also set the geometry of the helix itself, which is why a G-C-rich helix is a slightly different shape from an A-T-rich one.

- **Exam hook.** G-C equals 3 hydrogen bonds and A-T equals 2 is on his own list of precise numbers he stated, from Lectures 1 and 3. The melting-temperature **direction** is his fact and he linked it explicitly to PCR primer design: more GC means a higher Tm. **The G-C = 3 / A-T = 2 bond counts are the reliable numbers here.** The temperature example itself is inconsistent between his notes, so do not put a number on it.

4.3.4 What is a gene?

- **What it is.** A **gene** is one recipe. It is a stretch of DNA on a chromosome that carries the instructions for building one product. For most genes that product is a protein. For some genes it is a functional RNA, such as tRNA, rRNA or one of the small regulatory RNAs Dr. Wong covered in Lecture 1.

- **How it works / Why it matters.** The flow of information is the **central dogma**, and it runs in one direction. DNA is copied into RNA by **transcription**. That RNA is then used to build protein by **translation**. The RNA copy of a gene is called **mRNA**, and it is a photocopy of the recipe sent out to the factory floor. A gene is not read all the time. Whether it is on or off depends on how tightly the DNA is packed and on which proteins are present to read it, which is the subject of Lecture 5. That is also why two people with the same gene can be different, and why the same gene can be a driver in one cancer and harmless in another. Knowing the name of a gene is not enough. You also need to know what the gene does, where it is switched on, and what happens when it is broken. The clinical examples in this course all depend on that extra step: *K-RAS* only predicts a failure of cetuximab if you also know where it sits in the signalling cascade.

- **Exam hook.** Dr. Wong used the central dogma, DNA to transcription to mRNA to translation to polypeptide chain, as the spine of Lecture 1, and he returns to it in Lecture 5 to talk about control. The K-Ras story is the worked example of why a gene name alone is never a clinical answer. He did not set out a definition of a gene as a memorised line.

4.3.5 What is a chromosome?

- **What it is.** A **chromosome** is one printed manual: a single, extremely long piece of DNA together with the proteins that package it, condensed into a compact shape. A human body cell contains 46 chromosomes, in 23 matched pairs, one of each pair inherited from the mother and one from the father.

- **How it works / Why it matters.** A chromosome has a recognisable anatomy. The **centromere** is the narrowed waist, and it is the point where the spindle fibres attach during cell division, so it is how the cell pulls one copy of each chromosome to each end. The centromere divides the chromosome into a short **p-arm** and a long **q-arm**. The letter q is the long arm, so the band name q21.1 is read out as q-two-one-point-one, which Dr. Wong was careful to spell out on the Lecture 3 slide. Each end is capped by a **telomere**, which has three parts: a unique sequence right at the tip, then 100 to 300 kilobases of subtelomeric repetitive sequence, and then 3 to 20 kilobases of the repeated TTAGGG motif. Chromosomes are big enough to see. A technique called **karyotyping** stains them with Giemsa dye to produce **G-banding**, a pattern of light and dark bands that lets a trained eye read off large structural changes. Its limit is resolution. G-banding detects changes of roughly 0.5 to 10 megabases, so it finds extra chromosomes and large deletions, but it misses single-letter changes, which need sequencing.

- **Exam hook.** 23 pairs of chromosomes, the p-arm and q-arm with the centromere between them, and the resolution limit of G-banding are all on his list. Two details are worth keeping exactly. First, he was emphatic that q21.1 is q-two-one-point-one and not q-twenty-one-point-one. Second, he noted that the resolution figure appears as 5 to 10 megabases in one part of his teaching and 0.5 to 1 megabase in another, and that single-nucleotide changes always need sequencing.

4.3.6 How DNA packs into a chromosome - the bridge to Lecture 3

- **What it is.** The DNA in one human cell is about a metre long, and it has to fit inside a nucleus that is far smaller than a millimetre. It fits because it is wrapped and folded. The packaging ladder has standard steps.

- **How it works / Why it matters.** Step one is the **nucleosome**. DNA wraps around a core of eight histone proteins, two copies each of H2A, H2B, H3 and H4, with a linker histone H1 at the point where the DNA enters and leaves. One turn of DNA holds 147 base pairs, and the whole spool is about 10 nanometres across. Step two is the fibre. A chain of nucleosomes looks like beads on a string, about 11 nanometres thick, and it is packed into a 30 nanometre **chromatin fibre**, then extended at 300 nanometres, condensed at 700 nanometres, and finally a mitotic chromosome of about 1,400 nanometres, which is 1.4 micrometres. The whole packaged form of DNA with its histones is called **chromatin**. Step three is the switch. Tightly packed chromatin is called **heterochromatin**, and it is switched off, because the packing physically blocks transcription factors and enhancers from reaching the gene. Loose chromatin is called **euchromatin**, and it is being read. The state is not fixed. Nucleosomes slide along the DNA to expose or cover a gene, and two things push them: the intrinsic preference of a DNA sequence for a histone, and ATP-powered **chromatin remodelling complexes** that slide them deliberately. Two facts Dr. Wong added make this concrete. Histones are extremely well conserved, from fruit fly to human, which is why model organisms work. And histones are hard to crystallise because they carry a heavy positive charge, which is why getting the nucleosome crystal structure was a real achievement.

- **Exam hook.** He stated the diameters himself: 2 nanometres for the helix, 11 for beads on a string, 30 for the fibre, and 1,400 for the mitotic chromosome. The nucleosome numbers, 147 base pairs and about 10 nanometres, are on his list of precise figures. He also stated the nucleosome core structure was solved at 2.8 angstrom resolution, with below 2.0 angstrom being "very excellent". The concept he said you must understand is the one he put in those words: euchromatin is open and active, heterochromatin is condensed and silent, and it physically blocks access.

4.3.7 How DNA is copied - the bridge to Lecture 4

- **What it is. Replication** is the copying of DNA before a cell divides. Each daughter cell must receive a complete set of chromosomes, so the entire genome is copied, and it is copied fast.

- **How it works / Why it matters.** First, the rule. Replication is **semi-conservative**. Every new double helix contains one old strand and one newly made strand. Two rival models were on the table, semi-conservative and conservative, in which the old helix would stay whole and a completely new one would be made alongside. A Meselson-Stahl-type experiment with heavy and light isotopes settled it, and Dr. Wong showed the three candidate models on a Lecture 4 slide. Second, the team of enzymes. **Helicase** unwinds the double helix. **Topoisomerase** relieves the twisting strain that builds up ahead of the fork. **Primase** lays down a short RNA primer to give the polymerase a free end to start from. **DNA polymerase** adds the new letters and proofreads as it goes, using a 3' to 5' exonuclease activity to cut out a wrong letter. **Exonuclease** removes the primers afterwards, and **DNA ligase** seals the last gaps with a phosphodiester bond. Third, the direction problem. New DNA can only be built from 5' to 3', but the two templates are antiparallel. So one strand, the **leading strand**, is copied continuously, and the other, the **lagging strand**, is copied in short stretches called **Okazaki fragments**, each of which needs its own primer. Fourth, the scale. A human chromosome is enormous, so copying does not start from one point. It starts at thousands of **origins** at once, each opening a **replication bubble** with a fork running in each direction. Fifth, the ends. Because the last RNA primer on the lagging strand cannot be replaced, the end of a chromosome cannot be finished. This is the **end-replication problem**, and the result is that telomeres shorten by roughly 50 to 200 base pairs with every cell division. When they run out, the cell stops dividing, which is the **Hayflick limit**. **Telomerase**, an unusual enzyme that is a reverse transcriptase with

its own internal RNA template, can rebuild the ends. It is active in embryonic stem cells, in some adult stem cells, in germ cells, and, alarmingly, in about 90 percent of cancers.

- **Exam hook.** In his own words in Lecture 4: *"this is important, you need to understand the elongation happens from 5 prime to 3 prime"*, and *"I think one concept you'll need to understand is there's two different strands. So one is the leading strand, and the other one is the lagging strand."* His summary table gives the pairing: 5' to 3' for elongation, 3' to 5' for proofreading, and ligase sealing a 5' phosphate to a 3' hydroxyl. The thousands of origins and the two forks per bubble are his figures, as is the telomere shortening of roughly 50 to 200 base pairs per division and the telomerase activation in about 90 percent of cancers.

4.3.8 What is a mutation?

- **What it is.** A **mutation** is a change in the DNA sequence. It is a typo in the recipe. The word covers everything from a single letter changed to a whole chromosome rearranged.

- **How it works / Why it matters.** Three things can happen when a typo appears. It can do nothing, because the changed letter still codes for the same amino acid. It can change the protein slightly, which may be harmless, may be useful, or may change the protein's job. Or it can break the protein altogether, for example by creating an early stop signal. A typo only becomes real if it survives. Cells have repair machinery that catches most errors, and any change that is fixed or lost does not matter. A change that is copied into a daughter cell and passed down the family becomes a permanent difference, and that is the raw material of inherited disease and of cancer. The important clinical habit is to separate the two questions. Does the mutation exist? And does it matter? Dr. Wong makes this point repeatedly, most sharply in Lecture 3 with the difference between a **driver** change, which gives the tumour a growth advantage, and a **passenger** change, which is simply carried along.

- **Exam hook.** The driver versus passenger distinction is on his list of classic distractors, and the way he said to tell them apart is to integrate copy number profiles with mRNA expression, so that a copy gain that matches an increase in transcription points to a real driver. His worked example was NPM1 in leukaemia. He did not define "mutation" as an examinable line; the driver versus passenger distinction is the exam-facing version of it.

4.3.9 The kinds of mutation: one letter, or a few letters

- **What it is.** The smallest mutation changes a single letter. The next smallest adds or removes a few letters. Everything bigger is saved for the next block.

- **How it works / Why it matters. (a) A single letter changes.** This is a substitution, and it is called a **point mutation**. It can be **silent**, meaning the protein comes out exactly the same, or **missense**, meaning one amino acid changes, or **nonsense**, meaning the change creates a stop signal early and truncates the protein. The cleanest example in the course is **KRAS G12C**, where a single letter change turns one amino acid and leaves the RAS protein stuck in its active state, so the cell keeps dividing whether or not the receptor upstream is blocked. That is why cetuximab fails in K-Ras-mutant colorectal cancer, and why testing for K-Ras is a companion diagnostic. Dr. Wong also showed the NNK story from Lecture 3: the chemical in cigarette smoke alters a base, the altered base pairs wrongly, and the next round of copying fixes the mistake as a GC to AT change in the K-Ras gene. **(b) Letters are inserted or deleted.** If the number of letters added or removed is a multiple of three, the reading frame is preserved. That is an **in-frame** change, and the protein comes out the right length but with a piece missing; his slide showed an in-frame codon deletion producing MET-PHE-GLY-STOP instead of MET-LYS-PHE-GLY-STOP. If the number is not a multiple of three, the reading frame shifts. That is a **frameshift**, and everything downstream is misread. His slide showed a one-letter deletion in the third codon giving MET-LYS-LEU-ALA and running on, and a one-letter insertion right after the start codon giving an immediate stop. Frameshifts usually produce a short, useless protein. They can also produce **neoantigens**, new peptides that the immune system has never seen, and those are exactly what makes immunotherapy work in some cancers.

- **Exam hook.** The KRAS G12C example and the frameshift cartoons are straight from his Lecture 1 slides. The point he drew out from them is that a mutation matters because of what it does to the protein, and because a frameshift can hand the immune system a completely new protein fragment. He did not ask you to memorise

the codon strings on the slide, and he did not set out the labels silent, missense and nonsense as a memorised list; learn them as the three possible consequences of a one-letter change.

4.3.10 The kinds of mutation: whole genes, whole chromosomes, and chemical damage

- **What it is.** Bigger mutations take out a stretch of DNA, copy it, flip it around, or move it somewhere else. Bigger still change the number of whole chromosomes. And there are chemical injuries that turn into mutations when the cell copies the damaged DNA.

- **How it works / Why it matters. (a) Whole genes and whole chromosomes change. Deletions** remove a stretch, **duplications** copy it, **inversions** flip a stretch around, and **translocations** swap pieces between two chromosomes. Dr. Wong's point about inversions in Lecture 3 is the interesting one: an inversion removes no DNA and changes no gene dosage, yet it can change expression dramatically, because it moves a gene away from its enhancer or brings an enhancer next to the wrong promoter. **(b) The wrong number of chromosomes.** When the number of chromosomes is wrong, the condition is called **aneuploidy**, and it comes from **non-disjunction**, which is the failure of chromosomes to separate properly during meiosis.
  Non-disjunction in meiosis I gives two normal gametes and two abnormal ones, all of the same kind.
  Non-disjunction in meiosis II gives one normal gamete and three abnormal ones, of two different kinds. The results are conditions such as **trisomy 21**, which is Down syndrome, and Turner syndrome, which is 45,X. **(c) Copy number varies.** A **copy number variation** means the same stretch of DNA is present in different numbers of copies in different people. The amylase gene, AMY1, is the textbook example, with higher copy numbers in populations that have historically eaten a lot of starch. Dr. Wong flagged this as an idea rather than a mechanism list, and said so: *"I'm just give you a taste... don't be scared with all these mechanism"*. **(d) Chemical damage becomes a mutation.** Reactive oxygen species from the cell's own metabolism, and tobacco-specific nitrosamines such as NNK, attach themselves to a base and change its shape. The changed base then pairs wrongly, replication copies the error, and the error is fixed in the DNA. This is the chain from cigarette to cancer, and it is how the KRAS mutations in lung cancer arise.

- **Exam hook.** He named the meiosis I versus meiosis II distinction as one of his five exam-type questions, in his own words: *"you might need to distinguish the M1 or the M2 failure"*, and *"I think you need to distinguish these two situations here"*. The table to memorise is that meiosis I fails to separate homologous chromosomes and gives 2 normal plus 2 abnormal gametes, all abnormal ones the same, while meiosis II fails to separate sister chromatids and gives 1 normal plus 3 abnormal gametes, of two different kinds. Recombination happens in meiosis I and not in meiosis II. On inversions he said: *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."* He also excluded the copy-number mechanisms from the exam, so learn the amylase example as an idea rather than as a mechanism. The DNA adduct story is his Lecture 3 material and the bridge into the repair material of Lecture 4.

4.4 The twelve things to remember

1. A nucleotide is base plus sugar plus phosphate. Remove the phosphate and it is a nucleoside.
2. DNA has four bases, A, T, G and C. RNA swaps thymine for uracil.
3. The sugars and phosphates form the sugar-phosphate backbone. The bases hang off it and pair inward.
4. A always pairs with T, held by 2 hydrogen bonds. G always pairs with C, held by 3.
5. The two strands are antiparallel, and that is why one is copied continuously and one in pieces.
6. A gene is one recipe. A chromosome is the printed manual that holds thousands of recipes.
7. A human body cell has 46 chromosomes, in 23 pairs. Each has a centromere, a short p-arm, a long q-arm, and telomeres at the ends.
8. Packaging runs from a 2 nanometre helix, through the 147 base pair nucleosome, to the 30 nanometre fibre, to the 1,400 nanometre chromosome.
9. Euchromatin is open and being read. Heterochromatin is packed and silent. It is a switch, not a setting.
10. Replication is semi-conservative. Each new helix has one old strand and one new one.
11. Helicase, topoisomerase, primase, polymerase, exonuclease and ligase do the copying. New DNA is built 5' to 3' and proofread 3' to 5'.
12. A mutation is a typo. It may change nothing, change one amino acid, shift the reading frame, or rearrange a chromosome. What matters clinically is whether it is a driver.

4.5 Check yourself

Q1. Explain what a hydrogen bond does in DNA, state how many hydrogen bonds each base pair makes, and say what that tells you about melting temperature.

A1. First, the mechanism. A hydrogen bond is a weak sideways attraction between a slightly positive hydrogen atom and a slightly negative oxygen or nitrogen atom. In DNA, each base pair is held together by these bonds, so the two strands can be held together or pulled apart. Second, the counting. Adenine pairs with thymine using 2 hydrogen bonds, and guanine pairs with cytosine using 3. Third, the consequence, as a direction rather than a number. Separating the two strands, which is what happens when you heat DNA, means breaking these bonds. So a G-C-rich stretch needs more heat to separate than an A-T-rich stretch of the same length, which is the melting temperature, or Tm: more GC content means a higher Tm. That direction decides the annealing temperature of a PCR primer, which is why GC content matters at the bench. His notes also carry a board example with a specific temperature difference, but it does not reconcile with another figure in the same notes, so do not put a number on it. The same logic also explains why damaged bases are dangerous: if a chemical changes the shape of a base, the pairing rule breaks and a wrong letter gets copied in.

Q2. A patient has a K-Ras mutation and is being considered for cetuximab. Explain, at the level of DNA structure, why the mutation predicts the drug will fail.

A2. Start with the observation: the tumour has a single-letter change in the K-Ras gene, the G12C change, so one amino acid in the RAS protein is different from the normal one. The mechanism is at the protein level, but it is written in the DNA. RAS is a **transducer**. In its normal form it only relays a growth signal while the receptor above it, EGFR, is firing. The G12C change leaves RAS locked in its active state, so it signals whether or not the receptor is blocked. Cetuximab blocks EGFR, the receptor at the top of the cascade, so it removes the trigger but cannot remove a locked-on transducer downstream. The clinical meaning is that this is a **predictive** biomarker, not a prognostic one. The mutation tells you the drug will not work; it does not by itself tell you the patient is sicker. And the limit: you cannot make this call from the gene name alone. You need to know where the gene sits in the pathway, which is exactly the reasoning the exam expects.

Q3. Two people have the same rearrangement in chromosome 8, but one has a normal amount of that region's DNA and one does not. Explain how a rearrangement can change gene expression without changing how much DNA is present.

A3. The rearrangement to consider is an **inversion**, in which a stretch of the chromosome is cut out and put back in the opposite direction. The first point is what an inversion does not do: it removes no DNA and adds none, so the gene dosage, which is simply how many copies of the gene the cell has, is unchanged. The second point is what it does do. A gene does not read itself; it is read with the help of regulatory DNA, such as an **enhancer**, which can sit some distance away. Flipping a stretch can move a gene away from its enhancer, so the gene is no longer boosted. Or it can move an enhancer next to a different gene, so the wrong gene is boosted. The expression of both genes changes even though not one letter has been added or lost. That is Dr. Wong's point in his own words: *"you need to understand that some of these abnormalities can cause the loss of the genetic materials, but some does not."* Clinically, this is why a cytogenetics report showing an inversion is not automatically a finding of no consequence.

Q4. Why do we say replication is semi-conservative, and what would we expect to see if the conservative model were true instead?

A4. A semi-conservative copy means each new double helix contains one original strand and one newly synthesised strand. A conservative copy means the original helix stays completely intact and an entirely new helix is built beside it. The way to tell them apart is to label the original DNA and then follow the label through two rounds of copying. If replication is semi-conservative, the first round gives two hybrids, each with one old and one new strand, and the second round gives two hybrids plus two fully new molecules. If replication is conservative, the first round gives one completely old helix and one completely new one, and that old helix persists as a separate molecule in later rounds. Dr. Wong used the Meselson-Stahl-type isotopic labelling experiment to make exactly this distinction, and showed all three candidate models on a Lecture 4 slide. The reason it matters is that semi-conservative copying is what makes proofreading and repair possible in later rounds. Each new strand has an old strand to be checked against, and that is the basis of mismatch repair, which is a Lecture 4 topic and the basis of the MSI story.

Q5. Explain how a gene can be present and working in a cell but completely silent, and what changes that.

A5. Begin with the packaging, because silence is a packaging problem before it is a genetic one. DNA is wrapped around histone spools to make nucleosomes, and nucleosomes are packed into chromatin. Where the chromatin is tightly packed, the DNA is physically unreachable. Dr. Wong's phrasing was that heterochromatin physically blocks transcription factors and enhancers from reaching the promoter. So the gene is there and its sequence is perfectly intact. Nothing can get to it. Then the switch. The state is dynamic and reversible. Nucleosomes slide along the DNA to expose or cover the gene's promoter region, and two things drive the sliding: the intrinsic preference of a particular DNA sequence for a histone, and ATP-dependent chromatin remodelling complexes that move nucleosomes on purpose. Then the second layer of control. Even when the chromatin is open, the gene still needs the right proteins present to be transcribed, and that is the layer Lecture 5 is about. The clinical consequence is the one Dr. Wong developed in Lecture 3: a tumour can silence a tumour suppressor gene by methylating its promoter, and the brake is off even though not a single letter of the gene has changed.

CHAPTER 5

The cell and its parts

5.0 What this chapter is about

Everything in this course happens inside a cell. A biomarker is a molecule. A molecule is made or used by cells. A drug works on a cell. So before any pathway, mutation or therapy makes sense, you need a picture of what a cell is made of and what each part does. This chapter builds that picture from nothing. It starts with what a cell is, then the two basic designs of cell, then each internal compartment in turn. It ends with the dividing schedule of a cell, because that is the bridge into Lecture 2.

5.1 Words to know before we start

Term	In plain English
Cell	The smallest unit that is alive: a bag of water with a membrane, some DNA, and machinery to copy it.
Organelle	A small compartment inside a cell with one job, like a room in a factory.
Prokaryote	A cell with no nucleus. Its DNA floats loose in the cytoplasm. Bacteria are prokaryotes.
Eukaryote	A cell with a nucleus that holds the DNA. Animal, plant, fungal and yeast cells are eukaryotes.
Plasma membrane	The thin, flexible skin around the cell. It decides what may come in or go out.
Cytoplasm	The watery jelly that fills the cell, where most of the chemistry happens.
Nucleus	The control room. It stores the cell's DNA, the master instruction manuals.
Mitochondrion	The power station. It burns fuel with oxygen to make ATP, the cell's spendable energy.
Ribosome	The machine that builds a protein by joining amino acids in order.
Endoplasmic reticulum (ER)	The workshop beside the nucleus. Ribosomes on it fold and check new proteins.
Golgi apparatus	The packing room. It sorts finished proteins and posts them to their destination.
Lysosome	The recycling and disposal plant. It digests worn-out parts and engulfed germs.
Cytoskeleton	The rope-and-tube frame inside the cell. It gives shape, moves cargo, and splits the cell.
Cell cycle	The repeating schedule a cell follows to copy its DNA and then split into two.

5.2 The big picture

Picture a small factory. It has a boundary wall, a control room, a power station, a workshop, a packing room, a recycling bay and a delivery network. Everything the factory sells, it builds for itself, from its own drawings. That is a cell.

The wall comes first, because a factory needs to hold everything in. The **plasma membrane** is a flexible, oily skin built from **phospholipids** - molecules with a water-loving head and a water-hating tail. Put such molecules in water and they line up tail to tail, heads out, forming a **lipid bilayer** two molecules thick. That is why cell membranes form themselves without anyone assembling them. The membrane is also a filter and a set of doors. Some of those doors are **receptors**: proteins that sit in the membrane, catch one specific molecule from outside, and pass the news inward.

Inside is watery **cytoplasm**, where the chemistry of life happens. The control room is the **nucleus**, a separate compartment holding the cell's DNA. Think of DNA as the master instruction manuals - long printed documents written in only four letters. The manuals stay in the control room, and only authorised copies travel out. This separation is the whole difference between a **eukaryote**, which has a nucleus, and a **prokaryote**, a bacterial cell with no nucleus at all, which keeps its DNA loose in the cytoplasm and copies a recipe and builds the part in the same spot.

The **ribosomes** are the machines on the factory floor. Each one reads a copy of one recipe, an **mRNA** (messenger RNA) - a photocopy sent to the factory floor, sent out of the control room - and strings **amino acids** together to make a **protein**. A protein is a folded machine made from a bead necklace. The folding is what gives it its job.

The **endoplasmic reticulum (ER)** is the workshop next to the control room. Ribosomes sit on its rough surface, and new proteins are folded, checked and sometimes decorated with sugar groups (**glycosylation**) there. The **Golgi apparatus** is the packing room downstream: it sorts finished proteins, puts them into **vesicles**, and posts them either to the cell surface or out of the cell entirely. The **lysosomes** are the recycling and disposal plant - acidic bags full of enzymes that digest worn-out cell parts and engulfed bacteria. The **cytoskeleton** is the frame, the lifting gear and the delivery network of ropes and tubes. It also pins the cell membrane, moves cargo around, and pulls the cell in two.

Then there is the **mitochondrion**, the odd one out. It is the power station, and it is the reason Dr. Wong can talk about the **Warburg effect**, the strange habit cancer cells have of ignoring it. A mitochondrion also keeps its own small circle of DNA, about 17 kilobases carrying 37 genes. That happens because a mitochondrion began life as a free-living bacterium that was swallowed and kept. Because of that history, it is inherited from the mother only.

Finally, the whole factory runs on a schedule. The **cell cycle** goes G1, S, G2, M and then cytokinesis: grow, copy the DNA, check the copy, split the chromosomes, cut the cell in two. Every later topic in this course - cancer, stem cells, immune killing, gene therapy - happens at one of those steps.

5.3 Walkthrough

5.3.1 The smallest unit of life - a cell

WHAT IT IS	A cell is a tiny bag of water wrapped in a membrane, with genetic instructions inside and machines to read them. It takes in raw material, makes its own parts, and does a job. It reproduces by copying itself. Nothing smaller than this counts as alive.
HOW IT WORKS / WHY IT MATTERS	Two things make a cell a cell. First, it has a membrane, so it keeps its contents together and can control what crosses. Second, it has DNA, so its parts can be specified by instruction rather than by chance. Human cells are about 30 micrometres across. A human hair is about 50 to 100 micrometres wide, so two or three human cells would sit across it.
EXAM HOOK	He did not flag "what is a cell" as examinable. He did make it the opening move of Lecture 2, where he says diseases are diagnosed and targeted at the cellular level, and where isolating the right cell population is half the battle.

5.3.2 Two designs of cell - prokaryote and eukaryote

WHAT IT IS	A **prokaryote** has no nucleus. Its DNA sits as a loop in the cytoplasm, in a region called the **nucleoid**. Bacteria and archaea are prokaryotes. A **eukaryote** has a real nucleus, with the DNA inside, plus other membrane-bound rooms such as mitochondria and the Golgi apparatus. Animals, plants, fungi and yeasts are eukaryotes.
HOW IT WORKS / WHY IT MATTERS	The compartment changes the whole workflow. In a prokaryote, copying a recipe and building the part happen in the same place at the same time. In a eukaryote, the recipe is copied in the nucleus, edited, and then shipped out to the cytoplasm for building. That two-stage system is what allows eukaryotes to splice, edit and regulate their messages, and it is why human cells can make many different proteins from one gene. It is also why bacteria are so hard to make into medicine factories: they lack the machinery for folding and decorating a complex human protein.
EXAM HOOK	He used exactly this contrast in Lecture 5, topic 2, when he set up prokaryotic operons against eukaryotic gene control. He did not ask students to memorise the organelle list.

5.3.3 The plasma membrane - the front door and the sensor

WHAT IT IS	The **plasma membrane** is the cell's outer skin: a **lipid bilayer** with proteins floating in it. It is fluid, not rigid. It decides what enters and what leaves.
HOW IT WORKS / WHY IT MATTERS	Two features carry most of the clinical weight. First, it is a barrier, so a cell can hold a high internal concentration of something that would otherwise diffuse away. Second, membrane proteins do the talking. **Receptors** in the membrane bind one specific signal molecule and start a chain of events inside. If a tumour copies a receptor gene many times, the cell ends up with far more of that receptor on its surface - this is exactly what happens to **HER2** in breast and gastric cancer, and it is why an antibody that blocks that receptor works. Membrane proteins are also prime drug targets because a drug can reach them from outside the cell.
EXAM HOOK	He did not flag membrane structure itself. He did make the clinical version exam-relevant: HER2 amplification on the cell surface, and receptor occupancy as the parameter that decides dosing. He stated the occupancy thresholds as 1 % for a typical antibody, 2 % for Herceptin in breast cancer, and 10 % for Herceptin in gastric cancer. Plain meaning: the same drug needs to grab far more of the target in the stomach than in the breast before it does anything.

5.3.4 The nucleus - the control room and the master manuals

WHAT IT IS	The **nucleus** is a membrane-bound room at the centre of the cell. It holds the cell's **DNA** packed into **chromosomes**. Around the DNA is the **nuclear envelope**, pierced by **nuclear pores** so that messages can leave and proteins can enter.
HOW IT WORKS / WHY IT MATTERS	Because DNA is kept separate, the cell can control what gets read and when. The cell decides which genes to open, copies only those, and ships the copies out as mRNA. When something goes wrong in the nucleus, the consequences are large. A **mutation** is a typo in one of the manuals. A **driver** is a typo in an important word that gives the cell a growth advantage. A **passenger** is a typo on page 900 that travels along for free. Drivers cause cancer; passengers are noise. The nucleus is also where a virus such as hepatitis B keeps a permanent reservoir of its own DNA, and where that viral DNA can land next to a human gene and switch it on.
EXAM HOOK	He did not flag nuclear structure. The nuclear stories he did flag are the driver-versus-passenger distinction and the mutant-versus-wild-type comparison for K-Ras, where he told you that K-Ras was historically called undruggable and now has an FDA-approved small-molecule inhibitor. He also stated that p53 mutations are found in about 40 % of cancers. Plain meaning: the cell's own brake fails in roughly two tumours out of every five.

5.3.5 Mitochondria - the power station that keeps its own manual

WHAT IT IS	A **mitochondrion** is a bean-shaped **organelle** with two membranes. The inner membrane is folded into **cristae**, which is where the energy machinery sits. Inside the **matrix** is the mitochondrion's own DNA, a small circle of about 17 kilobases carrying 37 genes.
HOW IT WORKS / WHY IT MATTERS	The mitochondrion runs the **TCA cycle** and **oxidative phosphorylation**, which together make most of the cell's **ATP**. That is why cells with high energy demand - muscle, heart, brain, eye, hearing - suffer first when mitochondria fail. It is also the control centre of **apoptosis**, the cell's own death programme: the outer membrane decides whether the cell lives or dies. Three facts about mitochondria matter clinically, and Dr. Wong treats all three as exam content. First, **endosymbiotic origin**: the mitochondrion is a bacterium that was engulfed and kept. Second, **maternal inheritance**: an affected mother passes it to all her children, an affected father passes it to none. Third, the **threshold effect**: a cell can carry mutant mitochondrial DNA and still be healthy, and disease appears only when the proportion of mutant DNA exceeds what that tissue can tolerate. Tissues with high energy demand have a low threshold, so they fail first.
EXAM HOOK	This is Tier 1, his own words: *"this is you'll have to a 100 % understand, right? Not only the mutation within the mitochondrial DNA can cause the mitochondrial disease. Some mitochondrial disease can actually result from the mutation within the nuclear DNA."* And: *"Just make sure you understand that different mutation can cause the mitochondrial diseases, and all this mutation can cause the different inheritance patterns when you analyze this pedigree tree."* Plain meaning: if a pedigree looks mitochondrial, do not stop there - a nuclear gene can mimic it, and the inheritance pattern will be different.

5.3.6 Ribosomes - the machines on the floor

WHAT IT IS	A **ribosome** is a small machine made of RNA and protein. It reads an mRNA and joins **amino acids** together in the order the message specifies.
HOW IT WORKS / WHY IT MATTERS	The ribosome is where the language of four letters becomes the physical world of folded machines. Read every three letters, and you get one amino acid; that three-letter unit is a **codon**. Start codon AUG means methionine and marks the beginning. The stop codons UAA, UAG and UGA mark the end. The ribosome is also a deliberate target in modern medicine: an mRNA vaccine delivers a message and the patient's own ribosomes build the antigen; an in-vivo CAR-T approach delivers mRNA in **lipid nanoparticles** and the patient's T-cells make the receptor on their own surface.
EXAM HOOK	He asked you to remember the start and stop codons on the Lecture 1 slide. He also excluded the amino-acid table: *"don't memorize this... definitely not include in the exam."* Plain meaning: you need the idea of the genetic code, not the full list of 64 codons.

5.3.7 The endoplasmic reticulum and the Golgi apparatus - workshop and packing room

WHAT IT IS	The **endoplasmic reticulum (ER)** is a network of folded membranes attached to the nucleus. The **rough ER** has ribosomes stuck to it. The **smooth ER** makes lipids. The **Golgi apparatus** is a stack of flattened sacs further downstream.
HOW IT WORKS / WHY IT MATTERS	This is the cell's production line for anything leaving the cell. A protein with an **N-terminal signal peptide** is recognised by the ribosome as it is made, and is fed into the rough ER. There it is folded, checked, and often given **glycosylation** - decorated with sugar groups. A transport vesicle then carries it to the cis face of the Golgi, through the cisternae, out of the trans face, and into a secretory vesicle that fuses with the plasma membrane to release the protein outside. That whole route is the **secretory pathway**. It is the route used for antibodies, for hormones and for many medicines made by cells. He makes the point forcefully: remove the signal peptide and the protein never reaches the ER at all, so it never gets made properly, and your experiment produces nothing. The signal peptide also decides whether a protein ends up on the cell surface, inside a lysosome, or secreted into blood.
EXAM HOOK	Tier 2, his words: *"So remember, signal peptide is really, really important for your study."* He told the story of a colleague who lost four months of work by omitting it. He also warned against confusing it with the **N-terminal domain (NTD)** of a nuclear receptor, which is a completely different thing: that one is a regulatory region on a protein that binds DNA, not a postal address for the ER.

5.3.8 Lysosomes - the recycling and disposal plant

WHAT IT IS	A **lysosome** is a membrane-bound sac with a very acidic inside, packed with **enzymes** that break down large molecules.
HOW IT WORKS / WHY IT MATTERS	Lysosomes are where the cell recycles. They digest worn-out organelles, clear protein clumps, and destroy bacteria the cell has swallowed. They are also the recyclers in **autophagy**, the self-eating process in which a starving cell - for example one in the middle of a solid tumour with no blood supply - digests its own unneeded parts to get nutrients. It is a survival trick, and it is one reason a tumour core can keep growing in a hostile place. The failure mode is straightforward: if the recycling plant breaks, material accumulates, and the cell gets sick.
EXAM HOOK	He taught autophagy in Lecture 2, topic 9, as a survival mechanism in the hypoxic tumour core. He did not flag lysosomal storage diseases as examinable.

5.3.9 The cytoskeleton - the frame, the freight system and the dividing machine

WHAT IT IS	The **cytoskeleton** is a network of protein fibres through the cytoplasm. Three kinds matter: **microtubules** (hollow tubes), actin filaments (thin ropes) and intermediate filaments (rope-like, for strength).
HOW IT WORKS / WHY IT MATTERS	The cytoskeleton does four jobs. It gives the cell its shape and holds the membrane in shape. It acts as tracks and motors to move cargo around the cell. It builds the **mitotic spindle**, the ropes that pull duplicated chromosomes apart during division. And it forms the **contractile ring**, a tightening band of **actin** and **myosin** that pinches the cell in two during **cytokinesis**. The cytoskeleton also carries information: the Hippo pathway in Lecture 1 is switched by F-actin, the cytoskeleton branch, so the cell's shape and its growth-control pathway are physically linked.
EXAM HOOK	The contractile ring is Tier 1, his own words: *"Again, like I - I remember that I asked this question during the midterm in the first year, yeah. I think only half student can answer it correctly... So again this is the type of the question that I would try to include in the midterm."* The question is what you see when you block cytokinesis. See 5.3.11 for the answer.

5.3.10 The cell cycle - how one cell becomes two

WHAT IT IS	The **cell cycle** is the repeating schedule that lets a cell copy itself. It has four stages plus a split. **G1** is the growth phase, where the cell makes proteins and organelles and does its work. **S phase** is where the DNA is copied, so the cell ends up with two identical copies of every chromosome. **G2** is the checkpoint phase, where the copy is checked before division. **M phase** is mitosis, the division of the nucleus, where the duplicated chromosomes are pulled apart. **Cytokinesis** is the division of the cytoplasm, which follows mitosis. Cells that are not dividing rest in a side branch called **G0**.
HOW IT WORKS / WHY IT MATTERS	The schedule is driven by a two-part switch: **cyclins** and **cyclin-dependent kinases (CDKs)**. A CDK is the machine that does the work. A cyclin is the key that switches it on. Each cyclin rises, switches on its matching CDK, and the pair drives the next step. That pairing is the control point drugs attack. **Palbociclib** blocks CDK4/6 and so stops cells leaving G1. You can read the cycle from the bench: stain the DNA with **propidium iodide (PI)** and run the cells on a flow cytometer. Cells in G1 show 2N DNA content, cells in S show anything between, and cells in G2 or M show 4N. To line the cells up at the same starting point, researchers starve them of serum by dropping fetal bovine serum from 10 % to 0.5-1 %.
EXAM HOOK	He excluded the detailed pairings: *"you don't need to memorize all this cyclin or cyclin-dependent kinase, CDK."* He did exclude the detailed signalling cascade on the slide too: *"I don't memorize all this thing, I just highlight here in case some student are really interested in molecular biology."* The concept he wants is the four-stage shape of the cycle and the idea that a drug can freeze the cell at one stage.

5.3.11 When each part fails - the clinical organelle inventory

WHAT IT IS	Every compartment has a failure mode, and each one has a disease attached to it. The table below is the summary. The first six rows are Dr. Wong's own material. The last three are ordinary background biology included because the chapter scope asks for it.
HOW IT WORKS / WHY IT MATTERS	Read the table as a set of cause-and-effect stories.

Part	What it normally does	What failure looks like	Where it comes from
Plasma membrane	Controls entry and exit; carries receptors	Too many receptors on a tumour cell, or a signal that says "keep growing" when it should say "stop"	Lecture 2, HER2 amplification and PD-L1
Nucleus	Stores DNA; decides which genes are read	Mutations; a driver that gives growth advantage; p53 mutations in about 40 % of cancers	Lectures 2 and 3
Mitochondrion	Makes ATP; controls apoptosis	Mitochondrial disease: energy-hungry tissues fail first, and inheritance follows the mother	Lecture 2, topics 3 and 5
Ribosome	Builds proteins from the message	The cell cannot make the proteins it is told to make	Background biology
Rough ER and Golgi	Fold, decorate, sort and post outgoing proteins	A secreted protein never reaches the cell surface; a folding fault makes the cell sick	Lecture 2, topic 4, on signal peptides
Lysosome	Digests and recycles	Material piles up because the recycling plant is broken	Background biology
Cytoskeleton	Shape, transport, spindle, dividing	The cell cannot pull itself apart; blocked cytokinesis leaves one cell with two nuclei	Lecture 2, topics 7 and 9
Centrosome and spindle	Organises the chromosomes for division	Chromosomes fail to separate properly	Lecture 2, slides 35 and 36
Cell-cycle control	Times the stages; stops when there is damage	Uncontrolled division - the definition of cancer	Lecture 2, topic 6, and p53 in about 40 % of cancers

EXAM HOOK	The contractile-ring question is the one he named first. Here is his own framing, and it is worth learning as a template: *"if you're a genetic consultant"* style questions aside, for this one he says: treat dividing cells with an inhibitor of the contractile ring, or knock out its expression. What do you see? The answer is that **mitosis** and **cytokinesis** are independent processes. Mitosis - the chromosome separation - proceeds normally, because it is done by microtubules, not by the contractile ring. Cytokinesis - the pinching of the cytoplasm - does not happen. So you get a **binucleated cell** - two nuclei and 4N DNA content - and on a flow-cytometry DNA histogram you see a 4N peak instead of separate 2N and 4N peaks. He added a consequence that explains a clinical observation: because organelles and mitochondria are partitioned at random during cytokinesis, two siblings with the same mutation can have very different severity.

Figure 5.1 - Inside a cell



A cut-away animal cell with every compartment named in plain words: the membrane that lets things in and out, the nucleus that keeps the master manuals, the mitochondrion that makes energy, the ribosomes that build protein, the ER workshop, the Golgi packing room, the lysosome recycling plant, and the cytoskeleton frame.

5.4 The 12 things to remember

1. A cell is alive because it has a membrane that holds it together and DNA that specifies its parts.
2. A prokaryote has no nucleus. A eukaryote does. Everything else follows from that one difference.
3. The plasma membrane is a phospholipid bilayer, and it is also a set of receptors and doors.
4. The nucleus is the control room. It stores DNA and decides which recipes get copied.
5. The mitochondrion is the power station, it makes most of the ATP, and it keeps its own DNA because it used to be a bacterium.
6. The ribosome reads mRNA and joins amino acids. Three letters make one amino acid: that is a codon.
7. Rough ER to Golgi to plasma membrane is the secretory pathway, and the signal peptide is the postal address that starts it.
8. Lysosomes are the recycling plant. Autophagy is the cell eating its own spare parts to survive starvation.
9. The cytoskeleton gives shape, moves cargo, builds the spindle and forms the contractile ring.
10. The cell cycle is G1, S, G2, M, then cytokinesis. Cyclins are the keys; CDKs are the machines.
11. Blocking the contractile ring gives one cell with two nuclei and 4N DNA, because mitosis and cytokinesis are separate.
12. Every organelle has a failure disease, and each failure points at a different clinical story.

5.5 Check yourself

Q1. A bacterium has no nucleus. Why does that make it hard to use a bacterium to manufacture a human antibody?

Q2. You add a drug that blocks the contractile ring to a culture of dividing cells. What do you see under the microscope, and what does the DNA-content plot from flow cytometry look like?

Q3. Mitochondria carry their own DNA. Where did that come from, and how does it show up in a family tree?

Q4. Put the compartments a secreted antibody passes through on its way out of the cell in the correct order.

Q5. A patient has a normal gene sequence but the protein never reaches the cell surface. Name two different faults that could cause this, and say which one a deleted signal peptide would cause.

Answers.

A1. Reasoning: a human antibody is a eukaryotic protein, and eukaryotes are separated into compartments. In a prokaryote there is no nucleus and no ER-Golgi secretory pathway, so the machinery that folds, assembles and glycosylates a complex protein is missing. Bacteria are excellent for making simple proteins and bad at making complex glycosylated ones. The clinical version of this is that mammalian cells, such as **CHO cells**, are used when glycosylation matters, and they need **codon optimisation** to read a human gene well.

A2. Reasoning: the contractile ring is actin plus myosin, and its job is **cytokinesis**, the division of the cytoplasm. Mitosis is done by microtubules and is independent of it, so chromosome separation still happens normally and each half gets 2N. But the cell cannot pinch. The result is a **binucleated cell**: one cell containing two nuclei, with 4N DNA content. On a PI-stained flow-cytometry histogram you see a single 4N peak rather than separate 2N and 4N peaks. Note the clinical ripple: mitochondria are partitioned randomly at cytokinesis, so siblings with the same mutation can differ in severity.

A3. Reasoning: the mitochondrion is the **endosymbiotic** case - it began as a free-living bacterium that an ancestral cell engulfed and kept. That is why it has two membranes, its own circular DNA of about 17 kilobases with 37 genes, and its own ribosomes. Because the cell's own machinery destroys paternal mitochondria after fertilisation, inheritance is **maternal**: an affected mother passes the condition to all her children, an affected father passes it to none, and it appears in every generation. The twist he insists on: mitochondrial disease can also be

caused by mutations in **nuclear** DNA, and those follow ordinary Mendelian patterns instead.

A4. Reasoning: the sequence runs cytosol to rough ER to Golgi to secretory vesicle to plasma membrane and out. Step by step: the ribosome on the **rough ER** makes the protein because the **signal peptide** addresses it there; the protein is folded, checked and given **glycosylation** in the ER - the sugar groups are added there; a **transport vesicle** carries it to the **cis face** of the **Golgi**; it moves through the cisternae to the **trans face**; a secretory vesicle buds off and fuses with the plasma membrane, releasing the antibody outside the cell. Note the order matters: glycosylation happens in the ER, sorting happens in the Golgi.

A5. Reasoning: there are two classic routes. First, the protein may be made but never delivered, if the **signal peptide** is missing or broken. That is the answer to the second half of the question: a deleted signal peptide means the protein is never recognised as belonging to the secretory pathway, never enters the rough ER, and is therefore never folded, glycosylated, sorted or exported. Second, the protein may be made and delivered but fail inside the cell, for example a folding fault in the **rough ER** or a sorting fault in the **Golgi**. A signal peptide deletion therefore blocks expression of secreted and membrane proteins entirely - his postdoc lost four months of work that way.

CHAPTER 6

How a cell actually works

6.0 What this chapter is about

Chapter 5 gave you the rooms. This chapter gives you the working day. It starts with **metabolism**, which is simply how a cell turns food into usable energy. Then it follows the information all the way from a gene to a working protein - the **central dogma** - using the factory line picture. After that it explains how a cell receives a message and switches a response on and off, how it decides whether to divide or to die, and how cells talk to each other and to the immune system. Almost every drug and every biomarker in this course is one of those jobs being interrupted.

6.1 Words to know before we start

Term	In plain English
Metabolism	All the chemical reactions in a cell. Breaking things down for energy, and building things up.
Glycolysis	Breaking glucose into smaller pieces without using oxygen. The first step of energy production.
ATP	The cell's spendable energy. Made in the mitochondria and used to power everything else.
Central dogma	The rule that information runs one way: DNA to RNA to protein.
Transcription	Copying one recipe from DNA into a working message (RNA).
Translation	Reading that message and building the protein it describes.
Codon	A three-letter unit on the mRNA. Each one names one amino acid.
Gene expression	The whole process of turning a gene on, making its message, and making its protein.
Enzyme	A protein that makes a chemical reaction go faster.
Kinase	A protein that attaches a phosphate group to another protein. Usually a switch.
Phosphatase	A protein that removes a phosphate group. It is the other half of the switch.
Receptor	A protein, often on the cell surface, that catches one specific signal molecule.
Signal transduction	The relay of a message from the outside of the cell to somewhere inside it.
Apoptosis	Programmed cell death. The cell dismantles itself on purpose.

6.2 The big picture

Start with fuel. A cell takes in sugar and fat and protein, and through **metabolism** it breaks them down to release energy, then rebuilds them into the cell's own parts. It needs raw material as well as energy, and in a cell that is about to divide, raw material matters more. One step in this breakdown is **glycolysis**: glucose is split into smaller molecules without oxygen. When oxygen is available, most cells go further and use the **mitochondrion** to burn the products properly, which yields far more **ATP**, the cell's spendable energy. ATP is the cell's money: it pays for building, moving, copying and signalling. Here is where the Warburg effect comes in. Cancer cells often choose glycolysis even when oxygen is available. On his Lecture 1 slide the numbers are stark: a resting cell with oxygen makes about 36 molecules of ATP per glucose, and without oxygen only 2. A proliferating cancer cell, with or without oxygen, sends about 85 % of its pyruvate to lactate, only about 5 % into mitochondria, and nets roughly 4 molecules of ATP. The inefficient route looks like a mistake. It is not. The half-broken sugar is perfect building material, and a cell about to divide a billion times needs building material far more than fuel.

Now the information half. The rule is the **central dogma**, and it runs one way: DNA to RNA to protein. Think of DNA as the master instruction manuals, locked in the control room. A **gene** is one recipe. Copying a recipe out is **transcription**, and the copy is **mRNA**, a photocopy of one recipe sent to the factory floor. Reading the photocopy and building the machine is **translation**. The machine is a **protein**: a folded machine made from a bead necklace. The ribosomes do the building. The mRNA is read in threes, and each three-letter **codon** names one amino acid.

Then the protein has a life of its own. It folds into a shape, and the shape is the job. It may be decorated with sugar groups in the ER, sorted in the Golgi, and shipped out through the secretory pathway. It may become a receptor, an enzyme, a structural piece, or a signal.

And the cell is listening the whole time. A **receptor** in the membrane catches a specific signal molecule. The message is relayed inward by **signal transduction**, usually through **kinases** that attach a phosphate group and **phosphatases** that remove it. That pair is a light switch. A phosphate group carries charge, so adding one changes a protein's shape and pushes it into an active state; taking it away turns the switch off. The cell can therefore respond in seconds and then recover.

Two huge decisions follow. First, divide or die. A cell with damaged DNA is supposed to stop, or to commit suicide by **apoptosis**. A tumour is a cell that ignored that instruction. Drugs exploit it: CDK4/6 inhibitors freeze the cell cycle, and bispecific antibodies that carry both a tumour marker and a death receptor trigger apoptosis only where the tumour is.

Finally, cells talk. They send **cytokines** to their neighbours, they display fragments of their own proteins on the surface as **MHC** so that immune cells can inspect them, and they receive instructions from the immune system. The most important conversation in modern cancer therapy is the checkpoint: **PD-L1** on the tumour cell binds **PD-1** on the T-cell and switches the T-cell off. An anti-PD-1 antibody blocks that handshake and switches the killer back on. That is why a tumour with many neoantigens, called a **hot tumour**, responds far better than a cold one.

6.3 Walkthrough

6.3.1 Metabolism - how the cell handles its food

WHAT IT IS	**Metabolism** is everything the cell does chemically. Two directions at once: **catabolism**, breaking big molecules down to release energy, and **anabolism**, building big molecules from small ones using that energy.
HOW IT WORKS / WHY IT MATTERS	Sugar is the main starting fuel. It is broken down by **glycolysis** into pyruvate, and then fate depends on oxygen. With oxygen and healthy mitochondria, pyruvate goes into the TCA cycle and oxidative phosphorylation, giving roughly 36 ATP per glucose. Without oxygen, only 2 ATP. But a cell also needs raw material, and the split-up sugar is a better building block than the burned-up version. This is exactly why cancer cells behave oddly, the **Warburg effect**, also called **aerobic glycolysis** because the fermentation happens despite oxygen being present. The tumour gains building material at the cost of efficiency.
EXAM HOOK	He used the Warburg effect as the entry point to Lecture 1 and as the reason metabolomics became a useful way to find cancer biomarkers. He did not ask you to memorise the pathway. He explicitly excluded it: *"Please don't memorize all these chemical structure... They would not be included in my exam."* Plain meaning: understand the trade-off, not the chemistry.

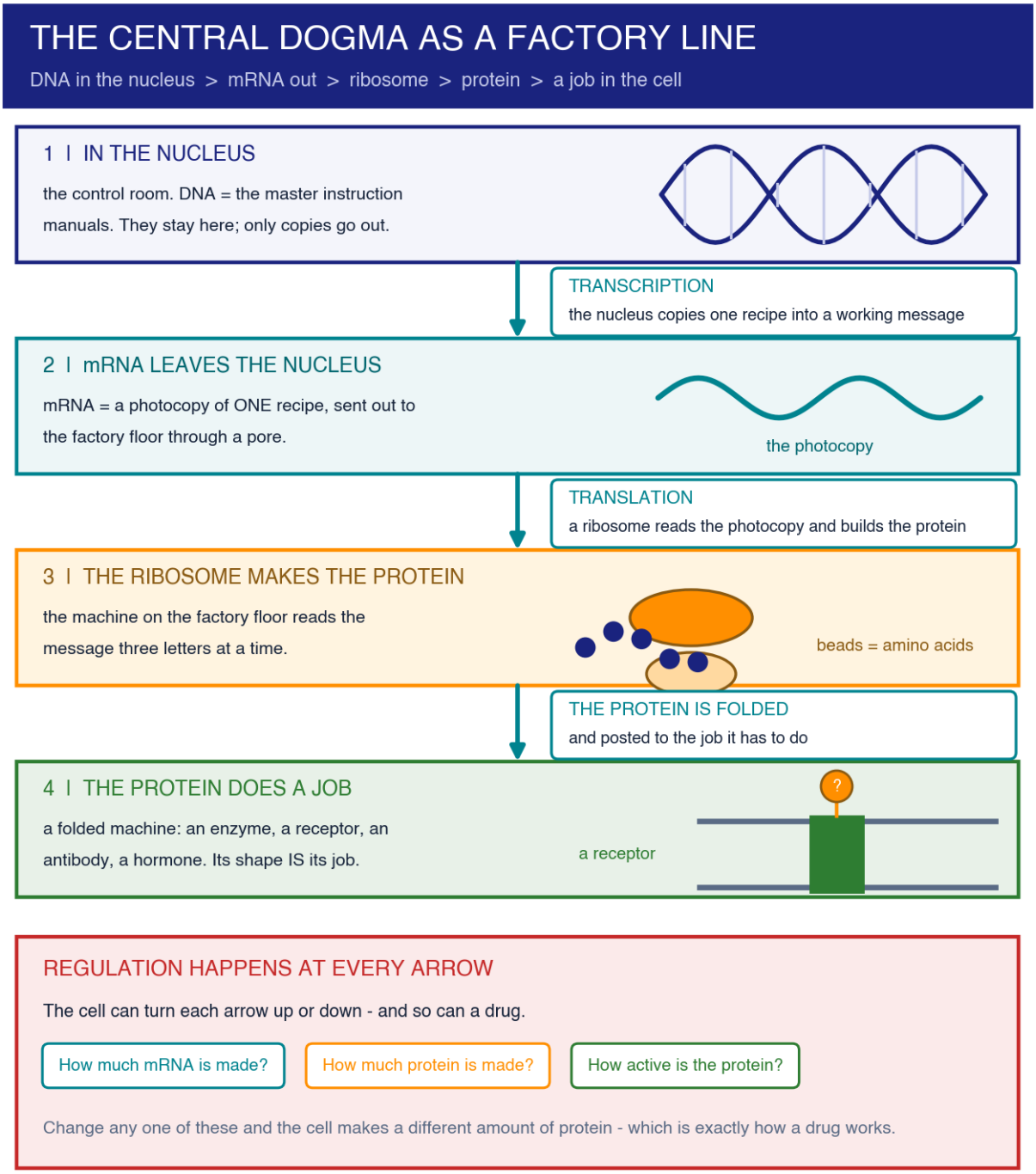
6.3.2 Energy - ATP, the cell's money

WHAT IT IS	**ATP** (adenosine triphosphate) is a small molecule with three phosphate groups in a row. The cell spends energy by breaking off the last phosphate, and it recharges ATP by putting it back.
HOW IT WORKS / WHY IT MATTERS	Every job the cell does is paid for in ATP. Moving molecules, building a protein, copying DNA, opening a channel. A mitochondrion is the biggest ATM: it takes the fuel from sugar and reloads the ATP. This is why the mitochondrion's failure is so dangerous, and why a tumour that switches off oxidative phosphorylation is really switching off its main ATM. It is also why **protein folding** costs the cell energy - the cell has to spend ATP to make the chain fold into the right shape, which is one reason a misfolded protein factory burns fuel and produces nothing.
EXAM HOOK	He stated the ATP yields on the Lecture 1 slide: about 36 molecules with oxygen, 2 without, and about 4 for a cancer cell running aerobic glycolysis. Plain meaning: the cancer cell is running at roughly one-tenth the energy efficiency of a normal aerobic cell, on purpose, because it wants the raw material. He did not ask you to derive these numbers.

6.3.3 The central dogma - a factory line with three arrows

WHAT IT IS	The **central dogma** is the direction rule for genetic information. It runs one way: DNA is copied into RNA, and RNA is read to build protein. Information does not flow backwards from protein to DNA.
HOW IT WORKS / WHY IT MATTERS	There are three arrows, and each one is a step where the cell can turn the output up or down. Arrow one, DNA to RNA, is **transcription**. Arrow two, RNA to protein, is **translation**. Arrow three is what the protein then does in the cell. Because the arrows are separate, each is a separate point of control, and each is a separate opportunity for a drug. This is the single most useful picture in the course, because every later lecture adds a new way of controlling one of these arrows.
EXAM HOOK	He taught the central dogma in Lecture 1, topic 13, and revisited it in Lecture 5, in the promoter and gene-structure topics. He did not list it among the five questions he named as exam-type, but it is the frame that every one of those questions sits inside.

Figure 6.1 - The central dogma as a factory line



DNA stays in the nucleus. A photocopy of one recipe, the mRNA, is carried out to the factory floor. A ribosome reads it and builds a folded protein. The protein then does a job in the cell. Every arrow is a point where the cell can turn the output up or down, and therefore every arrow is a point where a drug can work.

6.3.4 Transcription - making the photocopy

WHAT IT IS	**Transcription** literally means "writing across". An enzyme, **RNA polymerase**, reads along a gene on the DNA and builds a complementary RNA strand. In eukaryotes that first copy is **pre-mRNA**.
HOW IT WORKS / WHY IT MATTERS	The enzyme does not start anywhere. It is recruited to a **promoter**, a specific DNA sequence just upstream of the gene. On the Lecture 5 slide, a complex of general transcription factors assembles at the **TATA box** about 30 bases upstream, and RNA polymerase II is added. Control sequences called **enhancers** and **silencers** can sit thousands of bases away, and DNA-bending proteins loop them onto the promoter. So a gene's output depends on a small DNA word (the promoter) plus distant words (the enhancers) plus the proteins that read them. The pre-mRNA is then edited: **introns** are cut out by the spliceosome and **exons** are joined, which is **splicing**. If the cell chooses different exons, one gene makes several different proteins - **alternative splicing**.
EXAM HOOK	The lac operon is his worked example of transcription control in Lecture 5, topic 5: strong transcription needs lactose present, so that allolactose can bind the **Lac repressor** and lift it off the operator, and glucose low, so that cAMP rises and **CAP-cAMP** can recruit RNA polymerase. Two conditions, not one. He also made the point that *"not all the enhancer are located on the upstream of the sequence"* - some sit downstream of the gene.

6.3.5 Translation - building the machine

WHAT IT IS	**Translation** is the ribosome reading the mRNA and joining amino acids to make a chain. The code is read in threes: each **codon** names one amino acid, and the chain grows as the ribosome moves along the message.
HOW IT WORKS / WHY IT MATTERS	Two details matter for this course. First, the code is not universal in practice: different organisms prefer different codons for the same amino acid, which is called **codon-usage bias**. If you put a human gene into a hamster cell line without **codon optimisation**, the protein comes out badly or not at all. He made this point explicitly: *"So remember that when you use the CHO cell, you need to do the codon optimization."* Second, a mistake in the reading frame is catastrophic. If a mutation adds or removes a number of letters that is not a multiple of three, the reading frame shifts and everything downstream is wrong - that is a **frameshift**, and it usually produces a premature stop and a truncated protein.
EXAM HOOK	Frameshifts are his Lecture 1, topic 15, and they matter because they create **neoantigens** - protein fragments the immune system has never seen, which are ideal targets for immunotherapy. He also stated the start codon AUG (methionine) and the stop codons UAA, UAG and UGA.

6.3.6 What the protein does afterwards - the job

WHAT IT IS	A translated chain is not yet a working protein. It folds into a specific three-dimensional shape, and that shape is what lets it do one job.
HOW IT WORKS / WHY IT MATTERS	The chain is folded in the cytoplasm, or in the rough ER if it carries a signal peptide. **Chaperone** proteins help it fold correctly. Many proteins are then modified - **glycosylation** adds sugar trees, and **phosphorylation** adds phosphate groups. Some proteins are cut in two by enzymes, and that cut is the switch that activates them. The finished protein may end up as an **enzyme** that speeds a reaction, a **receptor** in the membrane, a **cytokine** that is released to signal, an **antibody** that binds a target, or a structural piece such as a part of the cytoskeleton. Its job, its location and its lifetime are all set by the journey.
EXAM HOOK	He excluded the chemistry of folding and the amino-acid R-group table from the exam, but the concept matters for everything else: *"don't memorize this... definitely not include in the exam."* What is examinable is the clinical consequence, which he taught in Lecture 1, topic 7: a single chemical change, such as **deamidation** of asparagine to aspartate at high pH, can alter how well a therapeutic protein binds its target and therefore change its efficacy.

6.3.7 Receiving a message - receptors, kinases and the on/off switch

WHAT IT IS	A **receptor** is a protein that recognises one specific signal molecule and changes shape when it binds. **Signal transduction** is the relay of that news through the cell to the nucleus.
HOW IT WORKS / WHY IT MATTERS	The relay is usually a chain of **kinases**, each one switching on the next by adding a phosphate group, and the switch is reversible because **phosphatases** remove it. The phosphate group carries charge, so adding it changes the protein's shape and its activity. That is the on/off switch. Two examples carry the whole course. First, the **Hippo pathway**: cell-to-cell contact activates MST1, which activates LATS1, which phosphorylates **YAP**; phosphorylated YAP is held in the cytoplasm, so the growth programme stays off. When the pathway breaks, unphosphorylated YAP enters the nucleus and drives growth. Second, **NF-kappaB**: it is normally held in the cytoplasm by **IkappaB**. When lipopolysaccharide (LPS) is detected, IkappaB is degraded, NF-kappaB moves into the nucleus, and the inflammatory genes are switched on. NF-kappaB then makes the inflammatory proteins that keep reinforcing it - a **positive-feedback loop** that makes the response fast. The order of the response is timed: **TNF-alpha** peaks at 4-6 hours, then IL-1beta, then IFN-gamma, then IL-2 and IL-15.
EXAM HOOK	The cascade order LPS to TLR4 to IKK to IkappaB to NF-kappaB to the nucleus is one of his published MCQ shapes. The exact cytokine order is explicitly excluded: *"don't worry about all these cytokines. I'm just want to show you that different responses can be observed in different times."* He did ask you to remember the Hippo phosphorylation logic, and he said the Hippo pathway is a subject of pharmacological targeting, with druggable inhibitors in development.

6.3.8 Deciding to divide, or to die

WHAT IT IS	A cell checks its work before it copies its DNA. If the DNA is damaged, it should stop and repair. If it cannot be repaired, it should die by **apoptosis**, the controlled dismantling programme.
HOW IT WORKS / WHY IT MATTERS	Apoptosis has two routes. The **intrinsic** route starts inside the cell, usually at the mitochondria: stress signals push the balance of the Bcl-2 family, cytochrome c is released, and a cascade of **caspases** is activated. The **extrinsic** route starts outside: a **death receptor** on the cell surface is triggered by its ligand, which starts the caspase cascade directly. Both routes converge on the same executioners, and the cell is taken apart in an orderly way and swallowed by its neighbours. When this fails, the cell keeps dividing and you have cancer. He noted that p53 mutations appear in about 40 % of cancers: the brake is broken. Two drug classes exploit the same logic. **CDK4/6 inhibitors** such as palbociclib freeze the cell in G1 and stop the cycle. A **bispecific antibody** is built to carry two recognition heads at once, one for a tumour-specific antigen and one for a death receptor, so that apoptosis fires only where the tumour is.
EXAM HOOK	He presented the bispecific antibody BI 905711 in Lecture 2 as the clean clinical example, using his own **CDH17** target: apoptosis happens only in tumour tissue, because only tumour tissue has both markers present. He also made apoptosis the connection to the **Warburg effect** and to mitochondrial control of cell death. He did not ask you to memorise the caspase cascade.

6.3.9 Talking to neighbours and to the immune system

WHAT IT IS	Cells communicate by sending molecules and by displaying molecules. They also physically touch, using adhesion molecules to grab each other and to let immune cells slip through tissue.
HOW IT WORKS / WHY IT MATTERS	Three mechanisms carry the clinical weight. First, **cytokines** are released into the space between cells and bind receptors on their targets; the NF-kappaB programme makes a whole cascade of them in a timed sequence. Second, **MHC** molecules on the cell surface display short protein fragments, so that T-cells can inspect what a cell is made of; this is how a mRNA vaccine works - the delivered message is translated, the protein is chopped into peptides, and the peptides are presented for T-cell and B-cell activation. Third, **cell adhesion molecules** control movement through tissue. On his Lecture 2 slide the white-blood-cell journey is in four steps: rolling, using **P-selectin**, **E-selectin** and **integrins**; firm adhesion through **ICAM-1** and **VCAM-1**; squeezing between endothelial cells through **PECAM-1**; and then following a chemical gradient, **MCP-1**. On top of that sit the **immune checkpoints**. **PD-L1** on the tumour cell binds **PD-1** on the T-cell and switches the T-cell off into a state called anergy. The tumour is invisible. An anti-PD-1 antibody blocks that handshake and puts the killer back to work.
EXAM HOOK	He tied the checkpoint story to the **hot tumour** versus **cold tumour** idea: *"So guys, you definitely remember that for the tumor that present many neoantigen and can be treated with the immunotherapy, we call this is a hot tumor, yeah. And not all the solid tumor is a hot tumor."* A tumour with defective mismatch repair accumulates frameshift mutations, produces many neoantigens, becomes hot, and responds to anti-PD-1. He called the dMMR-versus-pMMR survival figure one of the most important figures in modern medicine: *"I can for sure that the presenter will include this figure in his presentation deck."*

6.4 The 10 things to remember

1. Metabolism has two halves: breaking down for energy and building up for new parts.
2. ATP is the cell's money. The mitochondrion is the biggest ATM.
3. The Warburg effect is a strategy, not a mistake: the inefficient route also supplies building material.
4. The central dogma runs one way: DNA to RNA to protein.
5. Transcription copies a recipe out of the nucleus. Translation builds the machine on the factory floor.
6. One gene can make several proteins, because splicing can join exons in different combinations.
7. A signal is relayed inward by kinases and switched off by phosphatases. The phosphate group is the switch.
8. A damaged cell should stop dividing or die by apoptosis. A tumour is a cell that ignored that.
9. Cells talk with cytokines, with MHC display, and with physical adhesion.
10. PD-L1 on the tumour switches the T-cell off; an anti-PD-1 antibody switches it back on. That is why hot tumours respond and cold ones do not.

6.5 Check yourself

Q1. Put the three steps of the central dogma in order and describe each one in factory language.

Q2. A cancer cell has plenty of glucose and plenty of oxygen. Why might it still choose to ferment rather than use its mitochondria?

Q3. What is the difference between a kinase and a phosphatase, and why is that difference so useful to a cell?

Q4. A tumour cell keeps dividing when it should not. Name the two ways a cell normally stops dividing or dies, and say which drug class acts on each.

Q5. A tumour cell has a lot of PD-L1 on its surface. What happens to a T-cell that meets it, and what drug reverses that?

Answers.

A1. Reasoning: the direction rule is DNA, then RNA, then protein. Step one, **transcription**: inside the nucleus, RNA polymerase copies one recipe out of the DNA as mRNA, a photocopy sent to the factory floor. Step two, **translation**: a ribosome on the factory floor reads the photocopy three letters at a time and joins amino acids into a chain, which folds into a working machine. Step three, the job: the finished protein acts - as an enzyme, a receptor, an antibody, a cytokine, or a structural part. Each step is a separate control point, which is why each is a separate drug target.

A2. Reasoning: because efficiency is not the goal. Fermenting glucose to lactate gives only about 4 ATP per glucose instead of about 36, but it leaves the half-broken sugar molecule behind as raw material for nucleotides, lipids and amino acids. A cell that is about to divide many times needs far more building material than fuel, so it switches to the **aerobic glycolysis** of the **Warburg effect** even with oxygen present. This is a deliberate growth strategy, and it is also why metabolomics became a useful way to find cancer biomarkers.

A3. Reasoning: a **kinase** adds a phosphate group, using ATP; a **phosphatase** removes it. The phosphate group carries charge, so adding it changes the protein's shape and pushes it into an active state, and taking it away turns it off again. The usefulness is speed and reversibility: the cell can respond to a signal within seconds and then reset within seconds, which a permanent change could not allow. The **Hippo pathway** is the model: MST1 activates LATS1, which phosphorylates **YAP**, and phosphorylated YAP is trapped in the cytoplasm so the growth programme stays off; phosphatase activity flips that decision.

A4. Reasoning: the two ways are to stop dividing and to die. Stopping is done by the cell-cycle controls: **CDK4/6 inhibitors** such as palbociclib freeze the cell in G1, and this is exactly the drug class he describes in the cell-cycle slide. Dying is done by **apoptosis**, triggered either from inside the cell via the mitochondria or from outside via a

death receptor, and both routes converge on caspases. The **bispecific antibody** acts on the second route: it carries two heads, one for a tumour-specific antigen and one for a death receptor, so apoptosis fires only in tissue that has both - his BI 905711 example built on CDH17.

A5. Reasoning: **PD-L1** on the tumour cell binds **PD-1** on the T-cell, and that handshake switches the T-cell off into anergy. The immune system can see the tumour but will not attack it, which is immune evasion. An **anti-PD-1** antibody blocks the handshake and releases the brake, restoring the T-cell's killing. Whether that works depends on the tumour: a **hot tumour**, loaded with neoantigens because its mismatch repair is broken, responds; a cold one does not. He called the dMMR-versus-pMMR survival figure one of the most important figures in modern medicine.

CHAPTER 7

The plain-English glossary

7.0 Where the glossary is

The complete glossary for this book - every technical term, in one alphabetical list, with a plain-English meaning and the chapter it belongs to - is in **Part 7, section 13.1 (Master glossary)**. It is the only glossary you need. Use this chapter's few lines to get oriented, and Part 7.1 whenever a term stops you mid-sentence.

End of Part 1 Foundations, Chapters 5 to 7. Part 2 begins with Lecture 1, Introduction to Biomolecules.

The complete glossary for the whole book - every technical term, in one alphabetical list, with a plain-English meaning and the chapter it belongs to - is in **Part 7, section 13.1 (Master glossary)**. It is the only glossary you need; you only need to reach for it if you want to revise the foundations vocabulary before you start the lectures.

PART B

The foundation glossary

This is the complete vocabulary of the course: every term in the lectures, with a plain-English meaning and the place it does its work. **582** of the **905** entries belong to the foundation chapters you have just read - a "Ch" number in the last column means the term does its work in this book. The other **323** are forward references: the lectures will use them, and they are here so that you can look one up before you meet it. Read it two ways - as a dictionary when a word stops you, and as a revision list, by covering the middle column and saying the meaning out loud before you look.

Terms are alphabetical, ignoring capital letters and punctuation. Where a term has more than one common spelling, the usual one is used.

Term	In one sentence	Where
14-3-3	A scaffold protein that pins phosphorylated YAP in the cytoplasm and lets the whole complex be degraded.	Ch 7,Lect 1,Lect 5
2D-PAGE	Two-dimensional protein electrophoresis, separating first by charge and then by molecular weight.	Ch 2, Ch 11, Lect 1
3D reconstruction	Stacking optical sections back together to rebuild the three-dimensional shape of a cell.	Ch 9, Ch 11, Lect 2
3 prime untranslated region (3 prime UTR)	The stretch of an mRNA that comes after the stop codon and is never turned into protein, but that still carries all the regulatory sites for that message.	Ch 7, Lect 5
4-1BB	One of the two co-stimulatory signals built into a CAR; the other one is CD28.	Ch 9, Lect 2
53BP1, RIF1 and Shieldin	The three factors that push the cell towards NHEJ and hold HR back.	Ch 6, Lect 4
accuracy	The rate at which a diagnostic test gives the right answer, the overall performance of sensitivity and specificity weighed together.	Ch 11, Lect 1
Acetylation	Adding an acetyl group onto a histone loosens the chromatin, so the gene underneath it can finally be read.	Ch 4, Lect 3, Lect 5
Actin	One of the two proteins of the contractile ring, the one that pinches the cell in half; also the track along which cargo is moved inside the cell.	Ch 8, Lect 2
Activation energy	An energy threshold that every chemical reaction has to climb over, and enzymes exist to push it down.	Ch 2
Activator	A protein that binds the enhancer and helps pull the transcription machinery over to the promoter.	Ch 7, Lect 5
Active site	The pocket on an enzyme where the substrate slots exactly into place, and where the chemical reaction happens.	Ch 2
adaptive CNV	A difference in copy number that is the result of adaptation, not of a defect.	Ch 3, Ch 10, Lect 3
adenine	One of the four main bases of DNA and RNA, written A.	Ch 3, Lect 1
Adenosine	What is left of ATP once you cut off the phosphate tail: one base plus one sugar.	Ch 9
adenovirus	A viral vector for getting genes into cells.	Lect 2
adhesion and activation	The adhesion and activation step, after an immune cell has rolled along the vessel wall.	Ch 9, Lect 2
Adoptive cell therapy	Take cells out of the patient, modify or treat them, and put them back in to go after the tumour.	Ch 9,Lect 2,Lect 3
ADP (adenosine diphosphate)	What is left of ATP after one phosphate has been spent; put the phosphate back and it is recharged.	Ch 1
Aerobic glycolysis	Turning sugar into lactate even when oxygen is plentiful - the pathway cancer cells prefer.	Ch 10, Lect 1
AF-1 and AF-2	The two activation regions on a nuclear receptor that recruit co-activators.	Ch 7, Lect 5
AGO	The protein that carries a microRNA inside the silencing complex.	Ch 7, Lect 5
Alkylation	A kind of damage that glues a small chemical blob onto a DNA base, after which the base pairs with the wrong partner.	Ch 6,Lect 3,Lect 4
Allele	A different version of the same gene; you carry two of them, one from each parent.	Ch 3, Lect 3

Term	In one sentence	Where
Allogeneic	Cells taken from a donor rather than from the patient themselves, so they are an off-the-shelf product.	Ch 9, Lect 2
Allolactose	Allolactose is an isomer of lactose, and it is the molecule that actually knocks the Lac repressor off the operator and lets the lac operon run.	Ch 7,Lect 5
Allostery	A different molecule binds somewhere other than the main working site, yet changes the shape and the activity of that main site.	Ch 2, Lect 1
Alpha helix (alpha-helix)	The drill-shaped helix a protein chain usually takes, held up by hydrogen bonds between neighbouring turns.	Ch 2
alternative end joining (a-EJ)	The secondary pathway that can rejoin broken ends without needing much homologous sequence.	Ch 6, Lect 4
Alternative polyadenylation	Choosing to put the poly-A tail on at one position or at another, and in that way changing how long the 3' UTR ends up being.	Ch 7, Lect 5
Alternative splicing	One gene uses different combinations of exons to build several different proteins.	Ch 7, Lect 5
Amino acid	The twenty small molecules that make up proteins, each carrying a side chain that decides how the chain folds.	Ch 2
Amp	The ampicillin resistance marker on a plasmid, used to select the bacteria that took the plasmid up.	Ch 1, Ch 3, Lect 4
AMP (adenosine monophosphate)	The level left over after ATP has spent two of its phosphate groups.	Ch 1,Ch 3,Lect 4
Anabolism	The building half of metabolism: spend energy to assemble large molecules.	Ch 1
Anaphase	The stage of mitosis in which the chromosomes split up from each other and travel in opposite directions towards the two poles.	Ch 8, Lect 2, Lect 3
AND-gate	A switch that only fires when two inputs are present at the same time; synNotch CAR uses it so it does not kill by mistake.	Ch 7, Lect 5
Anergy	The state a T cell ends up in when a checkpoint signal switches it off: there is a signal, but no work gets done.	Ch 9,Lect 2
Aneuploidy	A cell whose chromosome number is wrong, usually because segregation failed during meiosis.	Ch 3, Lect 3
Angstrom	The unit for measuring atomic distances; a crystal structure at 2.8 A counts as good, and anything below 2.0 A counts as excellent.	Ch 2, Ch 4, Lect 3
Annealing	The process by which two complementary single strands of DNA pair up again as the temperature comes down.	Ch 3, Lect 4
Antibody	A Y-shaped protein made by B cells that recognises one antigen and then either blocks it or tags it for removal.	Ch 2, Ch 9, Lect 1, Lect 2
Antibody-dependent cellular cytotoxicity (ADCC)	The Fc portion of an antibody grabs an NK cell and orders it to kill whatever the antibody is stuck to.	Ch 2,Ch 9,Lect 1,Lect 2
Antibody-drug conjugate (ADC)	An antibody used as a delivery van, carrying a toxic drug only to the place where it recognises its target.	Ch 10, Lect 1, Lect 4
Antigen	Something that the immune system identifies as foreign and prepares to deal with.	Ch 2, Ch 9, Lect 1, Lect 2
Antigen-presenting cell	A cell that lays out fragments of protein on its surface so T cells can inspect them.	Ch 6, Lect 2, Lect 4
Anti-inflammatory mediator	The class of molecules that turns inflammation off once the danger has passed.	Lect 5
Antiparallel	The two strands of DNA run in opposite directions: one from 5' to 3', and the other from 3' to 5'.	Ch 3
Anti-PD-1 antibody	A drug antibody that blocks the PD-1 checkpoint and puts the T cell back on duty.	Ch 4, Ch 6, Ch 9, Lect 2
Anti-PD-L1 antibody	It blocks PD-L1 on the tumour cell, and the effect is the same as anti-PD-1.	Ch 9, Lect 2
Anti-TERT	An antisense oligonucleotide that degrades TERT mRNA, so the cell can no longer keep rebuilding the ends of its chromosomes.	Ch 5,Ch 7,Ch 12,Lect 4
APAF-1	The adaptor protein that assembles cytochrome c and procaspase-9 into the apoptosome.	Ch 8, Lect 2
APC	The first brake to fail on the multi-step road to colorectal cancer.	Lect 4
APOLO	A long non-coding RNA that forms an RNA-DNA hybrid with DNA.	Ch 7, Lect 5
Apoptosis	Programmed cell death: the cell takes itself apart neatly instead of bursting.	Ch 8, Lect 2
Apoptosome	The wheel-shaped structure assembled after cytochrome c is released, and it is what starts caspase 9.	Ch 8, Lect 2
Apo-repressor	A repressor with no ligand bound to it, and therefore one that cannot hold on to the operator.	Ch 7,Lect 5

Term	In one sentence	Where
AP site	The gap left behind when a base is cut out and only the sugar backbone remains.	Ch 6, Lect 4
arginine	A positively charged basic amino acid.	Ch 2, Lect 1
aristolochic acid I (AA-I)	An aristolochic acid from a Chinese herbal medicine, and one of the four carcinogens Dr. Wong names.	Lect 3
AR-V7	A shortened version of the androgen receptor produced by a splicing error, which is permanently active and which resists enzalutamide.	Ch 7, Lect 5
Arylamination	The chemical step in which an activated carcinogen precursor attaches an aromatic amine group to a DNA base.	Lect 3
asparagine	The amino acid that readily undergoes deamidation.	Ch 1, Ch 2, Lect 1
aspartic acid	One of the two products you get when asparagine is deamidated.	Ch 1, Ch 2, Lect 1
Atezolizumab	Atezolizumab is an immune checkpoint inhibitor that goes after PD-L1 rather than after PD-1.	Lect 3
ATM	The first sensor kinase to be switched on after DNA damage, and it mainly answers double-strand breaks.	Ch 6, Lect 4
Atom	The smallest unit of an element; a molecule is what you get when atoms are bonded together.	Ch 1
Atomic nucleus	The dense centre of an atom, and the place that holds the protons and neutrons.	Ch 1
ATP (adenosine triphosphate)	The energy the cell can spend; cut the terminal phosphate off it and the power comes out.	Ch 1
ATR	The other sensing kinase, the one that answers replication stress and single-stranded DNA.	Ch 6, Lect 4
AT-rich	A stretch of DNA in which the proportion of A, adenine, and T, thymine, is high.	Ch 3, Ch 11, Lect 3
AUC (area under the curve)	A number between 0 and 1 that summarises how well a test can tell two groups of people apart, and higher is better.	Ch 4,Lect 1
Autologous	Taken from the patient themselves and given back to the same patient.	Ch 9, Lect 2
Autophagy	The cell eats the parts of itself it does not need, and gets nutrients back in return.	Ch 8, Lect 2
Avery, MacLeod and McCarty	The three who proved in 1943, using DNase and protease, that the transforming factor was DNA.	Ch 3, Lect 1
Basal transcription	The baseline transcription that keeps running before any activator has arrived.	Ch 7, Lect 5
Base	One of the four letters of DNA: A, T, G and C; in RNA, U takes the place of T.	Ch 1,Ch 3
Base excision repair (BER)	The pathway that repairs a single damaged base: cut that base out and then put a new one in its place.	Ch 6, Lect 4
Base pairing	The rule for pairing bases in DNA: A pairs with T, G pairs with C.	Ch 3, Lect 1
Bcl-2 family	The group of proteins that decide whether the mitochondria release the apoptotic signal.	Ch 8, Lect 2
Benzo(a)pyrene	A tobacco carcinogen that forms an adduct with DNA after the cell has activated it.	Lect 3
Beta sheet (beta-sheet)	Several protein chains pulled apart and lying side by side, a shape held in place by hydrogen bonds between them.	Ch 2
BI 905711	A tetrahedral bispecific antibody that recognises TRAILR2 and CDH17 at the same time.	Ch 8, Lect 2
bid	The apoptosis signal protein that caspase-8 cuts into tbid.	Ch 8, Lect 2
Binucleated cell	A cell with two nuclei inside it and a DNA content of 4N.	Ch 8, Lect 2
Biomarker	A biomarker is a molecule you can measure in a sample and that tells you something about the disease or about how a treatment is working.	Ch 11, Lect 1, Lect 2, Lect 4
Biomolecule	Molecules that a living system makes: carbohydrates, lipids, proteins and nucleic acids.	Ch 1,Lect 1
Bispecific antibody	An antibody with two different binding heads, so that it can hit two targets at the same time.	Ch 8, Lect 2
Bisulfite sequencing	The gold standard test for methylation: unmethylated C turns into U, methylated C stays as it is.	Ch 4, Ch 11, Lect 3
Boundary elements	A chromatin boundary that fences off the active region from the silent one.	Ch 4, Lect 3
BRAF	A kinase sitting in the MAPK signalling pathway, and an early event on the serrated colorectal cancer route.	Ch 10, Lect 4
branched linker	A forked linker that carries two different drugs at the same time.	Ch 10, Lect 4
branch migration	The step that extends the stretch of DNA that has matched up.	Ch 6, Lect 4

Term	In one sentence	Where
BRCA1 and BRCA2	Lose these two genes and the cell can no longer repair broken chromosomes by homologous recombination.	Ch 6,Ch 10,Lect 4
BRD4	A protein that reads the acetylation mark and drags the transcription machinery over, and it is a current drug target.	Ch 4, Lect 3
Breakthrough Therapy Designation	The FDA's fast track for drugs with breakthrough therapeutic potential.	Ch 12, Lect 1
Breathomics	The study of the chemical content of exhaled breath, used to see inside the body.	Ch 10, Ch 11, Lect 1
calcium ions	Calcium ions inside the cell, one of the second messengers.	Ch 7, Lect 1
CA loop	The small loop left on the new strand when the polymerase slips while copying a repeated sequence.	Ch 6, Lect 4
cAMP (cyclic adenosine monophosphate)	A small second messenger; its concentration goes up when glucose is low, and it lets CAP pull RNA polymerase onto the promoter.	Ch 7, Lect 5
Cancer-associated fibroblast (CAF)	Fibroblasts that the tumour recruits and reprograms, so that they spend their time building the stroma around it.	Ch 9, Lect 2
CAP	The protein that binds cAMP and helps RNA polymerase onto the lac promoter.	Ch 7, Lect 5
Cap (5 prime cap)	The chemical cap on the front end of an mRNA: it protects the message and calls the ribosome over to collect it.	Ch 7,Lect 5
CAP-cAMP (cAMP-CAP)	The CAP and cAMP pairing that sits on the lac promoter and turns transcription up.	Ch 7, Lect 5
Carbohydrate	One of the four great biomolecules; mainly the fuel of the cell, and also structural material such as cellulose.	Ch 1,Lect 1
Carbon	The central element of the organic chemistry of life.	Ch 1
CAR (Chimeric Antigen Receptor)	A synthetic receptor fitted onto a T cell, so that the T cell learns to recognise a tumour protein and then kill that cell.	Ch 9, Lect 2
cardiotoxicity	The toxicity a drug does to the heart.	Ch 8, Lect 2
CAR-NK	A CAR put on a natural killer cell instead of on a T cell.	Lect 3
Carrier	A person carrying one problematic copy of a gene who does not themselves get ill.	Lect 3
Caspase	Enzymes that cut up other proteins; the caspase family is what actually carries out apoptosis.	Ch 8, Lect 2
Caspase 8 and caspase 9	The upstream starters of the two apoptosis pathways, in charge of the extrinsic and the intrinsic route respectively.	Ch 6, Ch 8, Lect 2
Catabolism	The breakdown half of metabolism: it takes big molecules apart and releases energy.	Ch 1
cccDNA	The covalently closed circular DNA that HBV forms inside the nucleus.	Lect 2
CCR2b	The chemokine receptor that lets monocytes find a tumour by following CCL2, which is MCP-1.	Lect 2
CCR4	The chemokine receptor that recognises CCL17, and one of the missing receptors on Dr. Wong's list.	Lect 2
CD28	The co-stimulatory signal in a CAR, telling the T cell that it has been legitimately switched on.	Ch 9, Lect 2
CD3 zeta (CD3z)	The signalling tail of the T cell receptor, used inside a CAR to switch the T cell on.	Lect 2
CD45	A general surface marker for white blood cells, used to exclude immune cells from a sample.	Ch 9, Lect 2
CD47	The "do not eat me" signal on the surface of a tumour, which makes the phagocytes stand down.	Lect 2
CD73	The enzyme that turns AMP into adenosine, and makes the area around a tumour immunosuppressive.	Lect 2
CDH17 (Cadherin-17)	CDH17 is the cell surface protein Dr. Wong's laboratory has studied for twenty years, and it is the tumour-targeting end of his bispecific antibody.	Ch 8, Lect 2
CDK4/6 inhibitor	It blocks the kinase that lets a cell leave G1, so the cell cycle stops; palbociclib is one of them.	Ch 8,Lect 2
CDKN2A	The gene for p16; when its promoter is methylated the cell loses one of its cycle brakes.	Ch 4, Lect 3
CDR1, CDR2 and CDR3	The three hypervariable loops in the variable region of an antibody, which together form the antigen-binding site.	Ch 2, Lect 1
CEA	A fetal antigen and a blood marker that rises in colorectal cancer.	Ch 10, Lect 1, Lect 4

Term	In one sentence	Where
Cell adhesion molecule (CAM)	An adhesion protein on the cell surface that decides who is allowed through the tissue.	Ch 9, Lect 2
Cell cycle	The timetable of G1, S, G2, M plus cytokinesis, the schedule by which a cell copies itself.	Ch 8,Lect 2
Cell cycle arrest	The cell stops at a checkpoint and does not divide; this is exactly what a CDK4/6 inhibitor produces.	Ch 8, Lect 2
Cell-free DNA (cfDNA)	DNA fragments drifting about in the blood; a tumour sheds some of itself into them.	Ch 4, Lect 3
Cell-type specificity	The same gene is switched on only in certain cell types, and switched off in all the others.	Ch 7, Lect 5
Cellular memory	The cell remembers the state of a switch long after the signal that set it has gone.	Ch 4,Lect 3
Cellulose	The structural polysaccharide that makes up plant cell walls.	Ch 1
censored	A patient who drops out during follow-up, or who has not yet had the event.	Ch 10, Lect 1
Central dogma	The rule that genetic information flows one way: DNA, then RNA, then protein.	Ch 7, Lect 1, Lect 5
Centromere	The narrow waist in the middle of a chromosome, where the spindle fibres grab hold.	Ch 3,Lect 3
Centrosome	The organelle that organises the microtubules and builds the spindle.	Ch 8, Lect 2
Cetuximab	An anti-EGFR antibody; it works in K-Ras wild-type rectal cancer and fails in patients whose K-Ras is mutated.	Ch 10, Lect 1
Chaperone	Proteins that help another protein fold correctly and stop it from clumping into a mess.	Ch 2
CHD	One of the four families of chromatin-remodelling complex, and it moves nucleosomes by reading histone marks.	Ch 4, Lect 3
Checkpoint	A checkpoint in the cell cycle that confirms the earlier work is finished before it lets the cell proceed.	Ch 8, Lect 2
Chemokine	A signalling protein that makes white blood cells move towards one particular place, by following a concentration gradient.	Ch 9,Lect 2
chemotaxis	Cells moving along a chemical concentration gradient.	Ch 9, Lect 2
CHO cell	A hamster cell line used to manufacture human therapeutic proteins, because it puts on the right sugars and puts them on in the right way.	Ch 1, Ch 2, Ch 7, Lect 1, Lect 5
Cholesterol	A lipid wedged into membranes that keeps them fluid at low temperature.	Ch 1
Chromatin	DNA plus the proteins that package it; the manual together with its binding.	Ch 3, Lect 3
Chromatin fibre	The 30 nanometre fibre you get when nucleosomes are packaged up again, one step above the beads on a string of DNA.	Ch 4,Lect 3
Chromatin remodelling complex	An ATP-powered machine that pushes nucleosomes along DNA, exposing genes or covering them up.	Ch 4, Lect 3
Chromosome	A well-printed manual of the genome, neatly bound.	Ch 3, Lect 3
CIMP	CpG island methylator phenotype: a tumour-level pattern in which methylation is spread widely across many promoters instead of being confined to a single gene, and which picks out one particular group of colorectal cancers.	Ch 4,Lect 3
Cip/Kip	One of the two big families of CDK inhibitors; its members are p21, p27 and p57.	Ch 8, Lect 2
Circular DNA	DNA that is a ring rather than a line; mitochondrial DNA is a circle of about 17 thousand bases.	Ch 3, Ch 8, Lect 2
circulating surrogate marker	A stand-in measurement that can be taken from the blood.	Ch 10, Ch 11, Ch 12, Lect 4
Circulating tumour cell (CTC)	Cancer cells that have broken off from a tumour and are now travelling around in the blood.	Ch 9, Lect 2
Cis face	The receiving end of the Golgi apparatus, where the vesicles coming in drop anchor and start being worked on.	Ch 8,Lect 2
Cis-regulatory element	A DNA sequence such as a promoter, enhancer or silencer, which only regulates the gene on the very DNA it sits on.	Ch 7, Lect 5
cisternae	The flat sacs stacked up in layers inside the Golgi.	Ch 8, Lect 2
Co-activator (nuclear receptor)	The protein summoned by an activated nuclear receptor to help drive transcription.	Ch 7, Lect 5
Codon	Three letters on the mRNA that specify one amino acid.	Ch 2, Lect 1
Codon optimisation	Rewriting a gene in the codon spelling that the host cell prefers, so that the protein comes out of the host factory easily and in good shape.	Ch 1,Ch 2,Ch 7,Lect 5
Codon usage	Different species prefer different codons for the same amino acid.	Ch 1, Ch 2, Ch 7, Lect 5

Term	In one sentence	Where
Codon-usage bias	Different species prefer different codons, so a gene that moves between species has to be adjusted.	Ch 1, Ch 2, Ch 7, Lect 5
Coenzyme	The small helper molecule an enzyme needs in order to work.	Background supplement (not lecture content)
Cofactor	A partner that an enzyme needs but which is not itself a protein, and it is very often a metal ion sitting in the middle of the reaction.	Background supplement (not lecture content)
Cold tumour	A tumour with few neoantigens, which responds badly to checkpoint inhibitors.	Ch 9, Lect 5
Co-localisation	Two proteins found in the same place in the same cell, proved by overlapping fluorescent staining.	Ch 9, Ch 11, Lect 2
Companion diagnostic	A test you have to do before the drug, because it decides whether this patient can have this drug at all.	Ch 11, Ch 12, Lect 2
Complement	A group of blood proteins that can kill cells on their own, with no antibody needed to point them at a target.	Ch 9,Lect 2
Compound	A chemical substance made by combining two or more elements.	Ch 1
Confidence interval	A range around a measurement, the range the true value is most likely to sit inside.	Ch 4, Lect 3
Confocal microscopy	A microscope that focuses on one thin optical section at a time, used to locate a labelled protein precisely.	Ch 9, Ch 11, Lect 2
Consensus sequence	The DNA "letter" that turns up most often in a real genome, when you line up many sequences of the same family and look at each position in turn.	Background supplement (not lecture content)
Constitutively active	Staying switched on without needing an upstream signal.	Ch 10, Lect 1
Contact inhibition	Healthy cells stop dividing as soon as they bump into a neighbour.	Ch 1, Ch 7, Lect 1
Contractile ring	A ring built out of actin and myosin that pinches a cell which is in the middle of dividing into two, so that the cytoplasm ends up in two separate cells.	Ch 8,Lect 2
Copy number variation (CNV)	A stretch of DNA that has more copies than normal, or fewer.	Ch 3, Lect 3
Corepressor	The molecule that does the work for a repressor; in the trp operon the corepressor is tryptophan.	Ch 7, Lect 5
Co-repressor (nuclear receptor)	The protein summoned by a nuclear receptor to press transcription down, exactly the opposite of a co-activator.	Ch 7, Lect 5
Core promoter	The stretch of DNA directly in front of a gene, where the transcription machinery assembles.	Ch 7, Lect 5
Co-stimulatory domain	The section inside the CAR that sends out the second signal, the one that switches the T cell fully on rather than merely nudging it.	Ch 9,Lect 2
Covalent bond	The kind of bond where two atoms share a pair of electrons and hold each other together.	Ch 1
CpG island	A short stretch of DNA rich in "CG"; methylate it and the gene next to it usually falls silent.	Ch 4, Lect 3, Lect 5
CpG site	A cytosine with a guanine right next to it on the DNA; methylating that cytosine is the switch that silences the gene, and it is the basic unit of epigenetics.	Ch 4,Ch 11,Lect 3,Lect 5
CRISPR/Cas9	A gene editing system that cuts DNA at a chosen place and lets the cell's own repair machinery make the change.	Ch 6, Lect 2, Lect 4
Cristae	The folds that the inner mitochondrial membrane folds inward, where the ATP-making machine sits.	Ch 8, Lect 2
Crossing over	A deliberate swap of DNA between homologous chromosomes during meiosis, which reshuffles genetic variation.	Ch 8, Lect 2, Lect 3
Cryptic exon	A stretch of sequence that is normally not used, but becomes a real exon when splicing switches it on.	Ch 7, Lect 5
CSA and CSB	The two proteins that TC-NER uses to notice that the gene being read has got stuck.	Ch 6, Lect 4
CTC0	The group of patients in whom no circulating tumour cells can be found in the blood at all.	Ch 9, Lect 2
C-terminus	The very last end of a protein chain, where the last amino acid carries a free carboxyl group sticking out.	Ch 2
CTGF	Connective tissue growth factor, one of the target genes switched on by YAP through TEAD when Hippo is off.	Ch 7, Lect 1, Lect 5
CTLA-4	The checkpoint receptor that competes with CD28 for the co-stimulatory signal to T cells.	Ch 9, Lect 2, Lect 3
Cyclin	The protein that rises and falls through the cell cycle and switches on its matched CDK; it is the key, not the machine.	Ch 8, Lect 2

Term	In one sentence	Where
Cyclin-dependent kinase (CDK)	An enzyme that pushes the cell cycle forward one step at a time, and it only works when the matching cyclin switches it on.	Ch 8,Lect 2
cysteine	The amino acid with a thiol on it; two cysteines can be fastened together with a disulfide bond.	Ch 2, Lect 1
Cytochrome c	A protein that normally lives in the mitochondria and, once released into the cytoplasm, starts the intrinsic apoptosis pathway.	Ch 8, Lect 2
Cytokine	Small proteins that cells release to pass a message to the cells nearby.	Ch 7, Ch 9, Lect 5
Cytokinesis	The step after mitosis that divides the cytoplasm into two.	Ch 8, Lect 2
Cytoplasm	The watery jelly inside the cell, and almost every chemical reaction in the cell happens inside it.	Ch 7,Ch 8
Cytosine	One of the four bases in DNA, and the one that gets methylated.	Ch 3, Ch 4, Lect 3
Cytoskeleton	The network of fibres that gives a cell its shape, moves cargo, builds the spindle and pinches the cell in half.	Ch 8, Lect 2
cytotoxic payload	The poisonous part of an ADC that does the killing.	Ch 10, Lect 1, Lect 4
DAG	Diacylglycerol, the second messenger that stays on the cell membrane.	Ch 1, Lect 1
DAPI	A fluorescent dye that glows blue and stains cell nuclei specifically.	Ch 9, Ch 11, Lect 2
Daraxonrasib (Rasonque)	A RAS inhibitor: the median overall survival is 13.2 months, against 6.7 months for the chemotherapy it was compared with.	Ch 10,Lect 1
datopotamab-deruxtecan	An ADC that goes after TROP2.	Ch 10, Lect 4
Deacetylation	Take the acetyl group off, the chromatin tightens up again, and the gene is silenced.	Ch 4, Lect 3, Lect 5
Deamidation	The slow chemical change in which asparagine in a protein becomes aspartate, and it changes how well the protein works.	Ch 2, Lect 1
Death receptor	A receptor on the cell surface that starts apoptosis when an outside signal touches it.	Ch 8, Lect 2
Deletion	A stretch of DNA that has gone missing, taking a bit of the sequence with it.	Ch 3,Ch 10,Lect 3
Delta ligand	The molecule on the surface of a neighbouring cell that Notch listens to.	Ch 7, Lect 5
Denaturation	The process of opening the double helix into two single strands of DNA; also called melting.	Ch 3, Lect 1, Lect 3, Lect 4
Dendritic cell	The specialised scavenger among immune cells, which picks up antigens and lays them out for T cells to look at.	Lect 2, Lect 4
De novo methyltransferases	The enzymes that build the methylation marks from scratch.	Ch 4, Lect 3
Deoxyribose	The sugar in DNA, which has one oxygen atom fewer than the sugar in RNA.	Ch 3
deruxtecan	The family name for the DXd class of topoisomerase inhibitor payloads.	Ch 10, Lect 4
detergents	Molecules that keep hard-to-soluble membrane proteins holding their shape in water.	Lect 2
Developmental specificity	The same gene being switched on and off at different stages of the body's development.	Ch 7, Lect 5
Diagnostic biomarker	Something that tells you whether the patient has the disease right now or not.	Ch 11,Lect 1
Dicer	Dicer is the enzyme that cuts long RNA short; one piece of what it cuts becomes a microRNA.	Ch 7, Lect 1, Lect 5
Differentiation	A cell decides to become one particular kind of specialist cell.	Ch 8, Lect 2
Diffusion	Molecules spreading by themselves from where there are many to where there are few, at no energy cost.	Ch 1, Lect 1
Dimerisation	Two identical or different proteins joining up as a pair, which often makes them far better at binding DNA.	Ch 7,Lect 5
Diploid	Two full sets of chromosomes, one set from the father and one from the mother; human body cells have 23 pairs.	Ch 3, Ch 8, Lect 3
Direct reversal	A repair pathway that fixes the damage exactly as it stands, without cutting out a single nucleotide.	Ch 6, Lect 4
Disaccharide	Two sugar units joined together, like sucrose.	Ch 1, Lect 1
Disease relevance	How much this molecule actually matters to this disease.	Ch 12, Lect 4
dispersive	The third of the three models of DNA replication, in which the old strands are broken up and mixed into the new ones.	Ch 5, Lect 4
Disulfide bond	A covalent link between two cysteines that helps hold a protein's shape in place.	Ch 2, Lect 1
Diversity	The genetic differences between individuals of the same species.	Ch 6, Lect 2
dMMR	Deficient mismatch repair, meaning the mismatch repair machinery is broken.	Ch 9, Lect 2, Lect 4

Term	In one sentence	Where
DNA adduct	A chemical molecule stuck onto a DNA base, which may then be mismatched when the DNA is copied.	Ch 6,Lect 3
DNA-bending protein	A protein that bends DNA into a loop, so that a distant enhancer can reach its own promoter.	Ch 7, Lect 5
DNA-binding domain (DBD)	The highly conserved part of a nuclear receptor that recognises its own word in the DNA.	Ch 2, Ch 7, Lect 5
DNA Damage Response (DDR)	The cell's alarm system for DNA damage. It starts repair, and if the damage is bad enough, it starts cell death instead.	Ch 6, Lect 4
DNA (deoxyribonucleic acid)	The master manual, written with four letters, kept in the nucleus.	Ch 1, Ch 3
DNA fingerprint	A method for identifying a person by the differences in length of repeated sequences.	Ch 3, Lect 4
DNA glycosylase	The enzyme that cuts out a damaged base entirely, and the first step of BER.	Ch 6, Lect 4
DNA ligase	The enzyme that seals the last gap in a newly synthesised DNA strand.	Ch 5, Lect 4
DNA polymerase	The enzyme that builds a new DNA chain, and it can only add nucleotides in the direction from 5 prime to 3 prime.	Ch 5,Lect 4
DNA repair	The set of pathways that repairs the damage before it turns into a permanent mutation.	Ch 6, Lect 4
DNase	The enzyme that cuts DNA into pieces. It destroyed the transforming activity, and that is how DNA was identified.	Ch 3,Lect 1
DNMT1	A maintenance methyltransferase: once DNA has been copied, it copies the methylation pattern onto the new strand.	Ch 4, Lect 3
DNMT3A and DNMT3B	The de novo methyltransferases; they put the first methylation mark on a stretch of DNA.	Ch 4, Lect 3
dNTPs	The free nucleotide raw material used when DNA is synthesised.	Ch 5, Lect 4
donor plasmid	In molecular cloning, the plasmid that carries the DNA you want to put in.	Ch 3, Lect 4
donor template	The template DNA that supplies the correct sequence when you are doing a knock-in.	Ch 6, Lect 4
Dose escalation	The first stage of a trial, in which the dose is stepped up to find the safe ceiling.	Ch 12, Lect 2, Lect 4
Double helix	Two DNA strands twisted around each other, with the bases pairing up in the middle like the rungs of a ladder.	Ch 3
Double-strand break (DSB)	The most dangerous kind of DNA damage: both strands cut at the same position.	Ch 6, Lect 4
Down syndrome	Trisomy 21. Chromosome 21 comes in three copies instead of two.	Ch 3, Lect 3
Driver	A mutation that gives a cell a growth advantage. One of the typos in the important words.	Ch 3, Ch 10, Lect 3
Drosha	The enzyme that cuts the primary microRNA transcript into a short hairpin inside the nucleus.	Ch 7,Lect 1,Lect 5
Druggability	Whether this molecule can be turned into a drug that hits it without hitting anything else.	Ch 12, Lect 4
Drug resistance	A tumour stops responding to a drug that used to work. It can be caused by a new mutation, or by an isoform that has lost the binding site.	Ch 5, Ch 7, Lect 4, Lect 5
Drug target	The molecule a drug is designed to act on. Membrane proteins and kinases are the good examples.	Ch 10, Ch 12, Lect 1, Lect 2
Dual-payload ADC	An ADC with two different drugs hanging off the same antibody.	Ch 10, Lect 1, Lect 4
Duplication	A DNA segment that has been copied one extra time.	Ch 3, Lect 3
DXd	The drug payload that trastuzumab deruxtecan carries.	Ch 10, Lect 1, Lect 4
E2F	A transcription factor that is normally held down by pRb, and released when the cell is ready to divide.	Ch 8,Lect 2
ECM (extracellular matrix)	The scaffold of proteins and sugars that cells sit in and have to dig through in order to move.	Ch 9, Lect 2
EcoRI	A restriction endonuclease that cuts DNA at a specific position.	Ch 3, Lect 4
Ectoderm	Ectoderm. It becomes skin and the nervous system.	Ch 8, Lect 2
Edman degradation	The standard method for sequencing a protein, reading it off one amino acid at a time starting from the N terminus.	Ch 2
Effectors	The last link on the DDR signalling chain, the one that actually carries out the work.	Ch 6, Lect 4
efficacy	The measure of whether a drug really works, in the patients who should get it.	Ch 12, Lect 2, Lect 4
EGFR	The growth factor receptor at the top of the RAS chain. Blocking it is a standard anticancer strategy.	Ch 10, Lect 1, Lect 4, Lect 5

Term	In one sentence	Where
Electron	A negatively charged particle that moves around the nucleus and is responsible for every bond.	Ch 1
Electron shell	A fixed energy level outside the atomic nucleus. The outermost one decides what an atom can bond to.	Ch 1
ELISA	The standard test for measuring how much of a particular protein is in a fluid sample.	Ch 11,Lect 1,Lect 2
Endoderm	The endoderm, the innermost of the three germ layers. It becomes the lining of the gut and the lungs.	Ch 8, Lect 2
endogenous	A source of DNA damage that comes from inside the cell.	Ch 6, Lect 4
Endometrium	The endometrium, the lining of the uterus. Tamoxifen behaves differently there than it does in breast tissue.	Ch 7, Ch 10, Lect 5
Endoplasmic reticulum (ER)	The membrane network beside the nucleus, where new proteins are made, folded, checked and glycosylated.	Ch 2, Ch 8, Lect 2
Endosymbiotic origin	The idea that mitochondria began as bacteria that an ancestral cell swallowed and kept.	Ch 8, Lect 2
Endotoxin	The poisonous part of the outer membrane of a Gram-negative bacterium. It is the same molecule as LPS.	Ch 1,Lect 1
End-replication problem	DNA polymerase cannot copy the very end of a linear chromosome, so the telomere gets shorter every time the cell divides.	Ch 5, Lect 3, Lect 4
Energy	The capacity to do work. In a cell it comes from the net release of a metabolic reaction.	Ch 1
Enhanceosome	A cluster of proteins gathered on an enhancer and its promoter, which is how a distant enhancer gets wired to its own gene.	Ch 7, Lect 5
Enhancer	A stretch of DNA that increases transcription, no matter where it happens to sit relative to the gene.	Ch 7,Lect 5
Enzyme	A protein that is not used up itself but makes a chemical reaction go faster.	Ch 2
EpCAM	A surface protein used to pick tumour cells out of a blood sample.	Ch 9, Lect 2
episome	A DNA molecule that is not part of the host chromosome and can replicate on its own.	Lect 2
Epithelial-to-mesenchymal transition (EMT)	The change in which a cell loses its grip on its neighbours and becomes free to move and invade.	Ch 9, Lect 2
Epitope	The small patch on an antigen that the antibody actually touches.	Ch 2,Ch 9,Lect 2
EPO	Erythropoietin, the red blood cell growth factor switched on by HIF-1alpha.	Ch 7, Lect 5
ERBB2	The gene name for HER2.	Ch 10, Lect 1, Lect 5
E-selectin	One of the selectins, the one that lets white blood cells roll along the vessel wall.	Ch 9, Lect 2
Estrogen receptor (ER)	The nuclear receptor for estrogen. Tamoxifen acts on it.	Ch 7,Lect 5
Estrogen response element (ERE)	The stretch of DNA that the estrogen receptor dimer binds to.	Ch 7, Lect 5
Euchromatin	Loosely packed chromatin that is being transcribed.	Ch 3, Lect 3
Eukaryote	A cell with a nucleus. Animals, plants, fungi and yeast are all eukaryotes.	Ch 8, Lect 2, Lect 5
EXO	The proofreading segment of DNA polymerase that cuts out mismatched nucleotides.	Ch 5, Lect 4
exogenous	A source of DNA damage that comes from outside the cell.	Ch 6, Lect 4
Exon	The part of a gene that survives splicing and ends up in the mature mRNA.	Ch 7, Lect 5
Exon inclusion and exon skipping	The two opposite choices the splicing machinery makes when it builds a mature message.	Ch 7, Lect 5
Exonuclease	The proofreading activity of DNA polymerase: the part of the enzyme that cuts out a base it inserted by mistake.	Ch 5
Expansion cohorts	The group of patients in Phase 1B used to look at side effects and early signs of benefit.	Ch 12, Lect 2, Lect 4
Exportin-5	The protein that carries the short hairpin RNA out of the nucleus and into the cytoplasm.	Ch 7,Lect 1,Lect 5
Extension	The step where the polymerase adds new nucleotides along the template.	Ch 3, Lect 1, Lect 4
Extrinsic apoptosis pathway	The apoptosis pathway that starts outside the cell: a death receptor triggered by its ligand.	Ch 8, Lect 2
Ex vivo	Outside the body. Cells are taken out, worked on, and put back.	Ch 9,Lect 2
Fab fragment	The arm of an antibody, the part that grabs hold of the antigen.	Ch 2, Lect 1
FACS	The machine that sorts cells into different tubes according to their fluorescence.	Ch 9, Ch 11, Lect 2

Term	In one sentence	Where
FAP	Fibroblast activation protein: the marker found on the surface of activated fibroblasts.	Ch 9, Lect 2
Fatty acid	A string of carbons with a small acidic end. Whether there are double bonds in the chain decides if it is saturated or unsaturated.	Ch 1,Lect 1
Fatty-acid chain	The long carbon tail of a fatty acid, and whether it is saturated or not.	Ch 1
Fc fragment	The stem of the antibody, the part that recruits the rest of the immune system.	Ch 2, Lect 1
FDR	The false discovery rate, the proportion of false alarms among the results declared significant.	Ch 10, Ch 11, Lect 1
fetal bovine serum (FBS)	The cow serum used to culture cells, which supplies the growth factors.	Ch 8, Ch 11, Lect 2, Lect 4
first-to-invent	The filing principle used by the older American patent system.	Ch 12, Lect 4
FISH	The test that uses DNA probes rather than antibodies to count gene copies inside a cell, for example HER2.	Ch 11, Lect 2
Flow cytometry	A machine that counts cells with a laser and measures their size, their internal complexity and their fluorescence.	Ch 9,Ch 11,Lect 2
Fold (protein folding)	The process of a chain of amino acids twisting into a specific three-dimensional shape so that it can do its job.	Ch 2
Forward scatter (FSC)	In flow cytometry, the signal that reflects cell size.	Ch 9, Ch 11, Lect 2
FoSTeS	Fork stalling and template switching: a replication route that manufactures CNV.	Lect 3
Frameshift	A mutation in which the number of bases added or removed is not a multiple of three, so the reading frame shifts and the protein is usually truncated.	Ch 3,Ch 9,Ch 10,Lect 1,Lect 4,Lect 5
Freedom to Operate (FTO)	The patent risk check you do before you commercialise something.	Ch 12, Lect 2, Lect 4
Functional group	A small group of atoms on a molecule that gives that molecule its chemical character and reactivity.	Ch 1
Fv fragment	The smallest part of an antibody, the variable region alone. About 25 kDa.	Ch 2, Lect 1
G0	The side branch off the cell cycle where a cell rests, alive but not dividing.	Ch 8, Lect 2
G1	The growth phase of the cell cycle, before DNA replication.	Ch 8, Lect 2
G2	The checkpoint stage after DNA replication is finished and before the cell divides.	Ch 8, Lect 2
GAP-mediated GTP hydrolysis	The GTP hydrolysis that GAP promotes, which is the small G protein's own off switch.	Ch 10, Lect 1
Gating	Drawing a boundary on a flow cytometry plot so that one population can be picked out of many.	Ch 9, Ch 11, Lect 2
G-banding	The banded pattern of light and dark stripes a chromosome forms after staining, and it can expose large structural changes.	Ch 3, Lect 3
GC-MS	Gas chromatography mass spectrometry: it separates the chemicals in a sample first, and then identifies each of them as they come off.	Ch 10,Ch 11,Lect 1
GC-rich	A stretch of DNA with a high proportion of G (guanine) and C (cytosine) in its base composition.	Ch 3, Ch 11, Lect 3
Gene	A recipe: a stretch of DNA that specifies one product.	Ch 3
Gene expression	The whole process of switching a gene on, copying out the message, and building the protein.	Ch 7, Lect 5
Gene expression programme	The set of genes a particular cell type switches on and off, and what makes a T cell different from a dendritic cell.	Ch 7, Lect 5
Gene knock-in	Putting a working copy of a gene into the genome, which needs a precise repair pathway to land in the right place.	Ch 6, Lect 2, Lect 4
Gene knock-out	Making a gene stop working altogether, usually by letting CRISPR cut it and letting the cell's own repair machinery break the reading frame.	Ch 6,Lect 2,Lect 4
General transcription factor	A protein needed to start transcription on any gene, rather than only on one particular gene.	Ch 7, Lect 5
Genome	The complete set of DNA that one organism or one cell carries.	Ch 3
genomic instability	The state of a genome that has become unstable overall.	Ch 3, Ch 4, Lect 3
Genotype	The genetic make-up of a cell or a person: the version of each gene they carry.	Ch 7, Lect 3
Germ cell	A cell that will turn into a sperm or an egg, and that is how mutations get handed on.	Ch 4,Ch 8,Lect 3
GG-NER	The repair route that searches for damage anywhere in the whole genome.	Ch 6, Lect 4
Glucose	The monosaccharide that cells burn as fuel.	Ch 1, Lect 1
GLUT1	The glucose transporter that HIF-1alpha switches on, letting a cell keep working when there is no oxygen.	Ch 7, Lect 5

Term	In one sentence	Where
Glycine	The amino acid with the smallest side chain, and the most flexible one.	Ch 2, Lect 1
Glycogen	The form in which animals and fungi store glucose, a branched chain of sugar units.	Ch 1, Lect 1
Glycolysis	The first step of making energy: glucose is broken down into pyruvate in a set of ten steps, and no oxygen is involved at any point in it.	Ch 10,Lect 1
Glycosidic bond	The chemical link that joins one sugar unit to the next in a sugar chain.	Ch 1
Glycosylation	Attaching sugar groups to a protein, which usually happens in the endoplasmic reticulum.	Ch 7, Ch 8
Golgi apparatus	The packing room: it sorts finished proteins and sends each one to its destination.	Ch 8, Lect 2
GPCR (G-protein-coupled receptor)	A large family of membrane receptors, and the best family of drug targets there is.	Lect 2, Lect 5
granularity	How cluttered the inside of a cell is, and on a flow cytometer it is read off as side scatter.	Ch 9, Ch 11, Lect 2
granulocytes	Granulocytes: the group of cells that sit high on side scatter in a flow plot.	Lect 2
Granzyme	The enzyme that kills a target cell once it has got inside it.	Ch 9, Lect 2
Griffith	The scientist whose 1928 bacterial experiment showed that material taken from dead pneumococcus could turn harmless live bacteria into virulent ones, which was the first hint that something transferable was at work.	Ch 3,Lect 1
gRNA	The guide RNA that tells the CRISPR enzyme where to cut.	Ch 6, Lect 2, Lect 4
GTP-bound	The active state of a small G protein, when it is holding GTP.	Ch 10, Lect 1
guanine	One of the four main bases of DNA and RNA, written G.	Ch 3, Lect 1
Guide (lncRNA)	The way a long non-coding RNA carries chromatin-remodelling proteins to the exact words of DNA they are supposed to act on.	Ch 7, Lect 5
H1	The linker histone, which sits on the DNA where it enters and leaves the nucleosome core like a bookmark.	Ch 4, Lect 3
H2A K119ub	The mark left when lysine number 119 on histone H2A is ubiquitinated.	Ch 4, Lect 3
H2B K123ub	The ubiquitination mark on lysine 123 of histone H2B.	Ch 4, Lect 3
H3 K4me	The label that says lysine number 4 on histone H3 has been methylated.	Ch 4, Lect 3
H3K4me3	The mark that sits on active promoters.	Ch 4, Lect 3
H3 K79me	The methylation mark on lysine 79 of histone H3.	Ch 4, Lect 3
H3K9me3	A label that tells the cell a stretch of DNA ought to stay switched off.	Ch 4,Lect 3
H4 K12ac	The acetylation mark on lysine number 12 of histone H4.	Ch 4, Lect 3
H4 K8ac	The acetylation mark on lysine 8 of histone H4.	Ch 4, Lect 3
Haploid	One set of chromosomes, which is what a gamete is. A human gamete has 23.	Ch 8, Lect 3
Hayflick limit	How many times a human cell can divide before it is forced to stop.	Ch 5,Lect 3,Lect 4
HBV	Hepatitis B virus. Its integration into liver DNA is a prognostic marker.	Ch 6, Lect 2
HDAC inhibitor	A drug that blocks histone deacetylase, so acetylation piles up and the chromatin opens.	Ch 4, Lect 3, Lect 5
Heavy chain	The two longer chains of an antibody, which form the stem and part of the arms.	Ch 2, Ch 6, Lect 1
Helicase	The enzyme that unwinds the two DNA strands so that they can be copied.	Ch 5, Lect 4
Helper T cell	A T cell that releases cytokines, and in doing so tells the rest of the immune system what it ought to be doing next.	Ch 6,Ch 9,Lect 2
Hepatocellular carcinoma (HCC)	The primary cancer of the liver.	Ch 5, Ch 11, Lect 1, Lect 2
Hepatocyte	The liver cell, and the main working cell of the liver.	Ch 5, Lect 1
HER1, HER3 and HER4	The other members of the HER receptor family, which pair with HER2.	Ch 7, Ch 10, Lect 5
HER2	A receptor whose gene gets amplified in certain breast and gastric cancers, and the receptor that trastuzumab is designed to block.	Ch 7,Ch 10,Lect 2,Lect 4,Lect 5
Heterochromatin	Tightly packed, silent chromatin. It physically stops transcription factors from reaching the promoter.	Ch 3, Lect 3
Heterodimer	Two different proteins joined together, such as HER2 with HER3.	Ch 7, Lect 5
Heteroplasmy	The state in which normal and mutant mitochondrial DNA are mixed together inside one cell.	Ch 8, Lect 2
Heterozygous	Two different versions of the same gene, one from each parent.	Ch 3, Lect 3
HIF-1alpha	An oxygen sensor protein. When oxygen is normal it is made constantly and broken down just as constantly.	Ch 7,Lect 5

Term	In one sentence	Where
HIF-1beta	The half that binds HIF-1alpha in the nucleus and makes up the working hypoxia factor.	Ch 7, Lect 5
HIF-CAR	A CAR whose expression is controlled by an HRE, so the T cell only arms itself inside a hypoxic tumour.	Ch 7, Lect 5
High specific heat	The energy a substance needs to go up by one degree. Water has a high specific heat, so body temperature stays steady.	Ch 1
Hinge region	The flexible connecting piece between the DNA-binding domain and the ligand-binding domain of a nuclear receptor, the part that lets the two talk to each other.	Ch 9,Lect 5
Hippo pathway	The pathway that uses contact between cells to switch growth off. MST1 activates LATS1, and LATS1 phosphorylates YAP.	Ch 1, Ch 7, Lect 1, Lect 5
Histone	The protein that DNA is wound around to form a nucleosome.	Ch 4, Lect 3
Histone code	The system of histone marks that tells a cell whether a stretch of DNA is open or shut.	Ch 4, Lect 3
Histone modification	A chemical label on a histone, one that opens or closes the chromatin sitting around it, and that decides whether the gene underneath can be read.	Ch 4,Lect 3,Lect 5
Histone variants (H2A, H2B, H3 and H4)	The four proteins that make up the core of the nucleosome.	Ch 4, Lect 3
HNH	One of the two nuclease domains of Cas9, and it cuts the target strand.	Ch 6, Lect 2, Lect 4
Homodimer	Two identical proteins joined together. A dimer binds DNA far more strongly than a single protein.	Ch 7, Lect 5
Homoeologous recombination	Recombination between similar but not identical chromosomes, the mechanism that makes new combinations in polyploid crops.	Background supplement (not lecture content)
Homologous chromosome	One of the two matching chromosomes in a pair, one from each parent, and the ones that separate at the first meiotic division.	Ch 3, Ch 8, Lect 3
Homologous recombination (HR)	High-fidelity repair of a broken chromosome that uses the matching sister chromatid as a template. It only works in S and G2.	Ch 6, Lect 4
homologous template	The stretch of DNA that HR copies from.	Ch 6, Lect 4
Homology-directed repair	The precise repair route that uses a matching template. CRISPR knock-in needs exactly this.	Ch 6, Lect 2, Lect 4
Homoplasmy	Every copy of the mitochondrial DNA inside one cell is identical, whether they are all normal or all mutated.	Ch 8, Lect 2
Homozygous	Two identical versions of the same gene, one from each parent, with no hidden difference between them.	Ch 3, Lect 3
Hormone	A chemical messenger released by a gland into the blood, acting on cells far away.	Ch 7, Lect 5
Hot tumour	A tumour carrying a large load of neoantigens, and therefore a good responder to immunotherapy.	Ch 9, Lect 2, Lect 4, Lect 5
HP1	The protein that recognises H3K9me3 and spreads the methylation outwards in both directions.	Ch 4, Lect 3
HSP90	A chaperone in the cytoplasm that keeps the estrogen receptor folded until the hormone arrives.	Ch 7, Lect 5
HSPGs	Heparan sulfate proteoglycans, the large negatively charged molecules of the tumour matrix.	Lect 2
humanised mouse	An animal model with a human gene put into a mouse.	Ch 12, Lect 2
Human Protein Atlas	The public database he uses to nominate drug targets and to find out where a protein is actually made.	Ch 10, Lect 1
hydrodynamic focusing	The fluid trick that makes the cells pass through the laser beam one at a time in single file.	Ch 11, Lect 2
Hydrogen bond	The weak attraction between a hydrogen attached to an electronegative atom and another electronegative atom nearby.	Ch 1
hypermethylated	The state of being methylated too heavily.	Ch 4, Lect 3
hypomethylated	The state of being insufficiently methylated.	Ch 4, Lect 3
Hypoxia	Too little oxygen. It is the signal that switches HIF-1alpha on.	Ch 7, Ch 8, Ch 9, Lect 5
Hypoxia-responsive element (HRE)	The piece of DNA that switches a gene on when HIF-1 is active.	Ch 7, Lect 5
IkappaB (I-kappa-B)	The protein that holds NF-kappaB in the cytoplasm until some signal destroys it.	Ch 7, Lect 1, Lect 5
IkappaB kinase (IKK)	The enzyme that puts a phosphate on IkappaB, which is how IkappaB is marked for destruction.	Ch 1, Ch 7, Lect 5

Term	In one sentence	Where
ICAM-1	An adhesion molecule that lets white blood cells grip the wall of the blood vessel.	Ch 9, Lect 2
ideogram	A standardised diagram of one chromosome, with every band labelled.	Ch 3, Lect 3
IDO	An enzyme that starves T cells of an amino acid they need and switches them off.	Ch 9, Lect 2
IFN-gamma	The third cytokine in the NF-kappaB response, after TNF-alpha and IL-1beta.	Ch 4, Ch 7, Lect 5
IgG4	The antibody subclass that does not kill through ADCC and stays in the blood longer than IgG1.	Ch 2, Ch 9, Lect 1, Lect 2
IL-1beta	The second cytokine in the NF-kappaB response, coming after TNF-alpha.	Ch 7, Lect 5
IL-10	The cytokine that suppresses the immune response.	Ch 9, Lect 2
IL-15	A cytokine that comes after IFN-gamma in the NF-kappaB response and keeps NK cells alive.	Ch 9, Lect 2, Lect 5
IL-2	The cytokine that comes after IFN-gamma in the NF-kappaB response and drives T cell proliferation.	Ch 9, Lect 2, Lect 5
Immune checkpoint	The molecular brakes that stop immune cells from attacking. PD-1 on the T cell is the most important one.	Ch 9, Lect 2
Immune evasion	The state in which a tumour escapes an attack because the T cells have been switched off.	Ch 9, Lect 2
Immunofluorescence (IF)	Staining tissue with fluorescently labelled antibodies so you can see exactly where a protein is.	Ch 9, Ch 11, Lect 2
Immunohistochemistry (IHC)	Staining tissue with an antibody plus an enzyme that leaves a coloured deposit, so that you can see where the protein is.	Ch 9, Ch 11, Lect 2
Inclusion body	A hard, insoluble clump of misfolded protein. It appears when recombinant protein is made too fast or too early.	Ch 2, Ch 7, Lect 5
Indel	An insertion or a deletion in the DNA letters. If it is not a multiple of three, it causes a frameshift.	Ch 6, Lect 1, Lect 3
IND-enabling studies	The whole set of non-clinical studies a company does before it applies for an IND.	Ch 12, Lect 2
IND (Investigational New Drug)	The application a company has to file before it is allowed to test a drug in humans.	Ch 12, Lect 2, Lect 4
Inducible system	A system that stays switched off until the signal arrives.	Ch 7, Lect 5
Induction	Opening a gene up by adding an inducer (lactose in the lac system).	Ch 7, Lect 5
Inhibitor of apoptosis (IAP)	A protein that blocks the caspase executioners.	Ch 8, Lect 2
INK4	A family of inhibitory proteins that block CDK4/6 kinase and so stop the cell cycle.	Ch 8, Lect 2
INO80	One of the four families of chromatin remodelling complex.	Ch 4, Lect 3
insert	The piece of DNA you want, which is put into a plasmid.	Ch 3, Lect 4
Insulator	A DNA sequence that stops an enhancer from working on the wrong gene.	Ch 7, Lect 5
integration	The permanent insertion of a piece of HBV DNA into the host genome.	Ch 6, Lect 2
Integrin	A protein on the cell surface that grabs other cells and the matrix; it is how white blood cells hold on while they move.	Lect 2
Intermembrane space	The gap between the two membranes of the mitochondrion, where protons pile up to drive ATP synthase.	Ch 8, Lect 2
Intrinsic apoptosis pathway	The apoptosis pathway that starts inside the cell, usually at the mitochondrion.	Ch 8, Lect 2
Intron	The part of a gene that is cut out during splicing and never appears in the mature mRNA.	Ch 7, Lect 5
Inversion	A piece of chromosome is cut out and put back the other way round. No DNA is gained or lost, but the expression can change.	Ch 3, Lect 3
ion channels	Channels set into the cell membrane that let ions move in and out down their concentration gradient.	Lect 2
Ionic bond	The attraction between oppositely charged ions, like table salt.	Ch 1
IP	Intellectual property, the question of who owns the right to use a technology.	Ch 12, Lect 4
Ipilimumab	An anti-CTLA-4 checkpoint inhibitor; it works by releasing the T cell's brake.	Lect 3
iPSC (induced pluripotent stem cell)	An adult cell reprogrammed back to an embryo-like state, so that it can become almost any cell type.	Ch 8, Lect 2
IPTG	A synthetic inducer used to switch recombinant genes on in E. coli, and it is not metabolised.	Ch 7, Lect 5
iso-aspartic acid	The stereoisomer produced when asparagine is deamidated.	Ch 1, Ch 2, Lect 1

Term	In one sentence	Where
Isoelectric focusing	A method that separates proteins according to the pH at which they carry no charge.	Ch 11, Lect 1
Isoform	A slightly different version of a protein, made by the same gene choosing different combinations of exons.	Ch 7, Lect 5
Isotype	A version of an antibody class. IgG1 kills through ADCC; IgG4 does not.	Ch 2, Ch 9, Lect 2
ISWI	One of the four families of chromatin remodelling complexes.	Ch 4, Lect 3
Kaplan-Meier curve	A step-shaped curve showing how the probability of a patient surviving changes over time.	Ch 10, Lect 1, Lect 3
Karyotype	A picture of all the chromosomes of a cell, used to look for large structural abnormalities.	Ch 3, Lect 3
Karyotyping	Using Giemsa staining to produce G-banding, so that large structural changes can be read off directly.	Ch 3, Lect 3
KIBRA, MER and WILLIN	Proteins at the cell junction that recruit and switch on the Hippo kinase.	Lect 5
Kinase	An enzyme that puts a phosphate group onto another protein, usually switching it on.	Ch 1, Ch 2, Lect 1
K-Ras	The gene whose stuck-on mutant drives many cancers, including colorectal and pancreatic cancer.	Ch 10, Lect 1
Ku	The protein complex that is the first to grip the two broken ends of DNA in NHEJ.	Ch 6, Lect 4
KU and DNA-PKcs	The two members of NHEJ that protect the broken ends.	Ch 6, Lect 4
Lac operon	The classic bacterial switch, where several genes for using lactose are opened by one signal.	Ch 7, Lect 5
Lac repressor	A protein that sits on the operator and blocks the lac genes, until allolactose removes it.	Ch 7, Lect 5
Lactate	The molecule a cancer cell makes when it ferments sugar instead of burning it.	Ch 10, Lect 1
Lactose	The milk sugar that E. coli can use as an energy source, and the reason the lac operon exists.	Ch 7, Lect 5
LAG3	An inhibitory checkpoint receptor, and like PD-1 one of the brakes on a T cell.	Ch 9, Lect 2
Lagging strand	The strand that cannot be copied continuously and has to be built out of short Okazaki fragments.	Ch 5, Lect 4
Laser capture microdissection (LCM)	Cutting a clean region out of a tissue slide under the microscope, so that other cells cannot contaminate it.	Ch 9, Ch 11, Lect 2
LATS1	The kinase in the Hippo pathway that phosphorylates YAP and holds it in the cytoplasm.	Ch 7, Lect 1, Lect 5
LC-MS/MS	A method that cuts proteins into fragments, weighs them and reads out their sequence.	Ch 2, Ch 11, Lect 1
Leading strand	The strand that can be copied smoothly and continuously.	Ch 5, Lect 4
Ligand	The molecule that binds a receptor. It is the message, not the receiver.	Ch 7, Lect 5
Ligand-binding domain (LBD)	The part of a nuclear receptor that holds the hormone or the drug, and changes shape while doing it.	Ch 2, Ch 7, Lect 5
Light chain	The two shorter chains of an antibody, sitting at the tips of the arms.	Ch 2, Ch 6, Lect 1
linker	A short connecting peptide that joins two pieces of protein together.	Ch 9, Ch 10, Lect 2
Lipid	A family of molecules that fear water: they build walls, store energy, and also carry signals.	Ch 1, Lect 1
Lipid bilayer	The two-molecule-thick sheet that makes up the cell membrane.	Ch 1
Lipid nanoparticle (LNP)	A lipid bubble that carries mRNA into cells. Used for in-vivo CAR-T and for mRNA vaccines.	Ch 9, Lect 2, Lect 5
Lipoprotein	A particle wrapped in protein that carries lipids around the blood.	Ch 1
Liquid biopsy	Looking for tumour material in a blood sample instead of cutting a piece out of the tumour.	Ch 9, Lect 2
LncRNA (long non-coding RNA)	RNA longer than 200 nucleotides that does not make protein.	Ch 7, Lect 5
Log-rank test	The statistical test that compares two survival curves and produces a p value.	Ch 9, Ch 12, Lect 1, Lect 3
LPS (lipopolysaccharide)	A molecule on the bacterial outer membrane that, once recognised by TLR4, starts the NF-kappaB inflammatory cascade.	Ch 1, Lect 1, Lect 5
Luciferase	The reporter gene most often used in promoter experiments, because it gives off light.	Ch 7, Lect 5
Lysine	The amino acid that carries most of the histone marks. Its side chain is the handle the mark hangs on.	Ch 4, Lect 3
Lysosome	The acidic recycling and disposal plant inside the cell.	Ch 8, Lect 2

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Term	In one sentence	Where
Macromolecule	A very large molecule, such as a protein or a nucleic acid, built out of many small units.	Ch 1
Maintenance methyltransferases	The enzymes that copy the existing methylation marks onto the new strand.	Ch 4, Lect 3
manufacturing cycle	The five steps that carry CAR-T from the patient and back into them.	Ch 9, Lect 2
Mass spectrometry	The technology that weighs molecules and then identifies them from the pattern of fragments they make.	Ch 2, Ch 11, Lect 1
Maternal inheritance	The rule by which a mitochondrial disease passes from an affected mother to all of her children and from an affected father to none.	Ch 8, Lect 2
Matrix (mitochondrial matrix)	The liquid interior of the mitochondrion, where the TCA cycle runs and where mitochondrial DNA lies.	Ch 8
MCP-1	The chemokine that calls white blood cells to a particular place.	Ch 9, Lect 2
MDSCs	Cells the tumour recruits to turn the immune response down.	Ch 4, Ch 9, Lect 2
median overall survival	The median overall survival of a group of patients.	Ch 10, Ch 12, Lect 1
medicinal chemistry	The stage where a target is turned into a compound that is actually fit for the job.	Ch 12, Lect 2
MEI	Mobile element insertion, the mechanism by which a transposon creates a CNV.	Lect 3
Meiosis	The special division in a germ cell that makes four genetically different gametes.	Ch 8, Lect 2, Lect 3
Meiosis I and meiosis II	The first division separates homologous chromosomes; the second separates sister chromatids.	Ch 3, Ch 8, Lect 3
Melting temperature (Tm)	The temperature at which half of a DNA double helix has come apart.	Ch 3, Ch 11, Lect 1, Lect 3
Membrane protein	A protein sitting in the plasma membrane, often a receptor. Because a drug can reach it from outside the cell, it is an excellent drug target.	Ch 2, Ch 12, Lect 2
Mesoderm	The middle germ layer. It becomes muscle, bone and blood.	Ch 8, Lect 2
Metabolism	All the chemistry of the cell: breaking things down to get energy, and building things up.	Ch 1, Lect 1
Metabolite	A small molecule that the cell makes or uses up as it works.	Ch 10, Ch 11, Lect 1
Metabolite fingerprint	The pattern of small molecules that makes one sample different from another.	Ch 10, Ch 11, Lect 1
Metabolomics	The study of all the small molecules in a sample. A great many biomarkers are found this way.	Ch 10, Ch 11, Lect 1
Methylation	Adding a methyl group onto DNA, usually in order to silence a gene.	Ch 4, Ch 6, Lect 3, Lect 5
Methyl group	The tiny -CH3 label that DNA methylation puts onto cytosine.	Ch 4, Lect 3
MGMT	Its promoter methylation predicts that a brain tumour will respond well to alkylating chemotherapy.	Ch 4, Lect 3
MHC class I	The display cabinet that carries fragments of proteins from inside the cell out to the surface, where killer T cells can look at them.	Ch 6, Ch 9, Lect 4
MHC class II	The display cabinet that shows protein fragments taken in from outside the cell to helper T cells.	Ch 6, Lect 4
MHC (major histocompatibility complex)	The surface proteins that display a small fragment of protein so that T cells can inspect a cell.	Ch 6, Ch 9, Lect 5
Microarray	A chip carrying thousands of gene probes, used to measure which genes are switched on.	Ch 11, Lect 1
MicroRNA (miRNA)	A short RNA about 22 nucleotides long that blocks the translation of its target mRNA.	Ch 7, Lect 1, Lect 5
Microsatellite	Short repeating stretches of DNA. When mismatch repair fails their length changes, and that is how MSI is measured.	Ch 6, Lect 4
Microsatellite instability (MSI)	The change that short repeated DNA undergoes once mismatch repair has failed. An MSI-high tumour is a hot tumour.	Ch 6, Ch 9, Ch 10, Lect 4
Microtubule	A hollow tube made of protein. It forms the mitotic spindle, and it is also a track for cargo.	Ch 8, Lect 2
Misfolding	A protein that has not folded into the right three-dimensional shape. It is one of the risks of a therapeutic protein.	Ch 2, Lect 1, Lect 5
misincorporation	The polymerase putting the wrong base onto the new strand.	Ch 5, Lect 4
Mismatch repair (MMR)	The repair pathway that puts right the mistakes left behind after replication.	Ch 6, Ch 9, Lect 4
Missense mutation	A single-letter change that swaps one amino acid for another.	Ch 3, Lect 3
Mitochondrial DNA (mtDNA)	The small circular DNA inside the mitochondrion, about 17 kilobases with 37 genes. You inherit it from your mother.	Ch 8, Lect 2

Term	In one sentence	Where
Mitochondrial Eve	The one woman from whom all existing mitochondrial DNA is said to descend.	Ch 8, Lect 2
Mitochondrion	The power station. It makes most of the ATP, controls apoptosis, and keeps its own DNA.	Ch 8, Lect 2
Mitosis	The division of the nucleus, in which the replicated chromosomes separate.	Ch 8, Lect 2
Mitotic clock	The counting mechanism that turns cell division into a ceiling on how many times one cell may divide.	Ch 5, Lect 3, Lect 4
mitotic recombination	The exchange of chromosome segments that happens during mitosis.	Ch 8, Lect 3
Mitotic spindle	A rope of microtubules that pulls the copied chromosomes apart.	Ch 8, Lect 2
Mlh1	The mouse Mlh1 gene, used to build an animal model with MMR deficiency.	Ch 6, Lect 4
MLH1/PMS2	The second safety net of mismatch repair, the one that does the cutting.	Ch 5, Ch 6, Lect 4
MMR deficiency	The state in which the mismatch repair system has failed.	Ch 6, Lect 4
MOA	Mechanism of action, the evidence for what this compound actually does.	Ch 12, Lect 2
Molecular biology	The study of life at the level of molecules, above all DNA, RNA and protein.	Ch 1
Molecular cloning	The technique of putting a chosen piece of DNA into a small circular DNA and copying it in large numbers.	Ch 3, Lect 4
Molecule	Two or more atoms joined together in a fixed arrangement.	Ch 1
monocytes	Monocytes, which sit between the granulocytes and the lymphocytes on a flow plot.	Lect 2
Monosaccharide	A single sugar unit, such as glucose or fructose.	Ch 1, Lect 1
Monosomy	A chromosome that is present in one copy rather than two, written as 2n-1.	Ch 3, Lect 3
M phase	The part of the cell cycle in which the nucleus divides. Mitosis is the first half of it.	Ch 8, Lect 2
mRNA half-life	How long an mRNA survives before it is broken down. A short half-life means less protein.	Ch 7, Lect 5
mRNA (messenger RNA)	A photocopy of a recipe, carried from the nucleus to the factory floor.	Ch 2, Ch 3, Ch 7
MSH2/MSH6	The first safety net of mismatch repair, the one that recognises copying errors.	Ch 5, Ch 6, Lect 4
MSI-H	Microsatellite instability high, the label for a tumour full of unstable short repeats.	Ch 6, Ch 9, Lect 4
MSS	Microsatellite stable, which means the microsatellites stay stable.	Ch 6, Lect 4
MST1	The kinase at the start of the Hippo pathway. It activates LATS1.	Ch 7, Lect 1, Lect 5
Multinucleated cell	A cell with more than one nucleus, because the cytoplasm never divided.	Ch 8, Lect 2
Multiplex immunofluorescence	Staining several proteins at the same time, each one in a different fluorescent colour.	Ch 9, Ch 11, Lect 2
Multipotent	A cell that can become several cell types, but only within one family.	Ch 8, Lect 2
Multivariate logistic regression	The test that tells you whether a marker still gives independent information once several other factors have been taken into account.	Ch 4, Lect 3
Mutagen	Anything that causes a mutation, a chemical substance or a kind of radiation for instance.	Lect 3
Mutant splice site	A new splice site created by a single-letter change, which produces a shorter isoform than the original.	Ch 7, Lect 5
Mutation	A typo in the printed manual.	Ch 3, Lect 3
Myosin	The motor protein that builds the contractile ring together with actin.	Ch 8, Lect 2
NAHR (non-allelic homologous recombination)	Recombination between two similar but not identical stretches of DNA, which produces chromosomal rearrangements.	Lect 3
Natural killer cell (NK cell)	An immune cell that can kill tumour cells without being told which target to go for.	Ch 2, Ch 9, Lect 2
Necrosis	Uncontrolled cell death after an injury, in which the contents leak out and set off inflammation in the surrounding tissue.	Ch 8, Lect 2
Neoantigen	A fragment of protein the immune system has never seen, usually produced by a frameshift or by a new splice site.	Ch 9, Ch 10, Lect 1, Lect 4, Lect 5
Neutron	An uncharged particle that sits inside the nucleus. It adds mass without changing chemical properties.	Ch 1
NF-kappaB (nuclear factor kappa B)	A transcription factor held in the cytoplasm by I-kappa-B until a signal such as LPS moves it into the nucleus.	Ch 7, Lect 1, Lect 5
NICD	The part of Notch inside the cell, which is cut loose, moves into the nucleus and acts as a transcription factor.	Ch 7, Lect 5
Nick	A break in only one strand, with the other strand still intact.	Ch 5, Lect 4

Term	In one sentence	Where
Nivolumab	An anti-PD-1 checkpoint inhibitor.	Lect 3
NMR	Nuclear magnetic resonance. It is used to work out the shape of molecules in solution.	Ch 10, Ch 11, Lect 1
NNK	One of the tobacco-specific nitrosamines, which is activated by liver P450 and then binds to DNA.	Lect 3
NNN	One of the two tobacco-specific nitrosamines.	Lect 3
non-covalent	A way of bonding that does not involve sharing a pair of electrons.	Ch 4, Lect 3
Non-disjunction	Chromosomes failing to separate properly during meiosis, which produces an abnormal chromosome number.	Ch 3, Lect 3
Non-homologous end joining (NHEJ)	The repair route that needs no template when it fixes a broken chromosome, and therefore makes mistakes. It is also what builds antibody diversity.	Ch 6, Ch 9, Lect 4
Non-obviousness	The "non-obvious" requirement that a patent application has to get past.	Ch 12, Lect 4
Nonsense mutation	A single-letter change that creates a stop codon early, and the protein is cut short.	Ch 3, Lect 3
Notch	The receptor that lets one cell tell its neighbour to change its gene expression.	Ch 7, Lect 5
NOTCH2	One of the target genes switched on by YAP, through TEAD, when the Hippo pathway is off.	Ch 7, Lect 1, Lect 5
NotI	Another restriction endonuclease used in molecular cloning.	Ch 3, Lect 4
NPM1	His Lecture 3 worked example: you have to put CNV copy number together with mRNA expression before you can call it a driver.	Ch 10, Lect 3
N-terminal domain (NTD)	The regulatory region at the start of a nuclear receptor. It is not the same thing as a signal peptide.	Ch 2, Ch 7, Lect 5
N-terminal signal peptide	The short postal address at the very front of a protein that routes it to the endoplasmic reticulum and so into the secretory pathway.	Ch 2, Lect 2, Lect 5
N-terminus	The first end of a protein chain, where the amino acid keeps a free amino group.	Ch 2
Nuclear envelope	The double membrane that seals in the cell nucleus.	Ch 8, Lect 3
Nuclear pore	The doorway in the nuclear membrane, letting messages out and proteins in.	Ch 7, Ch 8
Nuclear receptor	A protein that binds a hormone or a drug inside the cell and then switches genes on in the nucleus.	Ch 7, Lect 5
Nuclear transcription factor	A protein that enters the nucleus and switches genes on or off there.	Ch 7, Lect 5
Nucleic acid	The class of biological molecules that includes DNA and RNA. It carries the information.	Ch 1
Nucleoid	The place where the DNA sits inside a bacterial cell, with no membrane around it.	Ch 8, Lect 5
Nucleolus	The dense region inside the nucleus where the parts of the ribosome are made.	Ch 8, Lect 2
Nucleoside	A base plus a sugar, with no phosphate. The building block before the phosphate goes on.	Ch 1, Ch 3, Lect 1
Nucleosome	DNA wrapped around a spool of histone protein. The basic unit of chromatin packaging.	Ch 3, Ch 4, Lect 3
Nucleotide	One base plus one sugar plus one phosphate. The building block of DNA and RNA.	Ch 1, Ch 3
Nucleotide excision repair (NER)	The pathway that repairs bulky damage such as a UV dimer, and it cuts out a whole stretch of the strand.	Ch 6, Lect 4
Nucleus	The control room. It keeps the DNA and decides which genes get read.	Ch 8, Lect 2
octamer	The structure at the core of a nucleosome, made of eight histone molecules.	Ch 4, Lect 3
Odds ratio	A way of expressing how many times more likely something is in one group than in another.	Ch 4, Lect 3
Okazaki fragment	A short stretch of DNA made first on the lagging strand and joined up later.	Ch 5, Lect 4
Oncogene	A gene that pushes the cell to divide once it has been switched on too far. It is a driver after it is mutated.	Ch 10, Lect 1, Lect 3, Lect 4
Operator	The stretch of DNA where the repressor sits. While the repressor is on it, the gene behind it cannot be transcribed.	Ch 7, Lect 5
Operon	A group of neighbouring genes switched on and off together by one shared control region.	Ch 7, Lect 5
Organelle	A compartment inside the cell that has one job, like a room in a factory.	Ch 8, Lect 2
Organoid	A small piece of human tissue grown from stem cells, used to test drugs.	Lect 2
origin of replication	The marked position on the DNA where replication starts.	Ch 5, Lect 4

Term	In one sentence	Where
orthology	Homologous genes in different species whose functions correspond.	Ch 4, Lect 1
Oxidation	Losing electrons, or gaining oxygen. It releases energy.	Ch 1
Oxidative phosphorylation	The process by which the mitochondria use oxygen to make most of the ATP.	Ch 8, Ch 10, Lect 2
p16	The tumour-suppressor protein encoded by CDKN2A.	Ch 4, Lect 3
p300/CBP	The proteins that HIF-1 recruits to help it drive transcription of its target genes.	Ch 7, Lect 5
P450	Cytochrome P450, the family of liver enzymes that turn drugs and toxins into more active molecules.	Lect 3
p50 and p65	The pair of NF-kappaB subunits that go into the nucleus and read the promoter of an inflammatory gene.	Ch 7, Lect 5
p53	The tumour suppressor protein that makes a cell carrying damaged DNA stop and not divide.	Ch 6, Ch 10, Lect 2, Lect 3, Lect 4
Palbociclib	A CDK4/6 inhibitor that stops the cell cycle at G1.	Ch 8, Lect 2
PAM	The short DNA sequence that the CRISPR enzyme requires to be present next to its target.	Ch 6, Lect 2, Lect 4
p-arm	The short arm of a chromosome, above the centromere.	Ch 3, Lect 3
PARP inhibitor	A drug that blocks a DNA repair enzyme, so it kills BRCA-deficient tumour cells while letting normal cells survive.	Ch 6, Ch 10, Lect 4
Pasha	The partner of the miRNA cutting machine, the one that lets Drosha recognise the hairpin to be cut.	Ch 7, Lect 1, Lect 5
Passenger	A hitchhiker mutation that gives no growth advantage of its own.	Ch 3, Ch 10, Lect 3
patent pool	A league of patent holders who license from each other.	Ch 12, Lect 4
paternal	On the father's side.	Ch 8, Lect 2
P-body	The place in the cytoplasm where silenced mRNA is gathered up and destroyed.	Lect 5
PCA (principal component analysis)	The method that squeezes thousands of measurements down onto a few axes, so that cells can be drawn as points.	Ch 10, Ch 11, Lect 1
PCR	Polymerase chain reaction, which copies a chosen piece of DNA into millions of copies so it can be measured.	Ch 3, Ch 11, Lect 1, Lect 3, Lect 4
PD-1	A checkpoint receptor on the T cell. When PD-L1 binds it, the T cell is switched off.	Ch 9, Lect 2, Lect 4
PD-L1	The protein on the tumour cell that binds PD-1 and switches the T cell off.	Ch 9, Lect 2, Lect 3, Lect 4
PD-L2	PD-L2, the second ligand for PD-1.	Ch 9, Lect 2
PECAM-1	Hold on to it and a white blood cell can squeeze out between two cells and leave the blood vessel.	Ch 9, Lect 2
Pembrolizumab	An anti-PD-1 checkpoint inhibitor. It is the drug for MSI-high tumours of the hot kind.	Lect 3
Penetrance	What proportion of the people carrying a mutation actually develop the disease. It is never 100%.	Ch 8, Lect 3
Peptide	A short chain of amino acids joined by peptide bonds.	Ch 2
Peptide bond	The chemical bond that joins one amino acid to the next along a protein chain.	Ch 2
Perforin	A protein that punches holes in the target cell so that granzyme can get in.	Ch 9, Lect 2
pH	The scale of acidity and alkalinity. Below 7 is acidic, above 7 is alkaline, and 7 is neutral.	Ch 1
PHD and FIH	The enzymes that add an oxygen group to HIF-1alpha, and that step is what marks it for destruction.	Ch 7, Lect 5
Phenotype	The way a cell or a person actually turns out.	Ch 7, Lect 5
Phosphatase	The enzyme that removes the phosphate group and switches the protein back off.	Ch 1, Ch 2, Lect 1
Phosphodiester bond	The chemical bond that joins adjacent nucleotides and forms the backbone of a DNA strand.	Ch 5, Lect 4
Phospholipid	It has a hydrophilic head and a hydrophobic tail, and that is exactly why a membrane can assemble itself.	Ch 1
Phosphorylation	Adding a phosphate group, and that usually means switching a protein on.	Ch 1, Ch 2, Ch 7, Lect 1
PI3K	The signalling kinase downstream of the receptor, the main carrier of the HER2 message.	Ch 10, Lect 2
pinhole	The small hole a confocal microscope uses to block out out-of-focus light.	Ch 9, Ch 11, Lect 2
PKC (protein kinase C)	A signalling enzyme that can switch on inflammation and growth pathways.	Lect 1, Lect 5

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Term	In one sentence	Where
Plasma membrane	The outer skin of the cell. A lipid bilayer that controls what gets in and out and carries the receptors.	Ch 8, Lect 2
plasmid	A small circular DNA in bacteria that can replicate on its own.	Ch 3, Lect 4
Pluripotent	A cell that can become almost any type of cell in the body.	Ch 8, Lect 2
PNCTR	A long non-coding RNA whose YYUCUY motif binds PTBP1 and so controls splicing.	Ch 7, Lect 5
POB-DNA	One of the DNA adducts formed by benzo(a)pyrene.	Lect 3
Point mutation	A change in a single letter of the DNA. It may be harmless, or it may be a driver.	Ch 3, Lect 3
POL	The segment of DNA polymerase that does the synthesis.	Ch 5, Lect 4
Polarity	One end of a molecule carries a little positive charge and the other a little negative, like a small magnet.	Ch 1
Pold1	The gene for DNA polymerase delta, the one that copies the lagging strand.	Ch 5, Lect 4
Pole	The gene for DNA polymerase epsilon, the enzyme that copies the leading strand.	Ch 5, Lect 4
Pole e/e Pold1 e/e	A double-mutant mouse that has lost the proofreading ability of both polymerases at once.	Ch 5, Lect 4
Polyadenylation	Adding the poly-A tail to the end of an mRNA. Which site you pick is itself a regulatory step.	Ch 7, Lect 5
Polyadenylation site (PAS)	The place on the RNA where the poly-A tail gets added. Choosing an early one or a late one changes the message.	Ch 7, Lect 5
Poly-A tail	The run of adenosines at the end of an mRNA. The longer the tail, the more translation.	Ch 7, Lect 5
Polycistronic mRNA	One RNA message that carries several genes at once, the way a bacterial operon does.	Ch 7, Lect 5
polymerase slippage	The polymerase slips when it copies a repeated sequence.	Ch 6, Lect 4
Polypeptide	A long chain of amino acids. Folded up, it is a protein.	Ch 2
Polysaccharide	A long chain of sugar units, such as glycogen, starch or cellulose.	Ch 1, Lect 1
Polyubiquitination	Stringing a chain of ubiquitin molecules onto a protein, which in this course's context is the label that leads to degradation.	Ch 7, Lect 5
Positive-feedback loop	The product of a reaction switches the same reaction on again, so the reaction amplifies rapidly.	Ch 4, Ch 7, Lect 3, Lect 5
Post-translational modification	A chemical change made to a protein after the chain has been made.	Ch 2, Lect 1, Lect 5
pRb	The protein that holds the cell cycle at G1 until the cell is ready to copy its DNA.	Ch 8, Lect 2
preclinical	The whole stretch of development work that comes before formal testing in humans.	Ch 12, Lect 2
Predictive biomarker	A marker that predicts whether a particular patient will respond to a particular treatment.	Ch 10, Ch 11, Lect 1
pre-IND	An informal meeting with the FDA before a company formally submits its IND.	Ch 12, Lect 2
Pre-initiation complex	The whole machine, made of general transcription factors plus RNA polymerase, that is assembled on the promoter before transcription starts.	Ch 7, Lect 5
Pre-mRNA	The first RNA copy of a gene, before it has been spliced.	Ch 7, Lect 5
pre-pre-IND	The first, very early sounding-out that comes before a pre-IND meeting.	Ch 12, Lect 2
Primary structure	The amino acid sequence of a protein, read from one end to the other.	Ch 2
Primase	The enzyme that lays down the short RNA primer that gives DNA polymerase a place to start.	Ch 5, Lect 4
Primer	A short RNA starting point that gives DNA polymerase something to attach to.	Ch 3, Lect 1, Lect 3, Lect 4
Primer annealing	The step where the primer, once cooled, finds its spot on the template and sticks.	Ch 3, Ch 11, Lect 1
Primer pair	The pair of primers PCR needs, one for each strand.	Ch 3, Ch 11, Lect 4
Priority Review	The priority review status the FDA gives to an application that represents a major therapeutic breakthrough.	Ch 12, Lect 1
Prognostic biomarker	A marker that tells you how the disease will go, whatever treatment you give.	Ch 10, Ch 11, Lect 1
progression-free survival	The time a patient survives without the disease getting worse.	Ch 9, Ch 12, Lect 2
Prokaryote	A cell with no nucleus. Bacteria and archaea are both prokaryotes.	Ch 8, Lect 2, Lect 5
Proliferative signal	A message that tells the cell to start copying its DNA.	Ch 10, Lect 2
Promoter	The word in the DNA that tells RNA polymerase where to start.	Ch 7, Lect 5

Term	In one sentence	Where
Proofreading	DNA polymerase checks its own work as it goes, cutting out the wrongly placed base and putting the right one in.	Ch 5, Lect 4
Propidium iodide (PI)	A dye that stains DNA, so once it is in a flow cytometer you know how much DNA each cell has.	Ch 11, Lect 2
Protease	An enzyme that cuts proteins open using its active site.	Ch 3
Proteasome	The protease machine that takes apart a protein once it has been given the recycling label.	Ch 7
Protein	A machine made by folding a string of amino acid beads; its shape decides what it does.	Ch 1
Protein domain	A stretch of protein that folds on its own and does one identifiable job.	Ch 2, Lect 1
Protein kinase	The enzyme that puts a phosphate group onto another protein, and ATP is its fuel.	Ch 1, Ch 2, Lect 1
Proteolytic cleavage	Cutting a protein once with an enzyme, and the fragment that comes off changes its identity.	Ch 7, Lect 5
Proteome	The whole set of proteins that a cell or a tissue has made at one moment.	Ch 2, Lect 1
Proteomics	The study of the whole set of proteins made by a cell, used to find biomarkers and drug targets.	Ch 11, Lect 1
Proton	A positively charged particle, and the ones in the nucleus decide the identity of the element.	Ch 1
P-selectin	One of the selectins that make white blood cells roll along the wall of a blood vessel.	Ch 9, Lect 2
P-value	How likely the data would look this extreme if the null hypothesis were true.	Ch 4, Ch 9, Ch 10, Lect 1, Lect 3
pyrimidine dimers	Adjacent pyrimidine bases stuck together by UV light.	Ch 6, Lect 4
Pyrogen	A substance that raises body temperature and brings on a fever.	Ch 1, Lect 1
Pyruvate	The product at the bottom of glycolysis, and where it goes decides what the cell does.	Ch 1, Ch 10, Lect 1
q-arm	The long arm of a chromosome, the side below the centromere.	Ch 3, Lect 3
qPCR	Quantitative PCR: it uses a fluorescent signal to measure how much of a particular stretch of DNA is in the sample.	Ch 11, Lect 1, Lect 4
Quaternary structure	The way several separately folded chains come together to make one working whole.	Ch 2
Quiescent	The state of being alive but not dividing, which is G0.	Ch 8, Lect 2
RAD51	The protein that winds itself around single-stranded DNA in HR and leads the way into the sister chromatid.	Ch 6, Lect 4
Rad51, Rad52 and RPA	The three players that carry out strand invasion.	Ch 6, Lect 4
Radiation	An outside source of damage that directly breaks both strands of DNA.	Ch 6, Lect 4
rcDNA	Relaxed circular DNA: the loose circular DNA HBV carries before it enters the nucleus.	Lect 2
Reactive oxygen species (ROS)	Highly reactive molecules produced by the cell's own metabolism, which put dents in DNA.	Ch 6, Lect 3
Receptor	A protein built to catch one specific signal molecule, and changed in shape by it.	Ch 7, Ch 8
Receptor occupancy	How much of the receptor has to be occupied by the drug before you get a clinical effect.	Ch 7, Ch 10, Lect 2, Lect 5
Receptor tyrosine kinase (RTK)	A membrane receptor that carries its own kinase activity, and HER2 is one.	Ch 10, Lect 2, Lect 5
recipient plasmid	In molecular cloning, the plasmid backbone that has been cut open and is ready to receive the insert.	Ch 3, Lect 4
recombinant chromatids	The chromatids that carry a new combination after a crossover.	Ch 8, Lect 2, Lect 4
recombinant plasmid	The recombinant plasmid formed once the gene of interest has been joined into the backbone.	Ch 3, Lect 4
Recombinant protein	A protein made for you by a host cell, after you have handed it a human gene.	Ch 7, Lect 5
recombination-signal sequences	The signal sequences at the ends of the V, D and J segments that tell RAG where to cut.	Ch 6, Lect 4
Reduction	Gaining electrons (or losing oxygen), which is how energy gets stored inside a molecule.	Ch 1
Regulatory gene	A gene that makes a switch protein, for example lacI.	Ch 7, Lect 5
Replication bubble	The bubble where the DNA is pulled open, with one replication fork running on each side.	Ch 5, Lect 4

Term	In one sentence	Where
Replication fork	The Y-shaped opening where the double helix is unzipped and copied as it goes.	Ch 5, Lect 4
replicons	A stretch of DNA that begins at a single origin and can be replicated on its own.	Ch 5, Lect 4
Reporter assay	You use a gene that glows to measure how active a given promoter really is.	Ch 7, Lect 5
Reporter gene	A substitute placed at the position of a real gene, used to measure how busy a promoter is.	Ch 7, Lect 5
Repressible system	On all the time, until its own product switches it off.	Ch 7, Lect 5
Repressor	A protein that sits on the DNA and blocks transcription.	Ch 7, Lect 5
resolution	The smallest change that a detection method can tell apart.	Ch 3, Ch 4, Lect 3, Lect 4
Reverse transcriptase	An enzyme that makes DNA using RNA as the template.	Ch 5, Lect 3, Lect 4
R-group (side chain)	The side arm on an amino acid that can change, and it decides how the whole chain folds.	Ch 2
Ribose	The five-carbon sugar in RNA; DNA uses its deoxy version.	Ch 3
Ribosome	The machine that reads the mRNA and joins amino acids together one by one.	Ch 8, Lect 2
RISC	The machine that uses a microRNA to switch off a target gene.	Ch 7, Lect 1, Lect 5
RNA-binding protein (RBP)	A protein that grabs hold of a stretch of RNA and changes how stable it is or how fast it gets read.	Ch 7, Lect 5
RNA interference (RNAi)	The older way of switching a gene off, using small RNA molecules.	Ch 8, Lect 2
RNA polymerase	The enzyme that copies a gene into RNA.	Ch 6, Ch 7, Lect 5
RNA polymerase II	The enzyme that copies protein-coding genes into precursor mRNA.	Ch 6, Ch 7, Lect 5
RNA (ribonucleic acid)	A whole family of molecules that read out and carry out the instructions in DNA.	Ch 1, Ch 3, Ch 7
RNase H	The exonuclease that shovels the RNA primer off the DNA.	Ch 5, Lect 4
RNA sequencing	The technology that reads all the RNA in a sample as sequences.	Ch 11, Lect 1
RNA vaccine	Deliver the mRNA and let the patient's own ribosomes build the antigen.	Lect 5
rolling	The first step, in which an immune cell rolls along the inside of a blood vessel.	Ch 9, Lect 2
Rough ER	The part of the endoplasmic reticulum that carries ribosomes, where new proteins are made and folded.	Ch 8, Lect 2
RPA	The protective protein complex that first covers the exposed single-stranded DNA.	Ch 6, Lect 4
rRNA	Ribosomal RNA, the bulk of the ribosome, and the part that lines the mRNA up.	Ch 8
RuvC	One of the two nuclease domains of Cas9, and it cuts the non-target strand.	Ch 6, Lect 2, Lect 4
RXR	The partner receptor that pairs with Type II nuclear receptors to form a heterodimer.	Ch 7, Lect 5
S-adenosyl methionine (SAMe)	The molecule that carries the methyl group, and the source of the methyl that goes onto DNA.	Ch 4, Lect 3
Sanger sequencing	The gold standard method for reading DNA, where a series of coloured peaks tells you which base is where.	Ch 3, Ch 11, Lect 3
SAV	The protein that carries the signal from MST to LATS in the Hippo pathway.	Ch 7, Lect 5
Scaffold	A molecule that ties two or more proteins together in the right place so that they can work as a team.	Ch 7, Lect 5
scFv	The smallest usable piece of an antibody, one heavy-chain variable region plus one light-chain variable region.	Ch 9, Lect 2
SCID (severe combined immunodeficiency)	Severe combined immunodeficiency: the failure to build a usable library of antibodies and T cell receptors.	Ch 6, Ch 9, Lect 2, Lect 4
SDS-PAGE	The standard gel electrophoresis for proteins, which lines the proteins up as a ladder by size.	Ch 2, Lect 1
Secondary structure	The local regular shape of a chain: the alpha-helix and the beta-sheet.	Ch 2
Second messenger	A small molecule inside the cell that carries the signal further along.	Ch 7, Lect 5
Secretory pathway	The outbound route, rough ER, then Golgi, then vesicle, then cell membrane, then outside the cell.	Ch 2, Ch 8, Lect 5
Seed sequence	The stretch of nucleotides 2 to 8 on a microRNA, the part that does the target recognition.	Ch 7, Lect 5
Selectin	A cell-surface protein that makes white blood cells roll along the walls of blood vessels.	Ch 9, Lect 2
Selective Estrogen Receptor Modulator (SERM)	A drug like tamoxifen: it blocks estrogen in the breast and slightly activates it in the endometrium.	Ch 7, Ch 10, Lect 5
Semi-conservative replication	Every new DNA double helix is made of one old strand and one new strand.	Ch 5, Lect 4

Term	In one sentence	Where
Senescence	The state in which a cell stops dividing but does not die.	Ch 5, Lect 3
Sensitivity	Of the patients who really have the disease, the proportion the test correctly picks out.	Ch 11, Lect 1
Sensors	The level of the DDR signalling chain whose job is to discover the damage.	Ch 6, Lect 4
Septic shock	The collapse that follows once sepsis overwhelms the body's own inflammatory response.	Ch 1, Lect 1
Serial deletion	You delete the promoter piece by piece and watch which deletion makes the function collapse, which tells you which part was actually doing the work.	Ch 7, Lect 5
Serum starvation	Dropping the serum in the medium from 10 % to 0.5-1 % so that the cells stop in G1.	Ch 8, Ch 11, Lect 2, Lect 4
sheath fluid	The fluid in a flow cytometer that squeezes the sample stream into a single thin line.	Ch 11, Lect 2
side scatter (SSC)	The scatter channel on a flow cytometer that reflects how complicated the inside of a cell is.	Ch 9, Ch 11, Lect 2
Signal peptide	The address label on the N-terminus of a protein that gets it delivered to the ER.	Ch 2, Lect 2
Signal transduction	Carrying a signal from the cell surface all the way to somewhere inside the cell.	Ch 7, Lect 5
Silencer	A stretch of DNA that turns transcription down, and it only works on the DNA molecule it happens to be sitting on.	Ch 7, Lect 5
Silent mutation	A mutation that changes a base without changing the protein.	Ch 3, Lect 3
siRNA	Small interfering RNA, a short double-stranded RNA used to switch off a specific gene.	Ch 7, Lect 1, Lect 5
Sister chromatid	The two identical copies of one chromosome that exist after DNA replication.	Ch 3, Ch 8, Lect 3
Small molecule	A chemical molecule the size of a drug, usually made by chemical synthesis rather than by biology.	Ch 10, Ch 12, Lect 4
Smooth ER	The part of the endoplasmic reticulum with no ribosomes attached, and it makes lipids.	Ch 8, Lect 2
SN-38	A topoisomerase-I inhibitor used as the cytotoxic payload of an ADC.	Ch 10, Lect 1, Lect 4
snRNP	A short piece of nuclear RNA plus its proteins; several of them together make up the splicing machine.	Ch 7, Lect 5
Sodium bisulfite	A reagent that turns unmethylated cytosine into uracil and leaves everything else alone.	Ch 4, Ch 11, Lect 3
solid-tumour barriers	The collection of obstacles that make CAR-T fail in a solid tumour.	Lect 2
Solvent	The liquid that other things dissolve in; for a cell, that liquid is water.	Ch 1
somatic recombination	Somatic recombination, which is the source of antibody diversity.	Ch 6, Lect 1, Lect 4
Specificity	Of the people who really are well, the proportion the test correctly clears.	Ch 11, Lect 1
S phase	The stretch of the cell cycle in which DNA is copied.	Ch 8, Lect 2
Spliceosome	The machine that cuts the introns out and joins the exons together.	Ch 7, Lect 5
Splicing	Cutting the introns out of pre-mRNA and joining the exons together to make mature mRNA.	Ch 7, Ch 9, Lect 5
Splicing factor	A protein that helps decide which exons get joined together.	Ch 7, Lect 5
Starch	The polysaccharide that plants use to store sugar, the stored form of glucose.	Ch 1
Start codon	AUG: the signal that starts translation, and at the same time the code for methionine.	Ch 1, Ch 2, Lect 1
STAT3	A transcription factor that, on his Lecture 3 slide, is what pushes PD-L1 expression up.	Ch 4, Lect 3
Stem cell	A cell that can copy itself and can also turn into a more specialised cell.	Ch 8, Lect 2
sticky end	The single-stranded overhang left behind when a restriction enzyme cuts.	Ch 3, Lect 4
Stop codon	UAA, UAG and UGA, where translation stops.	Ch 1, Ch 2, Lect 1
Strand	One of the two long chains that make up the DNA double helix.	Ch 3, Ch 5
Structural gene	A gene that makes the products the cell actually needs, like the enzyme that digests lactose.	Ch 7, Lect 5
structural tractability	Whether this molecule can be attacked, given its shape.	Ch 12, Lect 4
Structural variant	Large-scale rearrangements of the chromosome: deletion, duplication, inversion and translocation.	Ch 3, Lect 3
Substitution	One kind of structural variant: a piece of chromosome cut out and inserted into another chromosome.	Ch 3, Lect 3
Substrate	The thing an enzyme acts on.	Ch 2

Term	In one sentence	Where
Subunit	One of the chains that go into a protein made of several chains.	Ch 2
succinimide	The short-lived intermediate formed when asparagine is deamidated.	Ch 1, Ch 2, Lect 1
Sugar (carbohydrate)	One of the four big biological molecules, used mostly as fuel and also as building material.	Ch 1, Lect 1
Sugar-phosphate backbone	The backbone of one strand of DNA, made of sugars and phosphate groups alternating along it.	Ch 1
surrogate endpoints	Using a cheap and easy measurement in place of the endpoint you really want.	Ch 10, Ch 11, Ch 12, Lect 4
Suv39h	The enzyme that spreads the H3K9 methylation along the chromatin.	Ch 4, Lect 3
synNotch	A synthetic Notch, so that the presence of a neighbouring cell switches on one of your own genes.	Ch 7, Lect 5
Synthetic biology	Engineering new biological parts and switches.	Ch 7, Lect 5
Synthetic lethality	Two defects that the cell can live with one at a time but not together.	Ch 6, Ch 10, Lect 4
Tamoxifen	A drug that blocks estrogen in the breast and slightly activates it in the endometrium.	Ch 7, Ch 10, Lect 5
TAMs	Tumour-associated macrophages, the macrophages that gather round a tumour.	Ch 9, Lect 2
target identification	Finding, at the very start of drug development, the molecule the disease actually depends on.	Ch 12, Lect 2, Lect 4
Target nomination	The decision a pharmaceutical company makes about which molecule to go after.	Ch 12, Lect 1, Lect 4
TATA box	The key DNA word roughly 30 bases upstream of the promoter, where the transcription machine assembles itself.	Ch 7, Lect 5
tbid	The fragment cut out of bid by caspase-8, which links the extrinsic pathway to the mitochondrion.	Ch 8, Lect 2
TBP (TATA-box-binding protein)	The part of TFIID that recognises the TATA box and sits on it.	Ch 7, Lect 5
TCA cycle	The circular set of reactions in the mitochondrion that takes the fuel apart and loads the energy into NADH and FADH2.	Ch 8, Lect 2
T-cell	A lymphocyte of the immune system, split into helper, cytotoxic and regulatory kinds.	Ch 6, Ch 9, Lect 2, Lect 4
TC-NER	The repair route that fixes damage on the gene currently being transcribed.	Ch 6, Lect 4
TCR (T-cell receptor)	The recognition protein on the surface of a T cell, which reads a foreign fragment displayed by another cell.	Ch 6, Ch 9, Lect 4
TEAD	The DNA-binding protein that works alongside YAP in the nucleus.	Ch 7, Lect 5
Telomerase	A special enzyme that can rebuild the ends of chromosomes, and which most cancer cells switch back on.	Ch 5, Ch 12, Lect 3, Lect 4
Telomere	The repeated DNA tail at the end of a chromosome, which gets a little shorter with every division.	Ch 3, Lect 3, Lect 4
TEM	Transmission electron microscope.	Ch 5, Lect 2
Template strand	The strand of DNA that is being read during replication.	Ch 5, Lect 1
Teratoma assay	The gold standard for deciding whether a cell really is pluripotent: inject it into a mouse and see whether tissue from three germ layers grows.	Ch 8, Lect 2
TERC	The RNA template part of telomerase.	Ch 5, Lect 4
Tertiary structure	The three-dimensional shape a whole polypeptide chain takes once it has folded up.	Ch 2
TET	A family of enzymes that oxidise the methyl mark off, the executors of DNA demethylation.	Ch 4, Lect 3
TFIID	The first general transcription factor to arrive at the promoter, and it brings TBP with it.	Ch 7, Lect 5
TFIIH	The transcription factor that pulls the DNA near the promoter open so that RNA polymerase II can get going.	Ch 6, Lect 4, Lect 5
TGFbeta	The cytokine that a tumour uses to suppress the immune attack around it.	Ch 9, Lect 2
Therapeutic window	The margin between the lowest effective dose and the highest tolerated dose.	Ch 6, Ch 10, Lect 4
Threshold effect	Disease appears only when the proportion of mutant mitochondrial DNA rises above the tolerance limit of a particular tissue.	Ch 8, Lect 2
thymine	One of the four main bases of DNA, written T.	Ch 3, Lect 1
TIM3	An inhibitory checkpoint receptor, listed alongside PD-1.	Ch 9, Lect 2
TINCR	A long non-coding RNA that controls the stability of a message by pairing with the mRNA and then grabbing hold of Staufen 1.	Ch 7, Lect 5

Term	In one sentence	Where
TLR4	The cell-surface receptor that recognises LPS, and the first stop of the sepsis cascade.	Ch 1, Ch 7, Lect 1, Lect 5
TMZ	Temozolomide, an alkylating chemotherapy drug.	Ch 4, Lect 3
TNF-alpha	The first cytokine to appear in the NF-kappaB inflammatory response, peaking at 4 to 6 hours.	Ch 7, Lect 1, Lect 5
tobacco-specific nitrosamines	Carcinogenic precursors found only in tobacco.	Lect 3
tolerance	The tolerance the immune system learns for the body's own proteins.	Ch 6, Lect 1
Topoisomerase	The enzyme that cuts DNA ahead of the replication fork, lets it swivel, and puts it back, to release the twisting strain.	Ch 5, Lect 4
toxicology	The study of what a compound does to a whole animal.	Ch 12, Lect 2
Toxic pre-expression	Making the recombinant protein before the host is ready for it, which can kill the culture.	Ch 7, Lect 5
TRAIL	The ligand that starts apoptosis from an outside signal.	Ch 8, Lect 2
TRAILR2	The receptor for TRAIL, which is also called DR5.	Ch 8, Lect 2
Trans-acting factor	A diffusible protein that can act on a gene somewhere else, even on a different chromosome.	Ch 7, Lect 5
Transcription	Copying the recipe in the DNA into a working message that can be carried out of the nucleus.	Ch 7, Lect 5
Transcription factor	A protein that recognises a particular word in the DNA and helps or blocks transcription there.	Ch 3, Ch 7, Lect 5
Transcription start site	The base at which the new RNA copy really begins.	Ch 7, Lect 5
Transducer	A protein that converts one signal into another. RAS is one.	Ch 6, Ch 10, Lect 1
Trans face	The shipping end of the Golgi, where finished proteins are packed into vesicles.	Ch 8, Lect 2
Transforming principle	The name Griffith gave to the thing that turns dead bacteria into live ones, later shown to be DNA.	Ch 3, Lect 1
Translation	Reading the mRNA and assembling the protein exactly as the description says.	Ch 7, Lect 1, Lect 5
Translocation	A piece of one chromosome trading places with a fragment of another.	Ch 3, Lect 3
Transmembrane domain	The hydrophobic stretch of a protein that sits inside the membrane; take it away and the protein dissolves into the water and can be measured in blood.	Ch 9, Lect 2, Lect 5
transmembrane segments	The stretch of a protein that is buried in the membrane bilayer.	Ch 9, Lect 2
transmigration	The step in which an immune cell squeezes through the vessel wall into the tissue.	Ch 9, Lect 2
Transport vesicle	A small membrane bubble that carries cargo from one compartment to the next.	Ch 8
trastuzumab-deruxtecan	An ADC against HER2.	Ch 10, Lect 1, Lect 4
Trastuzumab (Herceptin)	An antibody drug against HER2, used in HER2-positive breast cancer and gastric cancer.	Ch 7, Ch 10, Lect 2, Lect 5
Tregs	Regulatory T cells: the brakes on the immune system.	Ch 9, Lect 2
Triglyceride	Three fatty acids joined to one glycerol, the way the body stores fuel.	Ch 1, Lect 1
Trisomy	A chromosome present in three copies rather than two, written as 2n+1.	Ch 3, Lect 3
Trisomy 21	Down syndrome: three copies of chromosome 21.	Ch 3, Lect 3
tRNA (transfer RNA)	The adaptor molecule that delivers amino acids to the ribosome and reads the codons.	Ch 7, Lect 1, Lect 2
Trop2	A cell-surface protein that is an antibody target in lung cancer and breast cancer.	Ch 10, Lect 1, Lect 4
Trp repressor	The protein that grabs the operator once it has bound tryptophan and switches off the tryptophan synthesis system.	Ch 7, Lect 5
Trp (tryptophan) operon	The tryptophan synthesis system: it makes its own tryptophan, and it decides when to stop making it.	Ch 7, Lect 5
Tumor microenvironment	The whole neighbourhood of a tumour: cancer cells, immune cells, fibroblasts and stroma all together.	Ch 9, Lect 2
Tumor suppressor	A gene whose normal job is to restrain division; lose it and the cell runs away.	Ch 4, Ch 10, Lect 3
Turner syndrome (45,X)	A woman with one X chromosome instead of two.	Ch 3, Lect 3
Type II nuclear receptor	Like the thyroid hormone receptor, already sitting on DNA with RXR, just waiting for a ligand to swap the team.	Ch 7, Lect 5
Type I nuclear receptor	Like the estrogen receptor, it is held in the cytoplasm by a chaperone until the ligand arrives and it can set off.	Ch 7, Lect 5
TZAP	The protein that shortens over-long telomeres and cuts off the t-loop.	Ch 5, Lect 4

Term	In one sentence	Where
Ubiquitin	A small protein chain that gets hung on another protein as the label that says "send this for destruction".	Ch 7, Lect 5
ultramutator	A cell whose mutation rate has been pushed up to an extreme level.	Ch 9, Lect 4
Ultraviolet light	An external injury that glues adjacent pyrimidine bases together.	Ch 6, Lect 4
Undraggable target	A target that no drug can attack, with K-Ras being the most famous example until very recently.	Ch 10, Lect 1
unmethylated	A cytosine that has not had a methyl group added to it.	Ch 4, Ch 11, Lect 3
Unsaturated fatty acid	A fatty acid with one or more double bonds in the chain, which makes it kinked and unable to pack tightly.	Ch 1
Untranslated region (UTR)	The two stretches of an mRNA that are not turned into amino acids: the 5' UTR before the start and the 3' UTR after the end.	Ch 7, Lect 5
uracil	The base that replaces thymine in RNA, written as U.	Ch 3, Ch 11, Lect 1, Lect 3
Variable region	The tip of the Y-shaped arms of an antibody, different in every antibody, and responsible for grabbing the antigen.	Ch 2, Ch 9
VCAM-1	The adhesion molecule that lets a white blood cell grip the wall of a blood vessel.	Ch 9, Lect 2
V(D)J recombination	Rearranging the V, D and J gene segments to make the diversity of antibodies and TCRs.	Ch 6, Ch 9, Lect 2, Lect 4
vector	The vehicle that carries DNA into a cell.	Ch 3, Lect 4
VEGF	Vascular endothelial growth factor, switched on by HIF-1alpha, which lets a tumour build its own blood vessels.	Ch 7, Lect 5
Vesicle	A small bubble wrapped in a membrane that moves cargo between the compartments of a cell.	Ch 8, Lect 2
VH	The variable region of the antibody heavy chain.	Ch 2, Lect 1, Lect 2
VHL	The protein that recognises a marked HIF-1alpha and hands it over to the proteasome for breakdown.	Ch 7, Lect 5
Vimentin	A structural protein inside the cell that leaks into the blood in liver cancer, and Dr. Wong's hands-on biomarker example.	Ch 2, Ch 11, Lect 1
VIP (vasoactive intestinal peptide)	A neuropeptide that can switch T cells off and help the tumour hide.	Background supplement (not lecture content)
VISTA	An inhibitory checkpoint sitting at the interface between the immune cell and the tumour cell.	Ch 9, Lect 2
VL	The variable region of the light chain of an antibody.	Ch 2, Lect 1, Lect 2
Warburg effect	A cancer cell ferments glucose into lactate even when there is oxygen, because what it wants is raw material rather than fuel.	Ch 10, Lect 1, Lect 2
Water	The solvent of the cell, whose unusual shape gives it a high melting point and a great dissolving power.	Ch 1
Watson, Crick and Franklin	The three people who built the double helix model, using Rosalind Franklin's X-ray images.	Ch 3, Lect 1
Western blot	The standard protein test: does this sample contain this particular protein?	Ch 11, Lect 1
Wild type	The normal, unmutated version.	Ch 5, Ch 10, Lect 1
XIST	A long non-coding RNA that silences an entire X chromosome; the classic example of an lncRNA.	Ch 7, Lect 5
X-linked dominant	An inheritance pattern carried on the X chromosome, in which it appears in every generation.	Lect 2
X-linked recessive	An inheritance pattern carried on the X chromosome that mainly affects males.	Lect 2
XPA	The protein that positions the NER enzyme team on the damaged DNA.	Ch 6, Lect 4
XPC	The protein that finds the damaged positions across the whole genome in GG-NER.	Ch 6, Lect 4
XPF and XPG	The two endonucleases that cut the damaged piece out in NER.	Ch 6, Lect 4
X-ray crystallography	The method that solves the precise three-dimensional atomic shape of a protein.	Ch 2
Yamanaka factors	Oct3/4, Sox2, c-Myc and Klf4 - the four factors that reprogram a fibroblast into an iPSC, the induced pluripotent stem cell.	Ch 8, Lect 2
YAP	The protein at the end of the Hippo pathway; when it is not phosphorylated it enters the nucleus and starts pushing growth.	Ch 7, Lect 1, Lect 5
ZAP70	The kinase that carries the signal inside a CAR.	Ch 9, Lect 2
ZFN	Zinc finger nuclease, the older protein-based method for cutting DNA at a chosen spot.	Ch 6, Lect 2, Lect 4

Term	In one sentence	Where
Zinc finger	A small fold held open by one zinc ion, which lets a DNA-binding domain grip the DNA.	Ch 7, Lect 5

905 terms: 582 belong to the foundation chapters 1 to 7, 323 are forward references for the lectures.

PART C

Check yourself

Every chapter ends with a "check yourself" block: a few questions, and the answers immediately below them, so you can mark yourself without a teacher. The self-test quiz carries the same material in multiple choice, 905 questions over the whole course, with an English version and a Traditional Chinese version and a switch at the top.

> **The self-test quiz**
>
> Self-test quiz - 905 questions, one for every term in the glossary. Filter by chapter or by question type, and play the term's own Cantonese recording next to the answer.

The quiz is a web page, because it loads its question bank with **fetch()**, which a browser will not do from a file on disk. To use it from this folder, serve the folder and open the quiz:

```
cd "/Volumes/Download/projects/Video lecture"
python3 -m http.server 8000
# then open http://localhost:8000/_site/quiz.html
```

The Cantonese narration of the whole book is 18 hours across 13 chapter albums and 905 term files, in **_cantonese/assets/**. The player is **_cantonese/index.html**.